

Fully-Integrated CMOS RF Power Amplifiers For Modern 802.11 Applications

Graduation Project Thesis

Submitted to the School of Engineering at Nile University
in Partial Fulfillment of the Requirements for
The Degree of Bachelor of Science
in Electronics and Communication Engineering

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June, 2025



CMOS POWER AMPLIFIERS FOR MODERN 802.11 APPLICATIONS

ECEN495

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List of Abbreviations

1. **RF**: Radio Frequency
2. **PA**: Power Amplifier
3. **TX**: Transmitter
4. **RX**: Receiver
5. **WLAN**: Wireless Local Area Network
6. **P1dB**: 1dB Compression Point
7. **IIP3**: Third-order intercept point
8. **PAE**: Power-added efficiency
9. **EVM**: Error Vector Magnitude
10. **ACPR**: Adjacent Channel Power Ratio
11. **LNA**: Low Noise Amplifier
12. **LO**: Local Oscillator
13. **Demod**: Demodulator
14. **Mod**: Modulator
15. **CMOS**: Complementary Metal-Oxide-Semiconductor.
16. **MCS**: Modulation & Coding Scheme.
17. **PVT**: Process, Voltage, & Temperature.
18. **SCE**: Short Channel Effects.
19. **CLM**: Channel Length Modulation.

Acknowledgments

I would like to give gratitude to everyone that helped me in the realization of this project, starting with my professor and supervisor, **Dr. Ahmed H. Madian**, who has always supported me and was one of the most humble and kind professors I have personally met, and who played an important role in my growth in the field of integrated circuits and systems. Additionally, I would like to sincerely thank **Dr. Ahmed Mahmoud**, who was pivotal in my growth during my undergraduate years and equipped me with foundations in electromagnetics that were crucial to the realization of this project, in addition to his insights. Furthermore, I'd like to thank **Dr. Mohammed Karmoose** & **Dr. Tamer Abu Elfadl** for their support and enthusiasm in answering various inquiries and for their insights pertaining to communication & microwave theory.

I am also extremely grateful to a handful of engineers, beginning with **Eng. Haytham**, whose support cannot be understated and who has helped me on several occasions through guidance and through resources that helped me transition from the theory to actual implementation. Additionally, I'd like to thank **Eng. Abdelrahman Ashraf, Eng. Abdelrahman Hany, Eng. Mohamed Abdallah, and Eng. Omar Samer**, who also provided resources and knowledge that helped me in this journey.

I would like to thank my **family members** for always being there for me, for their support and genuine care, and for the environment that helped me become the person I am today.

And most importantly, praise to Allah, for equipping me with the reasons and strength to carry on in life and the uncountable gifts bestowed upon me.

Abstract

The demand for increasing data rates is an integral motivator for advancements in 802.11 standards such as WiFi & WiGig, as well as wireless communication in general, as seen with the demand for 5G systems and the newly emerging 6G technology. And as the world is moving towards higher frequencies, more research is being conducted to realize transceivers that could enable such endeavors. One of the most pivotal blocks which make up such transceivers is the Power Amplifier.

In this work, two fully integrated CMOS power amplifiers were designed for the $2.4GHz$ & $5GHz$ bands. At a worst case corner of around 125°, simulations show an output $P1dB$ of $18.9dBm$ & $20.8dBm$ for the entire $2.4 \rightarrow 2.5GHz$ & $5 \rightarrow 5.8GHz$ bands respectively using the PDK's passives, with gains of $\geq 26.7dB$ & $\geq 19.9dB$ within the entire frequency bands of interest and with unconditional stability from $1Hz$ to $100GHz$. The two designs also show peak PAE of $\approx 20.8\%$ & $\approx 24.3\%$. For a 64-QAM, 40MHz, 802.11n signal, the two PAs showcased $\leq -22dB$ EVM at $P_{out} = 15.4dBm$ & $15.3dBm$ respectively. For a 256-QAM, 802.11ac signal, the $5GHz$ PA showed $\leq -37dB$ EVM at a $P_{out} = 11.3dBm$. Both envelope tests showed compliance with the spectral mask. The PAs also showed an output IIP_3 of $\approx 39dBm$ & $\approx 40dBm$ at the worst case corner.

The first iterations of the post-layout simulation without optimization show a degradation of around $-8dB$ in the gain, a minimal degradation in the $OP1dB$ of both PAs by around $1dB$ and a PAE drop of around $2 \rightarrow 4\%$ at the nominal corner for both designs while retaining stability over a wide range. The layout integrates all passives and occupies areas of around $1.50 \times 0.84 \approx 1.26mm^2$ & $1.73 \times 0.61 \approx 1.05mm^2$ for the $2.4GHz$ & $5GHz$ designs respectively. Nonetheless, the efficiency and linearity specs remain satisfactory and show promise for further tuning for the two PAs. Such a PA could target IoT devices and mobile device that make use of the 802.11n and 802.11ac WiFi standards.

Thesis Organization

The goal of this graduation project is to cap off and integrate the knowledge built throughout the major in the fields of integrated circuits & systems, communication, as well as microwave theory, to design a Fully Integrated RF Power Amplifier for 802.11 applications. This work will primarily make use of CMOS technologies due to the integrable & cost-effective nature of silicon technologies.
The main deliverables of this project include, but are not limited to, a successful design, simulation, & layout of a fully-integrated PA using TSMC 65nm technology. With the design and schematic process itself and going through the PA flow as the main focus.

The thesis is organized in the following manner:

- **Chapter One** serves as the introduction and motivation for the topic of this graduation project, as well as its possible impact as demanded.
- **Chapter Two** serves as a more thorough, overview of the main concepts relevant to this work, as well as previous works.
- **Chapter Three** details the methodology present and the flow in detail.
- **Chapter Four** showcases the schematic design process, results and analyzes the PA performance.
- **Chapter Five** showcases the layout process, extraction, and post-layout simulations.
- **Chapter Six** contains the final conclusions of this work with ideas for improvement.
- **Chapter Seven** contains the appendices that contain supplementary information and notes.

The main focus of this works involves two designs for the $2.4GHz$ & $5GHz$ bands. These two designs were made in order to target both bands for the WiFi protocol, thus serving standards such as 802.11n, 802.11ac, and 802.11ax.

An initial stage of a $60Ghz$ future design is also shown. While this is not the main project, certain aspects of this design were included as it showcases certain considerations in the design of mmWave class PAs and also targets another subset of 802.11 applications: the WiGig protocol, which serves standards such as 802.11ad and 802.11ay. It also deals with loadpull and sourcepull analyses. It's included in the appendix.

All tables and images present in this work are generated by the author of this documented unless otherwise noted and cited.

1 Introduction

The evolution of wireless communication was driven by an ever-increasing demand for higher performance. As WiFi is getting more and more adopted by mobile devices and mobile networks evolve from 4G to 5G and beyond (to newly emerging technologies such as 6G), there is an active area of research in achieving greater data rates and making use of higher frequency bands to keep up with demanding applications performance-wise. Such an endeavor necessitates the advancement of high-performance radio frequency (RF) transceivers, with power amplifiers (PAs) being a pivotal component in the transmitter of these systems [6].

This project aims to design a Power Amplifier for use in smart and mobile devices that employ extensive use of WiFi capabilities.

1.1 Problem & Motivation

One of the main challenges in wireless communication is achieving high efficiency, linearity, and compact integration of RF front-end circuits. The power amplifier (PA) is a fundamental block of the transmitter and plays a fundamental role in determining the overall system efficiency and preserving signal integrity.

However, designing high-frequency PAs (2.4 GHz and 5 GHz for 802.11 applications) with optimized performance is challenging due to trade-offs between efficiency, linearity, and output power [1]. And for even higher bands, the problem becomes exponentially difficult due to the problems caused by parasitic elements at these frequencies in addition to devices getting pushed to their limits at such high frequencies.

Furthermore, CMOS-based RF systems are the main selling point of this project. While III-V semiconductor technologies have been traditionally favored for their superior RF performance, CMOS & BiCMOS technologies are especially lucrative to utilize due to their cost-effectiveness and highly integrable nature [3]. A particular challenge of CMOS-based RF systems is the fact that they usually have lower maximum switching frequencies than their III-V counterparts, in addition to the many lossy effects of the substrate and various on-chip components. [3]

1.2 Economic & Industrial Impact

The demand for highly efficient and low-cost RF front-ends is growing in consumer electronics, IoT devices, and telecommunications infrastructure. The market for 5G RF front-ends has already exceeded the $15 billion mark in 2023 [7]. And naturally, power amplifiers are one of the most pivotal and power hungry blocks in the system and an area of active, extensive, research and developments.

RF Front-End Chip Market Growth and Industry Expansion [10] [13]

- There's rapid growth in the RF front-end chip market, and it is further projected to grow at a 12.4% CAGR through 2030.
- The miniaturization and integration of RF modules enable compact, efficient, and cost-effective wireless devices.
- The demand for efficient RF modules drives the expansion of sectors like consumer electronics, automotive, and industrial automation. IoT and cloud-based applications in particular greatly benefit from the development of such chips.
- Industries ranging from mobile communications all the way to automatic radar systems and wireless sensor networks for IoT applications can benefit from this. Even in healthcare applications where a high data rate between miniaturized devices is present.

Cost Reduction and Integrability [10] [12] [14]

- Miniaturized and integrated RF modules require less material and can be manufactured more efficiently.
- This makes advanced wireless technology accessible to a broader market at lower costs and with less area.
- The use of CMOS in general, the main selling-point of this project, is one of the key aspects justifying its use despite its comparatively poorer performance.

Device Performance, Efficiency, & Lifespan [11] [12] [14]

- Efficient RF modules, such as those using efficient PAs, can significantly reduce power consumption. For example, Broadcom's WiFi 7 modules cut energy use by 40% compared to previous generations, while Qualcomm's latest modules improve battery life by 50% and increase efficiency by 35%.
- Lower power consumption is critical for battery-powered devices (smartphones, wearables, IoT sensors), enabling longer operation and reducing total cost of ownership for end users and enterprises.

It should come as no surprise that such endeavors are extensively invested in by world-renown universities, multinational companies, and even governments and defense sectors of the largest powers.

1.3 Objectives

The primary objective of this project is to design, simulate, and layout a CMOS power amplifier for 802.11 Wi-Fi applications at the 2.4 GHz and 5 GHz bands.

- Develop a comprehensive understanding of power amplifier design principles.
- Analyze and optimize different Power Amplifier trade-offs, such as: efficiency, and linearity.
- Successfully design and implement a 802.11-compliant power amplifier.

This, technically, translates to the following:-

1. Design a CMOS Power Amplifier for WLAN applications.
2. Implement relevat impedance matching networks to maximize power delivery.
3. Perform extensive simulations in Cadence and/or ADS to design, layout, and test the PA.
4. Investigate the effects of different parasitics at higher frequencies.
5. Compare performance with previous PA works.

1.4 Project's Impact

This project contributes to the advancement of CMOS-based RF front-ends.

1.4.1 Technological Impact

- Enables cost-effective, high-performance PA design for WLAN applications.
- Provides a CMOS solution for future transceivers.

1.4.2 Economic Impact

- Lends itself to CMOS & Silicon technologies, which can be much more cost-effective than III-V Materials (e.g. GaN).
- Aligns with the need for fully integrated silicon solutions.

1.4.3 Societal Impact

- Contributes to high-speed connectivity in consumer and industrial applications.
- Supports advancements in IoT, smart cities, and next-generation wireless technologies.

2 Literature Review

Before a deeper dive into Power Amplifier Design as well as the different power amplifier challenges in CMOS technologies, the RF Transceiver as well as different architectures & considerations will be presented first alongside the standards of relevance.

2.1 Wireless Transmission

Wi-Fi communication—being wireless—suffers from challenges that are typical of wireless communication in general (frequency-dependent free path loss, fading, interference, and other channel effects that greatly degrade the signal), as well as challenges that are more unique to Wi-Fi given the data rates and applications it needs to accommodate. For example, a high peak-to-average-power ratio (PAPR) is due to the employment of OFDM as well as the need for an acceptable level of signal integrity and power to allow the receiver to correctly recover the signal, which typically employs digital modulation schemes that require high SNR and a low enough EVM (such as 16-QAM, 64-QAM, and higher). Modern Wi-Fi standards intensify the challenge. e.g., 802.11ax and 802.11be adopt 1024- and 4096-QAM, thus pushing the allowable error-vector magnitude (EVM) to -38 dB [21] [22]. And the problem is further compounded when higher frequencies are used. Two immediate limitations come to mind. The first of which is free-space-path-loss (FSPL). Assuming a transmitted and received power of P_T & P_R respectively.

$$FSPL = \frac{P_T}{P_R} = \frac{1}{G_T G_R}(\frac{4\pi d}{\lambda})^2 \implies FSPL_{dB} = 20\log_{10}(d_m) + 20\log_{10}(f_{Hz}) - G_{T(dB)} - G_{R(dB)} - 147.55 \tag{2.1}$$

Where: G_T & G_R represent the transmitter antenna gain & the receiver antenna respectively, λ & f are the wavelength and frequency, and d is the distance between the transmitter and receiver. Thus, path loss increases **quadratically** with an increasing frequency.

The second of which is concerned with the technology and system that enables wireless communication. In addition to the various hurdles semiconductor devices face in increasingly higher frequencies in terms of their capability to provide power gain, higher frequencies imply parasitics that become more nuanced at GHz range frequencies, and their effect is very prominent at mmWave frequencies. Furthermore, CMOS technology in particular has a notably lossy substrate and several loss mechanisms for inductors in particular. Thus, an RF transceiver needs to have certain characteristics to accommodate the various challenges.

2.2 Transceiver Architectures [2] [3]

Different RF Transceivers architectures are presented in the diagram below. A detailed discussion of transceiver architectures would demand several pages at least. However, the basic concepts & architectures are presented below.

Modern WLAN transceivers employ digital modulation schemes, which involve IQ modulation & demodulation, a baseband processor, and ADCs & DACs for A/D & D/A conversion to the mentioned digital processor, and the antennas & RF Front-ends, as well as other associated blocks, such as switches, filters, the mixer, and the local oscillator.

The main feature of an RX is an RX front-end (RX Antenna, LNA & different filters), followed by down-conversion (which could be done on several steps, as with heterodyne architectures) & demodulation, then the baseband processor. For a TX path, we would go in reverse: baseband processor, then modulation & up-conversion, then the TX front-end (filters, VGA/AGC, PA, & TX Antenna). Different architectures are presented below.

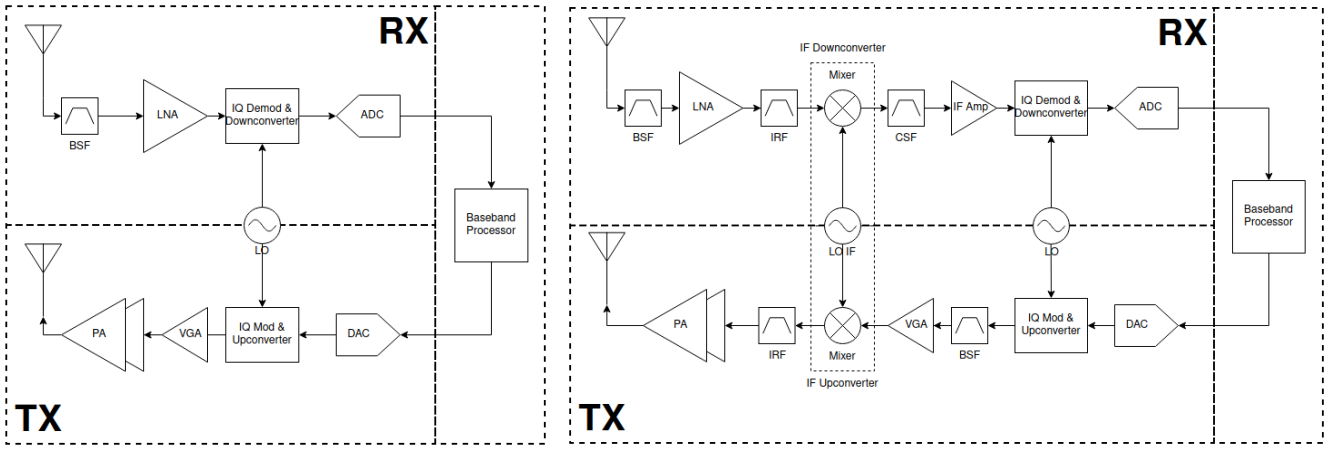


Figure 2.1: Homodyne Transceiver (Left), and Heterodyne Transceiver (Right) Examples.

Some of these include homodyne (direct-conversion) architectures, heterodyne & superheterodyne, Low-IF architectures, as well as Direct Sampling architectures. In the **homodyne architecture**, the RF signal is directly converted to baseband. However, they are traditionally impractical due to DC offsets, LO pulling, IQ imbalances and other problems. This made **heterodyne architectures** the main player due to superior performance as well as solving many issues of homodyne transceivers. However, they also have issues: the image problem, and the need for many filters, some of which are bulky and impractical. **Low-IF architectures** use an intermediate frequency (in the same way as heterodyne and superheterodyne architectures do), but with

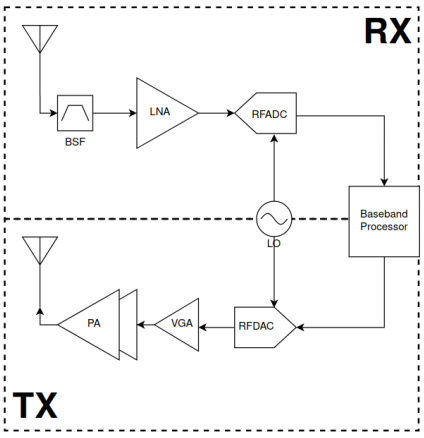


Figure 2.2: RF-Sampling Transceiver Example.

relatively lower frequency, thus combining the benefits of both homodyne and heterodyne. That said, homodyne architectures are resurfacing due to advancements in digital design which could solve many of their problems. Homodyne architectures have the benefits of simplicity, and a high level of integrability as well, as bulky SAW filters are not needed. Other emerging transceivers employ newer topologies, such as **direct-sampling architectures**, which make use of advances in data-converter and extensive employment of digital techniques for various functions, including up- and down-conversion, as well as modulation and demodulation.

As mentioned, the intricate details of transceivers lie beyond the scope of this text, but two key points are made. The first of which has to do with two important functions for the RX and TX paths.

- In the RX path, Friis' equation states that for cascade of N stages, each with noise figure NF_i & power gain G_i, the total noise factor is:

$$NF_{Tot} = NF_1 + \frac{NF_2 - 1}{G_1} + \frac{NF_3 - 1}{G_1 \cdot G_2} + ... + \frac{NF_n - 1}{G_{n-1}...G_2 \cdot G_1} = NF_1 + \sum_{i=2}^{n} \frac{NF_i}{\prod_{j=1}^{i-1} G_j} \tag{2.2}$$

  Which implies that the Noise figure and gain of the first stages (the LNA especially) are critical for the overall noise figure (and thus the sensitivity and dynamic range of the receiver).

- In the TX path, for a cascade of N stages each with a $P1dB_i$, $IIP3_i$ and power gain G_i, the overall $OP1dB$ and $OIIP3$ (for $IP1dB$ & $IIIP3$, the gains are in the numerator) are:-

$$\frac{1}{OP1dB_{Tot}} = \frac{1}{OP1dB_n} + \frac{1}{OP1dB_{n-1} \cdot G_n} + \frac{1}{OP1dB_{n-2} \cdot G_n \cdot G_{n-1}} + ... + \frac{1}{OP1dB_1 \cdot G_n...G_3 \cdot G_2} \tag{2.3}$$

  and

$$\frac{1}{OIIP3_{Tot}} = \frac{1}{OIIP3_n} + \frac{1}{OIIP3_{n-1} \cdot G_n} + \frac{1}{OIIP3_{n-2} \cdot G_n \cdot G_{n-1}} + ... + \frac{1}{OIIP3_1 \cdot G_n...G_3 \cdot G_2} \tag{2.4}$$

  Which implies that the compression points and gains of the last stages (the PA especially) are critical for the overall P1dB & IIP3 of the system (and thus the linearity of the transmitter).

The second point is that regardless of the transceiver architecture, the front-end module is an integral part of any wireless transceiver. And **the Power Amplifier remains one of the most pivotal blocks in the transmitter chain.** Not to mention, it's also the most power hungry block in the entire transceiver (and might even consume more than the rest of the transceiver. In fact, some cellular base stations, it was shown that it consumed $\approx 65\%$ of the total power! [20]) and any improvements in dissipated power benefits the transceiver as whole.

2.3 WLAN & 802.11 Standards

2.3.1 Overview

WLANs are computer networks that enable wireless communication within a localized area. They provide internet access and facilitate data exchange without physical cabling. The IEEE 802.11 family of standards is the dominant set of protocols in WLANs. A quick overview of them is presented.

2.3.2 802.11 Across the years

Standard	Releas Year	Frequency Bands	Max Theoret-ical Speed	Channel Band-width	Modulati Type	Modulatio Order	Features
802.11	1997	2.4 GHz	2 Mbps	20 MHz	DSSS FHSS	BPSK QPSK	-
802.11b	1999	2.4 GHz	11 Mbps	20 MHz	DSSS	CCK	-
802.11a	1999	5 GHz	54 Mbps	20 MHz	OFDM	BPSK to 64-QAM	-
802.11g	2003	2.4 GHz	54 Mbps	20 MHz	OFDM	BPSK to 64-QAM	Backward compatible with 802.11b
802.11n	2009	2.4/5 GHz	600 Mbps	20/40 MHz	OFDM	BPSK to 64-QAM	MIMO (up to 4 streams)
802.11ac	2013	5 GHz	1.3 Gbps	20/40/ 80/160 MHz	OFDM	BPSK to 256-QAM	MU-MIMO, Beamforming
802.11ad	2012	60 GHz	7 Gbps	2160 MHz	SC/OFDM	BPSK QPSK	Short-range, high throughput
802.11ax	2019	2.4/5 GHz	9.6 Gbps	20/40/ 80/160 MHz	OFDMA	BPSK to 1024-QAM	MU-MIMO, OFDMA
802.11ay	2021	60 GHz	176 Gbps	2 GHz	OFDM	BPSK to 64-QAM	MIMO, Channel Bonding

Table 2.1: Comparison of IEEE 802.11 WLAN Standards

2.3.3 802.11 RF Front-End Considerations

The above standards yield some pivotal considerations: -

- An increasingly **better EVM demands higher linearity**. This means an increasingly higher compression point (where the amplifier starts to "compress" and exhibit significant non-linearity, more detailed in section 2.5) is needed.
- OFDM also demands **higher compression points and output saturated power**, as its high PAPR means peaks will be several dB higher than the average of the waveform.
- The TX has to transmit the signals in a way that complies with the **densely-populated spectral mask** of the standard. The effect of different harmonics must also be suppressed to not interfere with other channels.
- Increasing frequencies means a signal that suffers from more attenuation in the wireless channel; thus, the RX has to be of better sensitivity to detect the degraded signal.
- Higher frequencies also mean that the increasingly larger bandwidth results in a larger noise floor, which in turn makes receiver specs tighter.

Furthermore, there are regulations that put hard constraints on the transmitter that must be obeyed. FCC Part 15 dictates that WLAN transmitters in the $2.4GHz$ & $5.8GHz$ ISM bands must have a maximum of $1W$ for digitally modulated devices (excluding antenna gain). PSD is also to be limited to $8dBm/3kHz$ and requires emissions $\leq 20dB$ below in-band power outside the channel. These regulations limit the Psat, and thus dictate the $P1dB$ and put a constraint that allows architects and designers to make the correct tradeoffs in the design of a transmitter system. [15] [16]

Higher symbol rates (which need lower EVMs) translate to higher data rates. A subset of an MCS index table (for the 802.11n, 802.11ac, and 802.11ax standards) provides an example of this depending on the spatial streams, coding rate, guard interval, and other parameters.

						OFDM (Prior 11ax)								OFDM (802.11ax)									
MCS Index						20MHz		40MHz		80MHz		160MHz		20MHz			40MHz			80MHz			160MHz		
HT	VHT	HE	Spatial	Modulation	Coding	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI
0	0	0	1	BPSK	1/2	6.5	7.2	13.5	15	29.3	32.5	58.5	65	8.6	8.1	7.3	17.2	16.3	14.6	36	34	30.6	72.1	68.1	61.3
1	1	1	1	QPSK	1/2	13	14.4	27	30	58.5	65	117	130	17.2	16.3	14.6	34.4	32.5	29.3	72.1	68.1	61.3	144.1	136.1	122.5
2	2	2	1	QPSK	3/4	19.5	21.7	40.5	45	87.8	97.5	175.5	195	25.8	24.4	21.9	51.6	48.8	43.9	108.1	102.1	91.9	216.2	204.2	183.8
3	3	3	1	16-QAM	1/2	26	28.9	54	60	117	130	234	260	34.4	32.5	29.3	68.8	65	58.5	144.1	136.1	122.5	288.2	272.2	245
4	4	4	1	16-QAM	3/4	39	43.3	81	90	175.5	195	351	390	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5
5	5	5	1	64-QAM	2/3	52	57.8	108	120	234	260	468	520	68.8	65	58.5	137.6	130	117	288.2	272.2	245	576.5	544.4	490
6	6	6	1	64-QAM	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	77.4	73.1	65.8	154.9	146.3	131.6	324.3	306.3	275.6	648.5	612.5	551.3
7	7	7	1	64-QAM	5/6	65	72.2	135	150	292.5	325	585	650	86	81.3	73.1	172.1	162.5	146.3	360.3	340.3	306.3	720.6	680.6	612.5
	8	8	1	256-QAM	3/4	78	86.7	162	180	351	390	702	780	103.2	97.5	87.8	206.5	195	175.5	432.4	408.3	367.5	864.7	816.7	735
	9	9	1	256-QAM	5/6	N/A	N/A	180	200	390	433.3	780	866.7	114.7	108.3	97.5	229.4	216.7	195	480.4	453.7	408.3	960.8	907.4	816.7
		10	1	1024-QAM	3/4									129	121.9	109.7	258.1	243.8	219.4	540.4	510.4	459.4	1080.9	1020.8	918.8
		11	1	1024-QAM	5/6									143.4	135.4	121.9	286.8	270.8	243.8	600.5	567.1	510.4	1201	1134.3	1020.8
8	0	0	2	BPSK	1/2	13	14.4	27	30	58.5	65	117	130	17.2	16.3	14.6	34.4	32.5	29.3	72.1	68.1	61.3	144.1	136.1	122.5
9	1	1	2	QPSK	1/2	26	28.9	54	60	117	130	234	260	34.4	32.5	29.3	68.8	65	58.5	144.1	136.1	122.5	288.2	272.2	245
10	2	2	2	QPSK	3/4	39	43.3	81	90	175.5	195	351	390	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5
11	3	3	2	16-QAM	1/2	52	57.8	108	120	234	260	468	520	68.8	65	58.5	137.6	130	117	288.2	272.2	245	576.5	544.4	490
12	4	4	2	16-QAM	3/4	78	86.7	162	180	351	390	702	780	103.2	97.5	87.8	206.5	195	175.5	432.4	408.3	367.5	864.7	816.7	735
13	5	5	2	64-QAM	2/3	104	115.6	216	240	468	520	936	1040	137.6	130	117	275.3	260	234	576.5	544.4	490	1152.9	1088.9	980
14	6	6	2	64-QAM	3/4	117	130	243	270	526.5	585	1053	1170	154.9	146.3	131.6	309.7	292.5	263.3	648.5	612.5	551.3	1297.1	1225	1102.5
15	7	7	2	64-QAM	5/6	130	144.4	270	300	585	650	1170	1300	172.1	162.5	146.3	344.1	325	292.5	720.6	680.6	612.5	1441.2	1361.1	1225
	8	8	2	256-QAM	3/4	156	173.3	324	360	702	780	1404	1560	206.5	195	175.5	412.9	390	351	864.7	816.7	735	1729.4	1633.3	1470
	9	9	2	256-QAM	5/6	N/A	N/A	360	400	780	866.7	1560	1733.3	229.4	216.7	195	458.8	433.3	390	960.8	907.4	816.7	1921.6	1814.8	1633.3
		10	2	1024-QAM	3/4									258.1	243.8	219.4	516.2	487.5	438.8	1080.9	1020.8	918.8	2161.8	2041.7	1837.5
		11	2	1024-QAM	5/6									286.8	270.8	243.8	573.5	541.7	487.5	1201	1134.3	1020.8	2402	2268.5	2041.7
16	0	0	3	BPSK	1/2	19.5	21.7	40.5	45	87.8	97.5	175.5	195	25.8	24.4	21.9	51.6	48.8	43.9	108.1	102.1	91.9	216.2	204.2	183.8
17	1	1	3	QPSK	1/2	39	43.3	81	90	175.5	195	351	390	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5
18	2	2	3	QPSK	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	77.4	73.1	65.8	154.9	146.3	131.6	324.3	306.3	275.6	648.5	612.5	551.3
19	3	3	3	16-QAM	1/2	78	86.7	162	180	351	390	702	780	103.2	97.5	87.8	206.5	195	175.5	432.4	408.3	367.5	864.7	816.7	735
20	4	4	3	16-QAM	3/4	117	130	243	270	526.5	585	1053	1170	154.9	146.3	131.6	309.7	292.5	263.3	648.5	612.5	551.3	1297.1	1225	1102.5
21	5	5	3	64-QAM	2/3	156	173.3	324	360	702	780	1404	1560	206.5	195	175.5	412.9	390	351	864.7	816.7	735	1729.4	1633.3	1470
22	6	6	3	64-QAM	3/4	175.5	195	364.5	405	N/A	N/A	1579.5	1755	232.3	219.4	197.4	464.6	438.8	394.9	972.8	918.8	826.9	1945.6	1837.5	1653.8
23	7	7	3	64-QAM	5/6	195	216.7	405	450	877.5	975	1755	1950	258.1	243.8	219.4	516.2	487.5	438.8	1080.9	1020.8	918.8	2161.8	2041.7	1837.5
	8	8	3	256-QAM	3/4	234	260	486	540	1053	1170	2106	2340	309.7	292.5	263.3	619.4	585	526.5	1297.1	1225	1102.5	2594.1	2450	2205
	9	9	3	256-QAM	5/6	260	288.9	540	600	1170	1300	N/A	N/A	344.1	325	292.5	688.2	650	585	1441.2	1361.1	1225	2882.4	2722.2	2450
		10	3	1024-QAM	3/4									387.1	365.6	329.1	774.3	731.3	658.1	1621.3	1531.3	1378.1	3242.6	3062.5	2756.3
		11	3	1024-QAM	5/6									430.1	406.3	365.6	860.3	812.5	731.3	1801.5	1701.4	1531.3	3602.9	3402.8	3062.5

Figure 2.3: MCS Table Example, created by F. Verges. [17].

2.4 Technologies & Materials [3] [4]

2.4.1 Transistor & Semiconductor Technologies

Bipolar junction transistors (BJTs) are a widespread active RF device due to their low cost, performance, and low noise characteristics. They are useful for amplifiers up to the low GHz range. BJTs are sometimes preferred over FETs at frequencies below 2-4 GHz due to higher gain, lower cost, and the possibility of biasing with a single power supply.

Heterojunction bipolar transistors (HBTs) operate similarly to BJTs but have a base-emitter junction made from a compound semiconductor material such as Gallium Arsendie (GaAs), Indium Phosphide (InP), or silicon germanium (SiGe). This structure offers improved performance at high frequencies, even in frequencies higher than 100 GHz.

Field effect transistors (FETs) are monopolar, with only one carrier type providing current flow through the device. They can take many forms, including MESFET, MOSFET, HEMT, and PHEMT. FET technology has been under continuous development for over 50 years, with GaAs MESFETs being widely used for microwave and millimeter wave applications. GaAs HEMTs are particularly useful for low-noise amplifiers due to their lower noise figures. Recently developed gallium nitride (GaN) HEMTs are also useful for high power RF and microwave amplifiers - the ability of such devices to have very high drain voltages compared to the drain current makees them particularly useful for Power Amplifiers. CMOS FETs are increasingly used for RF integrated circuits, offering high levels of integration at low cost and low power, especially as they downscale further and further, which has both benefits and challenges.

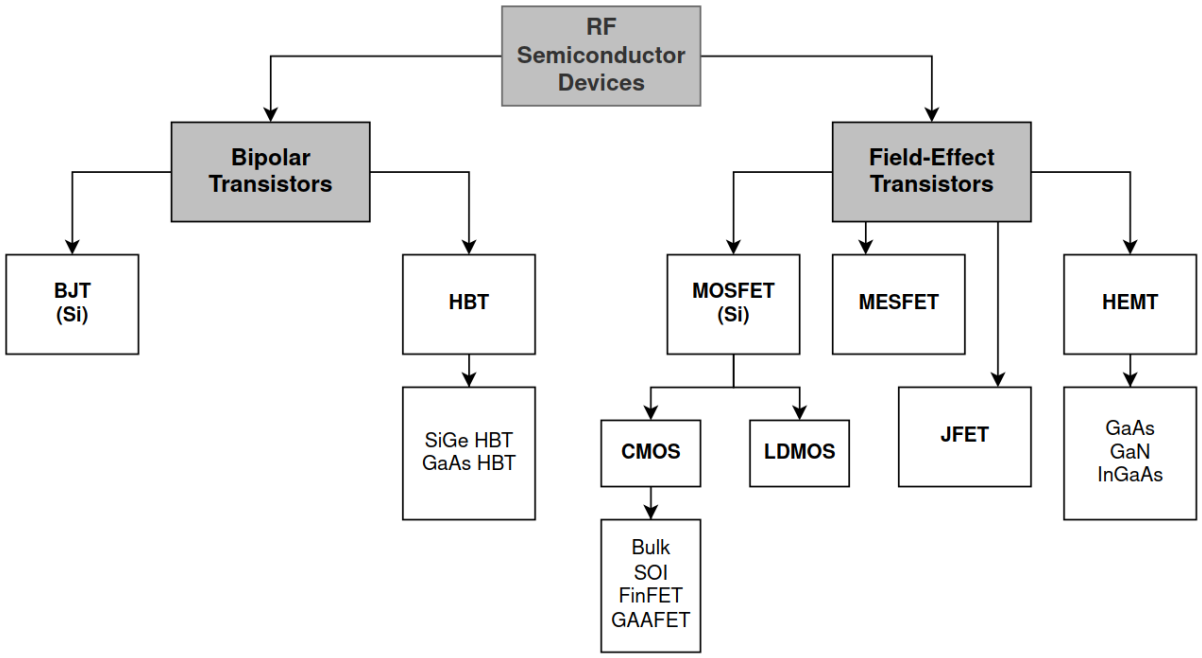


Figure 2.4: FET & BJT transistor families

2.4.2 CMOS Technology & PA Design Implications

Metal-oxide-silicon field-effect transistors (MOSFETs) were patented by Julius Edgar Lilienfeld in 1925 but were practically realized due to the world-changing contributions of Mohamed Atalla & Dawon Kang in the late 1950s to early 1960s. CMOS technologies quickly revolutionized the digital market. According to Moore's law (an observation, rather than a physical law), the number of transistors on a microchip doubles every two years, resulting in exponential growth in computing power and cost [23]. The highly integrable nature of CMOS technology gave birth to the landscape of integrated circuits as we know it today: not only were chips getting more powerful and smaller, but also cheaper.

This trend of downscaling is critical to our discussion. Analog design started shifting from BJTs to CMOS due to the increased speeds of MOS devices and low manufacturing costs. Additionally, the highly integrable nature of CMOS allows for analog and digital blocks to be on the same chip, offering sought-after performance and features, as well as lower packaging costs. [5] For RF transistors, downscaling also has benefits. The biggest of which has to do with the increased f_t & f_{max} of these devices. For bulk CMOS transistors, gate lengths of $65nm$ yield f_T above $150GHz$ and $f_{\max}$ values above $200GHz$ [24]. Additionally, since WiFi band frequencies are a fraction of the f_{max}, S-parameter gains are usually plentiful.

However, this trend of downscaling also has many problems for both analog and RF design. For analog design, transistors deviate from the square-law model, and many higher-order effects (e.g., short-channel effects, such as gate leakage and velocity saturation) start appearing as the devices scale down [5]. This significantly complicates the design process and makes PVT variations in fabrication unforgiving. For RF design, there are unique challenges. Mainly, the low breakdown voltage of CMOS devices, passive device modeling, and substrate losses. The latter two of which make the design flow harder and demand extensive use of post-layout analysis and EM solvers, with 3-D EM solvers being necessary for extremely high frequencies.

2.4.3 PA Design Implications

The above discussion has several consequences for PA design. Notably, the first one is that of the low breakdown voltage.

- Low breakdown voltage directly translates into an output power problem. This has direct impacts on linearity and might render CMOS unfeasible for certain applications that would otherwise demand very high output power (e.g. $> 1W$) with good linearity.
- The low breakdown voltage also means a high output power would necessitate higher current, which comes with a host of challenges in terms of possible heating, more voltage drop across the parasitic resistances of different inductors, and the need for relatively wide track widths to handle higher amounts of current.
- Notably, the optimum resistance R_{opt} will likely be very low, thus making the output matching very lossy and difficult, thus affecting the PAE.

- Additionally, since an optimum output impedance might not necessarily be a conjugate match, S_{22} will likely be poor. This issue is further compounded by the above point in deep submicron nodes. For example, assume a needed output power of $21dBm = 0.125W$ with a $V_{max} = 1.5V$ & $V_k = 0$ (for illustrative purposes). From the equations of a Class A PA, this means an $I_[max] = 0.67A \implies I_{DC} = 0.335$. Aside from the high current, this means $R_{opt} = \frac{V_{max}}{I_{max}} = 2.25\Omega$! Very far from 50Ω!

- Extra care must be taken in order to make sure that the breakdown voltage of the MOSFET's drain is not exceeded. For certain classes and depending on the configuration, the drain voltage of the device V_D can reach $2V_{DD}$ and even exceed $3V_{DD}$ for certain types of amplifiers (such as the class E). [3]

Another notable implication is that of the losses.

- Lossy circuits present losses that negatively affect the gain and the PAE of the PA. Post-layout simulations are crucial for analyzing and debugging the impacts of various parasitic elements on the overall performance. Additionally, the various losses of the passive elements mean that the matching networks will also have losses that degrade the efficiency.

- Additionally, the spiral inductors of CMOS technologies are also very lossy—typically with low Q factors—and need to be operated away from their self-resonance frequencies.

- The above point also has other consequences: the spiral inductor's finite Q means using a very large inductor for DC feeds (in order for the AC path to be of very high impedance compared to that of other impedances). For example, a 3nH inductor with $Q = 10$ used as an RF choke would present an RF impedance of $Z_L = j\omega L \approx j94.25\Omega$ at 5 GHz but with a parasitic resistance of $R = \frac{j\omega L}{Q} \approx 9.43\Omega$! This is a very high value and would mean a voltage drop of $0.943V$ for $I_{DC} = 100mA$!

- On the other hand, an 0.5nH inductor of the same Q factor—while it could serve its function as a choke without excessive parasitic resistance—will likely have relatively low impedance, appearing as a parallel impedance to that of the matching network. This means that it will have to be taken into account and that load-pull measurements, for example, need to be conducted with this lossy inductor at the output. Tuning and iterations will likely be needed to settle upon a satisfactory optimum.

- The wide range of effects that negatively affect the performance of inductors makes EM co-simulation indispensable, particularly at mmWave frequencies. This is compounded by the fact that some PDKs, such as TSMC 65nm, characterize inductors up to a certain frequency only, such as $20GHz$. And for very high frequencies,

In summary, 65nm technology is attractive for CMOS PAs that operate in WiFi bands as well as mmWave due to their high f_{mas} and good S-parameters in this frequency range, but the above considerations provide important guidelines for the design of such PAs.

2.5 Power Amplifiers

2.5.1 Definition

Power amplifier are circuits that make use of active devices (transistors) to convert DC-power into meaningful amounts of RF/microwave power for wireless transmission. It is different from other amplifier in that relatively high amounts of power it could output - which can easily exceed $100mW$ and even the $1W$ rangees, thus it could damage components and dissipate significant heat. Additionally, power amplifiers are **large-signal circuits**, so they are **nonlinear** by nature. This means that they design has unique considerations compared to other amplifiers as well.

2.5.2 Performance Metrics

When designing a PA for WLAN applications (e.g., 802.11ax), several key performance metrics are considered:

- **Output Power (Pout):** The RF power delivered to the load. Higher output power increases transmission range but also raises power consumption. This is one of the most important metrics and it is what makes a Power Amplifier so.

$$P_{out} \propto I_{max} V_{max} \tag{2.5}$$

  Where I_{max} and V_{max} denote that maximum current that can be provided by the FET as well, and the maximum drain voltage that the transistor could withstand (which is limited by the breakdown voltage of the device.). Note that for actual transistors, the voltage swing **will actually be** $V_{max} - V_k$**, where** V_k **denotes the knee voltage**, which is conceptually similar to the $V_{DS_{sat}}$ of a MOSFET.

- **1-dB Compression Point (P1dB):** Defines the maximum output power before nonlinear compression reduces gain by 1 dB. This is a very important metric for linearity, along with IIP3, which is to be discussed shortly. Typically, for a Class A PA, with a Knee votlage (V_k) and total width W_T: [3]

$$OP_{1dB} = \frac{I_{swing,pp} * V_{swing,pp}}{8} = (W_T \cdot J_{swing,pp}) \frac{(V_{max} - V_k)}{8} \tag{2.6}$$

- **Saturation Power (Psat):** The highest output power level before the PA enters strong compression. A well-designed PA ensures Psat is high enough to handle peak power demands. It's typically $2 - 3dB$ above the 1-dB Compression Point.

- **Third-Order Intercept Point (IP3):** Indicates linearity performance. A higher IIP3 value means better suppression of third-order intermodulation products. The third-order intermodula-

tion products are of particular importance as they usually lie very close to the tones of interest and filtering them is an arduous endeavor.

- **Power-Added Efficiency (PAE):** Represents how efficiently the PA converts DC power to RF power:

$$PAE = \frac{P_{\text{out}} - P_{\text{in}}}{P_{\text{DC}}} \tag{2.7}$$

A direct consequence of this equation is that for P_{in} levels that are low, the PAE will be low and will rise as Pin increases until it hits maximum. We are usually concerned with peak PAE, but the PAE at both the 1-db compression point as well as the linear output power will also be reported.

- **Drain Efficiency (η_{drain}):** sometimes used instead of PAE to describe efficiency of the PA. Where, $\eta_{drain} = \frac{P_{\text{out}}}{P_{\text{DC}}}$.

- **Power Gain (G):** Ratio of Output power to Input power: $G = \frac{P_{out}}{P_{in}}$. A pivotal metric. Usually, this will be tested using both Small-Signal tests as well as non-linear harmonic balance tests, as small-signal tests alone will not capture non-linear effects when the PA enters its compression.

- **Error Vector Magnitude (EVM):** Measures modulation accuracy and is a critical parameter to tests digital modulation schemes, especially higher orders like 64-QAM, 256-QAM & 1024-QAM.

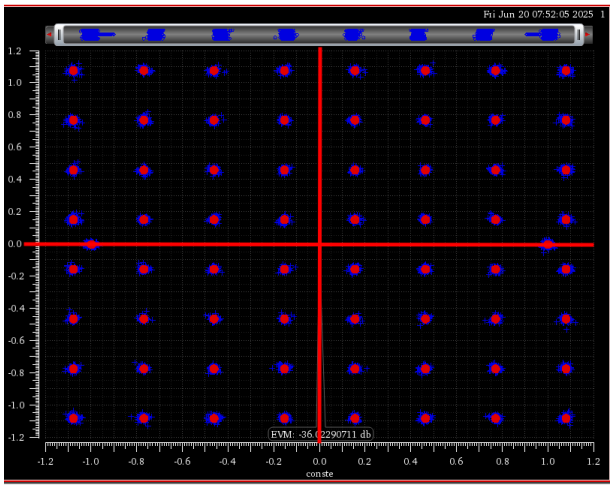


Figure 2.5: EVM example for a 64-QAM Constellation.

- **Adjacent Channel Power Ratio (ACPR):** The ratio of power transmitted into the adjacent frequency channels to the power that it is transmitted in the main channel. Used to check no excessive RF power is transmitted in adjacent channels as specified by the standard.

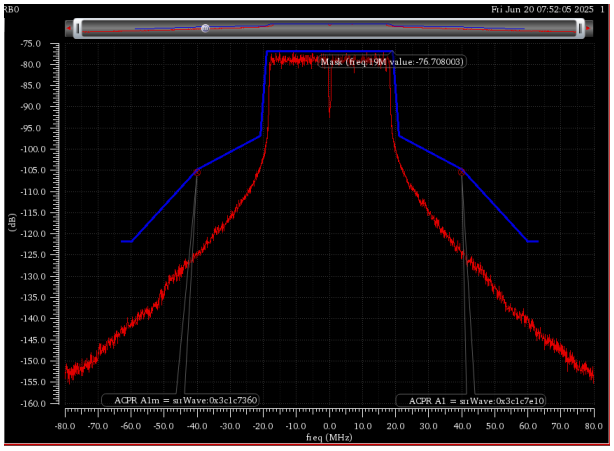


Figure 2.6: ACPR example mask & PSD.

- **Noise Figure (NF):** Quantifies the degradation in signal-to-noise ratio (SNR) due to the PA. A low NF is not as critical of a parameters as other parameters are for Power Amplifiers, but it should be controlled.

- **Stability:** PAs must achieve stability for proper operation. The **K-factor** is used to assess stability, where $K > 1$ to achieve unconditional stability.

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} \tag{2.8}$$

  Where $\Delta = |S_{22}S_{11} - S_{21}S_{12}| \leq 1$. Otherwise, if unconditional stability is not met, it's crucial to make sure that the load and source reflection coefficients of the amplifier are not in their unstable regions. More modern measures of stability, such as the μ and μ' factors, also exist and provide information about how "far away" an amplifier is from oscillation.

2.5.3 Power Amplifier Classes [1] [2] [3]

Conventional power amplifiers could be categorized into two large families: amplifying (transconductance amplifiers) and switching (wave-shaping amplifier).

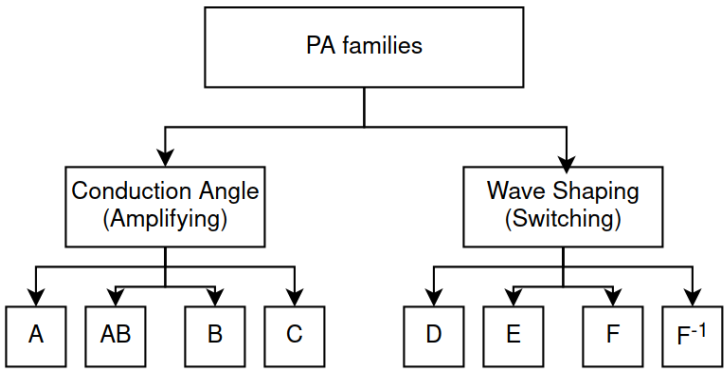


Figure 2.7: PA Classes

2.5.3.1 Amplifying Classes

For amplifying (V-I) PAs, the distinction is based on their **conduction angle**. Thus, amplifiers that have a 360° conduction angle are ON and conducting through the entire cycle and source the highest current and output power, but with efficiencies of only 50% max (and their low-frequency counterparts feature an efficiency of 25% max, as the drain current won't rise to $2V_{DD}$ in the absence of the RF choke inductor typically employed in RF designs).

True Class B amplifiers are biased at exactly the cutoff, so their conduction angle is 180°. They boost η_{drain} to a theoretical max of $\frac{\pi}{4} \approx 78.5\%$ at the expense of lower output power. Class C amplifiers are biased at below the cutoff point; thus, they have a conduction angle of $< 180°$ and can have theoretical drain efficiencies of up to 100%, but their strong non-linear features in addition to other issues (such as entering the reverse breakdown region) limit their use. A compromise between Class A & B amplifiers, the Class AB stage, has a conduction angle and maximum drain efficiency between those of Class A & B stages and has been the standard in WLAN PAs for its linearity-efficiency balance. The diagram below showcases the bias points of different amplifying classes on idealized FET characteristics.

The mechanism that helps Class B and C in general achieve better efficiencies is that of the changing DC current. The DC current I_{DC} increases only as needed by the input signal; thus, the DC supplied power is variable and accommodates the input signal, contrary to the case of Class A amplifiers, where the DC power and current are fixed regardless of whether the input signal is at a low value or at its peak. Class A amplifiers typically peak at higher values but boost very poor efficiency at lower input signals, as also reflected by its PAE.

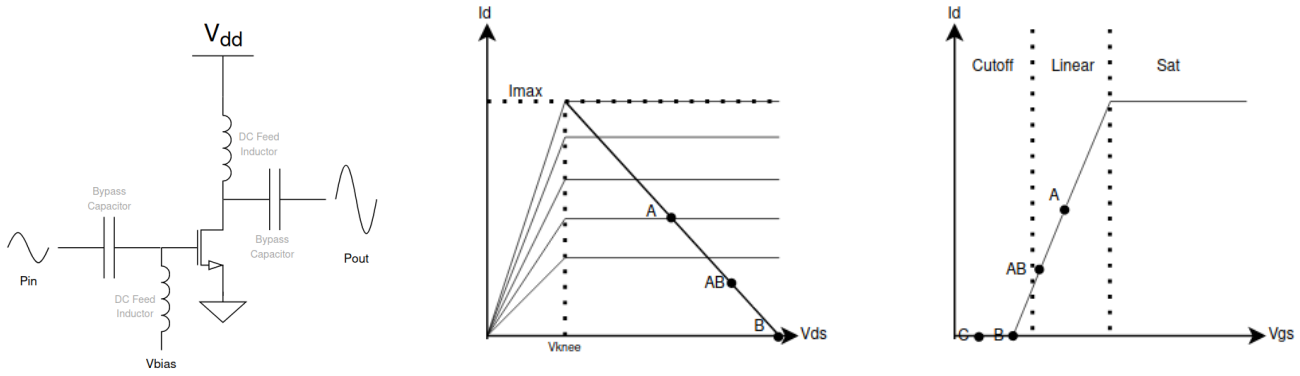


Figure 2.8: Basic RF CS Stage (Left). Biasing of (A, AB, B, C) classes (Right).

2.5.3.2 Switching Classes

On the other hand, switching PAs work by acting as switches (so that the output swings from a minima to a maxima in an ideally instantaneous switching time (e.g., a square wave drain output in the class D configuration), and harmonic filters are then employed to **"shape" the waveform** with the desired harmonics at frequencies of interest. This is contrary to classical configurations, where the input and output waveforms are sinusoidal or harmonics of sinusoids.

Since these devices conduct the maximum current when on (and the voltage is at a minimum) and ideally zero otherwise, the theoretical drain efficiencies can approach 100%, as ideally, the voltage and current waveforms do not overlap at the drain node. But they are not without their own challenges: similar to Class B amplifiers, they have lower current and output power relative to the Class A amplifier and have other challenges that also limit their maximum efficiencies (e.g., parasitics at the drain node and losses in the output network). The distinction between the classes depends on their configuration and their wave-shaping methods.

The class D amplifier features two switches in a push-pull configuration to get a theoretical square-wave waveform at the drain (with the input being a PWM signal), which is then filtered by an RLC network. They usually had challenges in higher frequencies due to the asymmetry between the low- and high-side transistors and power losses in the finite resistance of the transistors in the ON state (R_{ON} of the MOSFET); traditionally, it found more use in audio amplification. The class E amplifier—in its simplest topology—features only one transistor and a harmonic network at the output and helps alleviate some of the traditional issues associated with class D amplifiers by employing zero-voltage switching to minimize losses and employing one transistor (instead of two, where the high-side PMOS transistor often had decreased performance compared to the traditionally faster low-side NMOS), making it more suitable for RF applications. The input and drain voltages do not necessarily have to be nearly square for this class' operation. Class F and F^{-1} amplifiers work on a similar concept but have several harmonic terminations to shape the waveforms in a way that allows them to have increased efficiencies. For the class F, the voltage is nearly square and the current is sinusoidal, and vice versa for the F^{-1} class.

Notably, the class D amplifier regained use due to the down-scaling of CMOS, which provided similar performance between both the NMOS and POMOS counterparts, thus alleviating some of its traditional problems at very high frequencies. Class D amplifiers do find use nowadays at extremely high frequencies. [25]

2.5.3.3 PA Classes Summary

The following table summarizes key points.

Class	A	B	D	E	F	F^{-1}
V_{DD}	$\frac{V_{MAX}}{2}$	$\frac{V_{MAX}}{2}$	V_{MAX}	$\frac{V_{MAX}}{3.6}$	$\frac{V_{MAX}}{2}$	$\frac{V_{MAX}}{\pi}$
I_{DC}	$\frac{I_{MAX}}{2}$	$\frac{I_{MAX}}{\pi}$	$\frac{I_{MAX}}{\pi}$	$\frac{I_{MAX}}{2.06}$	$\frac{I_{MAX}}{\pi}$	$\frac{I_{MAX}}{2}$
$\frac{P_L}{(V_{MAX} \cdot I_{MAX})}$	0.13	0.13	0.16	0.16	0.16	0.16
R_{LOPT}	$\frac{V_{MAX}}{I_{MAX}}$	$\frac{V_{MAX}}{I_{MAX}}$	$\frac{2V_{MAX}}{\pi I_{MAX}}$	$1.075\frac{V_{MAX}}{I_{MAX}}$	$\frac{4V_{MAX}}{\pi I_{MAX}}$	$\frac{\pi V_{MAX}}{4I_{MAX}}$
$\eta_{d,max}$	50.00%	78.00%	100.00%	100.00%	100.00%	100.00%

Table 2.2: PA classes summary. [3]

2.6 Current Research

Significant research is being conducted to improve the performance of power amplifiers and overcome many of their challenges. Some of them come directly in lieu of the above discussion and are not limited only to power amplifiers, such as:

- The use of ever increasing frequencies to comply with the different emerging applications (e.g., mmWave bands of the 5G & 6G spectrum & WiGig 60 GHz license-free bands). For PAs, this translates to designing PAs at increasingly higher frequencies while overcoming many of the challenges (e.g. parasitics, passive element modeling & implementation) that come at these high frequencies.

- The investigation & usage of different materials (e.g. GaN) to achieve RF parameters of relevance (higher output power, higher power densities, lower noise figures, broadband performance). For PAs, this entails using different materials with superior performance to silicon for certain applications. Such materials, such as GaN, offer vastly superior performance in terms of output power, for example, as such devices can have operating voltages $> 10V$ & wattages of $> 10W$. [18]

- System Miniaturization and integration of different passives. E.g. fully integrated power amplifiers instead of using off-chip inductors.

- Improve bandwidth for use in higher frequencies. Broadband and ultra-broadband techniques are widely discussed topics aimed at meeting the demands of wireless systems that utilize multiple frequency bands. [19]

- The usage and employment of digital blocks that help in alleviating RF non-idealities (e.g., correcting DC offset in direct-conversion architectures). For PAs, this would entail the usage of digital predistorters, for example.

In order to realize high-performing PAs, there are several approaches that help improve output power, efficiency, and linearity. These techniques help realize the trends discussed above and are shortly presented. Each of these techniques would demand a well-detailed discussion & study by themselves.

- **Efficiency enhancement techniques**: Several techniques are used to enhance the efficiency of PAs, including, but not limited to: [1] [8]

  - Doherty PAs: which greatly improve efficiency at back-off levels, where the PAE typically peaks at a certain value but rapidly falls off with lower input powers as $PAE = \frac{P_o - P_i}{P_{DC}}$. This technique use a main and auxiliary/peaking, which allows the main PA to provide high PAE at back-off levels, and the peaking PA provides peak PAE at peak levels.
  - Envelope tracking techniques: where the supply voltage is dynamically adjusted according to the needed power output level, thus supplying less DC power when there's no need for a high power level and vice versa.

  - Use of switching-mode classes and newer topologies, which inherently offer better efficiency at the cost of linearity.

- **Linearity enhancement techniques**: Linearization techniques are presented to deal with the inherent non-linearity of power amplifiers and make efficient classes (that suffer from relatively more non-linearity compared to a Class A, for example) more linear. They include: [1] [9]

  - Digital Predistortion (DPD): where a predistortion block (more commonly implemented in the digital domain) pre-processes the signal or "distorts" it such that once the "distorted" signal passes through the power amplifier, the predistorted portions will "compensate" the compressed regions of the PA P_o-P_{in} characteristics.
  - Analog predistortion, feedforward, and feedback techniques precede digital distortion methods. However, they are seeing less employment in WLAN system with advances in digital design that make DPD appealing. Nonetheless, they still find use in specific applications.

- **Power boosting techniques**: Which allow for higher output power. They include: [3]

  - Stacked FET architectures: where a cascode-like configuration allows for an improved $P1dB$ by allowing a larger voltage at the drain of the output transistor, but with the swing equally shared among the two or more transistors to avoid breakdown.
  - Transformer Power combining: where the output power of several PA cells can be combined through the use of series-connected transformers.

2.7 Previous Works

The following survey shows the Saturated Power vs Frequency for different PA designs:- [36]

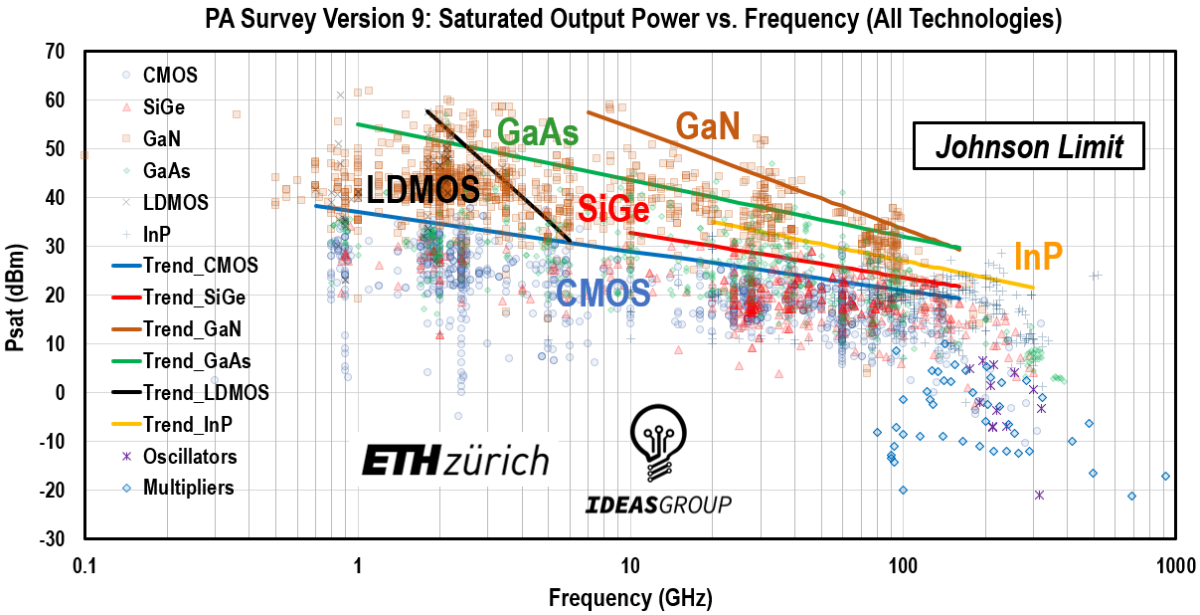


Figure 2.9: PA Survey Version 9, conducted by IDEAS group [36].

The relatively subpar performance of silicon based technologies is immediately appearent, especially in sub-THz frequencies. SiGe is slightly better, GaN and GaAs show by far the best promise. This agrees with the literature. [3]

2.7.1 CMOS PA Works

Metric	Tech	Freq (GHz)	Pout (dBm)	Max PAE	Psat (dBm)	P1dB (dBm)	EVM (dB)	Gain (dB)	Area (mm^2)	Vdd (V)
[26]	65nm	2.45	9.4 (avg) & 17.4 peak	-	-	17	-28.4 @ 9.4 dBm	-	1 × 1	3.3
[27]	65nm	2.4	-	25%	31.5	-	-25 @ 25.5dBm	12	1.3 × 0.9	3.3
[28]	55nm	2.4	19 (max)	32.7% (HP) & 10% (LP)	29.7	27.6 (HP) & 17.7 (LP)	-27 @ 19dBm	-	1.37 × 0.66	3.3
[29]	0.18µm	2.4	24.8	21.2%	-	-	-	-	1.4 × 0.72	3.3
[30]	40nm	2.442	-	-	29	-	-48.3 @ 18dBm	16	-	-
[31]	65nm	2.24	-	26.8%	26.1	25.4	-	21.8	1	2.2

Table 2.3: Comparison of different CMOS Power Amplifiers (2.4 GHz)

Metric	Tech	Freq (GHz)	Pout (dBm)	Max PAE	Psat (dBm)	P1dB (dBm)	EVM (dB)	Gain (dB)	Area (mm^2)	Vdd (V)
[32]	55nm	5	17.6	-	-	-	-35	20.6	0.76 × 1.26	5
[33]	65nm	5	21.0	51%	-	-	-	26	-	-
[34]	180nm	5.2	16.6	19.2%	27.6	23.5	-25	13.4	1.2 × 1.0	-
[35]	180nm	5 - 5.8	22.6	19.8%	23.1	18.6	-	-	-	-

Table 2.4: Comparison of different CMOS Power Amplifiers (5 GHz)

2.7.2 Commercial PA Products

Metric	Vendor	Freq (GHz)	EVM (dB)	P1dB (dBm)	Gain (dB)	Vdd (V)	Icq (mA)	Package Area (mm^2)
SKY85006-11 [37]	Skyworks	2.4	-43 @ 22 dBm (MCS11) -35 @ 23 dBm (MCS8)	31	35	5	340 @ 23dBm	3.8 x 2.8
SE2623L [38]	Skyworks	2.4	-35 @ 23 dBm (MCS8) -30 @ 26dBm	31	33	5	480 @ 26dbm	3.5 x 3.5
QPA5219 [39]	Qorvo	2.4	-35 @ 24 dBm (MCS8/9)	32	32	5	365	3.0 x 3.0

Table 2.5: Comparison of Different Power Amplifiers from SkyWorks and Qorvo (2.4GHz)

Metric	Vendor	Freq (GHz)	EVM (dB)	Psat (dBm)	P1dB (dBm)	Gain (dB)	Vdd (V)	Icq (mA)	Package Area (mm^2)
TQP5523 [40]	Qorvo	5Ghz	-37 @ 21 dBm (MCS9)	31	-	32	5.0	300 @ 21 dBm	4.0 x 4.0
SE5022T [41]	Skyworks	5GHz	-35 @ 15 dBm (256 QAM, 802.11ac)	-	25	32	3.6	195	3.0 x 3.0
SKY85408-11 [42]	Skyworks	5GHz	-43 @ 23 dBm (MCS11)	-	31	35	5.0	-	4.0 x 4.0

Table 2.6: Comparison of Different Power Amplifiers from SkyWorks and Qorvo (5GHz)

2.7.3 Other Relevant Works & Products

Metric	Vendor	Freq (GHz)	EVM (dB)	Psat (dBm)	P1dB (dBm)	Gain (dB)	Vdd (V)	Icq (mA)	Package Area (mm^2)
QPF4209 [43]	Qorvo	2.4	-47 @ 20 dBm (MCS13) –43 @ 22 dBm (MCS11)	32	30	33.0	5.0	300	3.0 x 3.0
QPF4530 [44]	Qorvo	5.0	-43 dB @ 15 dBm (MCS11)	-	20	30.0	3.3	152 @ $17dBm$	3.0 x 3.0
WLAN 7102C [45]	NXP	5.0	-43 dB @ 15 dBm (MCS10 /11)	-	-	30.5	3.85	335 @ $22dBm$	2.0 x 2.0

Table 2.7: Comparison of Different Power Amplifiers from Qorvo and NXP

2.8 Target Specs

For modern WLAN applications for use in both IoT devices and mobile devices, the following specs are targeted. The linear pout complies with what's typical for IoT Devices (from $0dBm$ to $10dBm$) & mobile devices ($10dBm$ to $23dBm$).

Metric	Design Target
Frequency band	$2.4GHz$ *or* $5GHz$
Linear P_{OUT}	10 to 15dBm
P_{1dB}	Typically $\approx 5dB \rightarrow 7dB$ higher than the Linear P_{OUT}
EVM	$\leq -22dB$ for 64-QAM — $\leq -35dB$ for 256-QAM — $\leq -43dB$ for 1024-QAM.
PAE_{MAX}	**Maximize** ($\geq 20\%$)
Small-signal gain	$\geq 20dB$

Table 2.8: Targeted Specs

The frequency band is specified by the standards. The output compression point, depending on the modulation order, could be anywhere from 5 to 8, depending on the modulation schemes and techniques used. Higher-order modulation schemes need a lower EVM, so they are operated further away from the PA's $P1dB$. Furthermore, OFDM's PAPR means that the compression point has to be high enough to allow acceptable transmission of the waveform peaks. Additionally, the power amplifier has to be stable over a very large range (much larger than its operating bandwidth), comply with the spectral mask, and be safe for operation transistor-wise and not exceed breakdown voltages in a manner that would destroy the device.

3 Methodology

3.1 Project Statement & Methodology Summary

The project focuses on the design, simulation, and layout of a CMOS power amplifier (PA) for IEEE 802.11 applications operating in the 2.4 GHz and 5 GHz bands. The design will be implemented using a 65 nm bulk CMOS process, aiming for high linearity and compliance with spectral mask and error vector magnitude (EVM) requirements. The Power Amplifier is to integrate all passives. This chapter outlines the design's approach and various key aspects of the process as a whole.

At a high level, this project entails:

1. **System-Level Specification Definition:**

   - Identification of key performance metrics, in this case, the Pout & P1dB. This will help determine the sizing, the bias points and give a good idea of the EVM at backoff levels from the compression point.
   - PAE will be maximized by attempting to bias the transistor at the optimum PAE-output power current density J_{opt}. The value of $0.3mA/um$ [3] is chosen as an initial value and tuned from there according to the stage (emphasis on higher output power for the output stage, more emphasis and less current density in the predriver).
   - For the above point, PAE can be further increased at the expense of linearity (as with using higher efficiency classes). This follows naturally from the above discussion: lower gate bias voltages than those of a class A would mean the transistor won't conduct for the entire cycle, or equivalently, the current density will be lower than the mentioned value of $0.3mA/um$.

2. **Initial Circuit Topology Selection:**

   - Selection of a suitable class for each stage (likely AB) based on efficiency vs. linearity trade-offs. For example, class A or AB for the output stages and class AB or other higher efficiency stages for subsequent stages, as they do not affect the $P1dB$ as much as the last stage.
   - Perform calculations for biasing and make extensive use of loadline theory in the design of the PA.
   - Using an accurate non-linear model of the device. e.g., A Cadence-ADS dynamic link is utilized in this project to use the PDK's device.
   - Knowing the passive devices of the PDK helps make the design process more streamlined. E.g., knowing the inductor's value and Q makes the design more accurate through integrating those passives in loadpull analysis and matching networks.

3. **Impedance Matching Network Design:**

    - Designing input and output matching networks to optimize gain and power transfer. (conjugate match for the input, output match for the output. Higher frequency designs might demand both source- and load-pull.
    - Using Smith charts and simulation tools to fine-tune impedance networks.

4. **Simulation and Optimization:**

    - Performing different linear and nonlinear analyses using cadence and/or ADS.
    - Performing the needed EM co-simulation using Sonnet (2.5D) or Ansys HFSS (3D).
    - Optimizing & tuning the PA design to meet specifications while balancing trade-offs.

5. **Layout and Parasitic Extraction:**

    - Implementing layout to investigate different parasitics and the effect of non-ideal components on performance.
    - Verification and extraction of parasitics.
    - Running post-layout simulations.

6. **Final Verification, integration and Tape-Out Preparation: (out of scope of this project)**

    - Validation against initial specifications.
    - Preparation of the final design for integration with other blocks & fabrication.

3.2 Approach & Methodology

The above summary leads us to the following points. The architecture and a more detailed account of the analyses and tools are presented.

3.2.1 System Architecture

The proposed power amplifier consists of two main stages and their respective matching networks.

1. **Input Matching Network:** An impedance matching network matches the 50Ω port to the input of the first stage.

2. **Driver Stage:** A lower-power, higher-efficiency amplifier that ensures sufficient gain and impedance matching to the main PA stage.

3. **An Interstage Matching Network:** An impedance matching network that maximizes the power delivery of the first stage while providing matching to the next.

4. **Power Stage:** A rather wide, high-linearity Class A amplifier designed to provide the required output power.

5. **Output Matching Network:** An impedance matching network that maximizes power delivery of the final stage and matches to the 50Ω output port.

Each of the two stages will implement inductive degeneration to improve the stability of the amplifier while limiting the value of such inductors in order to preserve the gain.

3.2.2 Simulation & Validation Plan

- **DC Analysis** to ensure proper large-signal & small-signal transistor operation and help with the initial sizing of the transistor by running VGS & VDS sweeps to determine optimal biasing points.

- **Transient Analysis** for stability and signal integrity verification, as well as to ensure that the transient drain voltage does not maximally stress the transistor and exceed its breakdown regions.

- **S-Parameter Analysis** To ensure proper small-signal gain is achieved, analyze the performance of the matching network and test the stability through the K- and/or Mu-factors.

- **Harmonic Balance Analysis** to evaluate power-added efficiency (PAE) and output power. This is an integral, crucial test of the project, as power amplifiers are non-linear and need to be tested using harmonic balance simulators. They are also integral for testing intermodulation and seeing the power level of out-of-band harmonics.

- **Envelope Analysis** to meet IEEE 802.11 linearity requirements by ensuring the needed EVM is achieved and to ensure that the waveform complies with the spectral mask.

- **Corner Analysis**: to analyze different corners to account for PVT variations. Particularly temperature.

- **EM Co-Simulation**: to ensure that the correct model of different parasitic elements (for example, spiral inductors) is accurate and suitable for our use cases, particularly after adding routing metals. Ideally, EM simulation of the entire PA should be done. Some devices are already characterized by the PDK and will be used as is.

- **Verification & Post-Layout simulations**: to resolve Design Rules Check (DRC) errors (while waiving others that lie outside of the scope of this project). A Layout-Vs-Schematic (LVS) test is also done to ensure proper connections. Parasitic extraction through PEX is finally conducted to perform post-layout simulations using the extracted circuit.

3.2.3 Resources and Tools

- **Schematic Design & Simulation Tools:** Cadence Virtuoso, Keysight ADS.
- **Layout & EM cos-imulation:** Calibre (for layout verification and extraction) & Sonnet (for EM simulations).
- **Technology Platform:** TSMC 65nm CMOS process
- **Other Supplementary tools:** Python. Matlab.
- **Reference Standards:** IEEE 802.11.

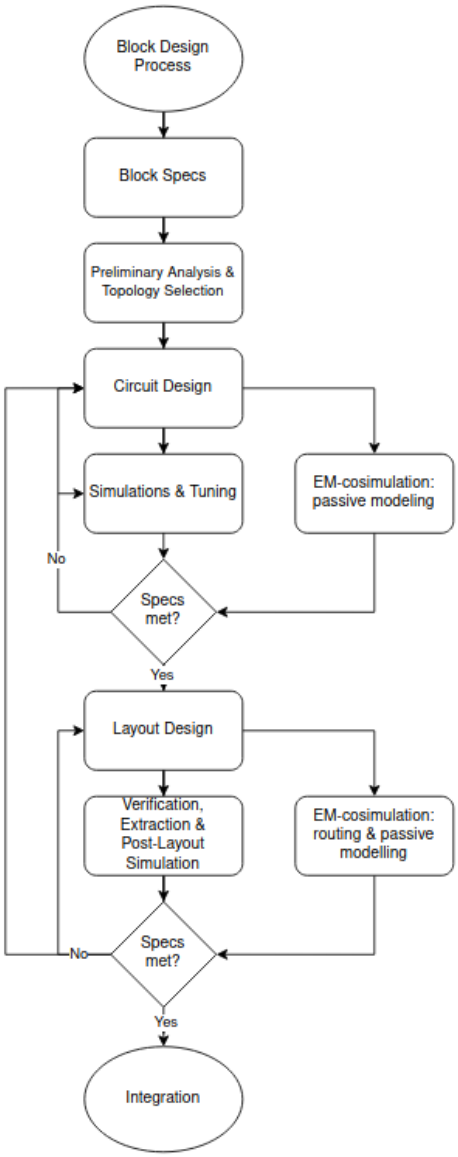


Figure 3.1: Simplified RFIC Design Flow.

4 Power Amplifier Design

This is the main focus of the thesis: the design process itself beginning from device and topology selection all the way to various analyses, while also integrating passive models.

4.1 Device Selection

An `nmos_rf` device model was used. This device—while it has lower voltage breakdown compared to other thick oxide devices—has the smallest channel length, has very high f_{max} (frequency at which unity power gain is achieved) values and thus higher gains at the WiFi frequencies. A testbench and simulation for pinpointing the f_{max} is shown below. However, a thick-oxide device might be more suitable when higher output powers are needed and space is not overly constrained (as several stages might be needed to compensate for its lower gains). The use of such a device might be more beneficial for an access point or more powerful mobile devices.

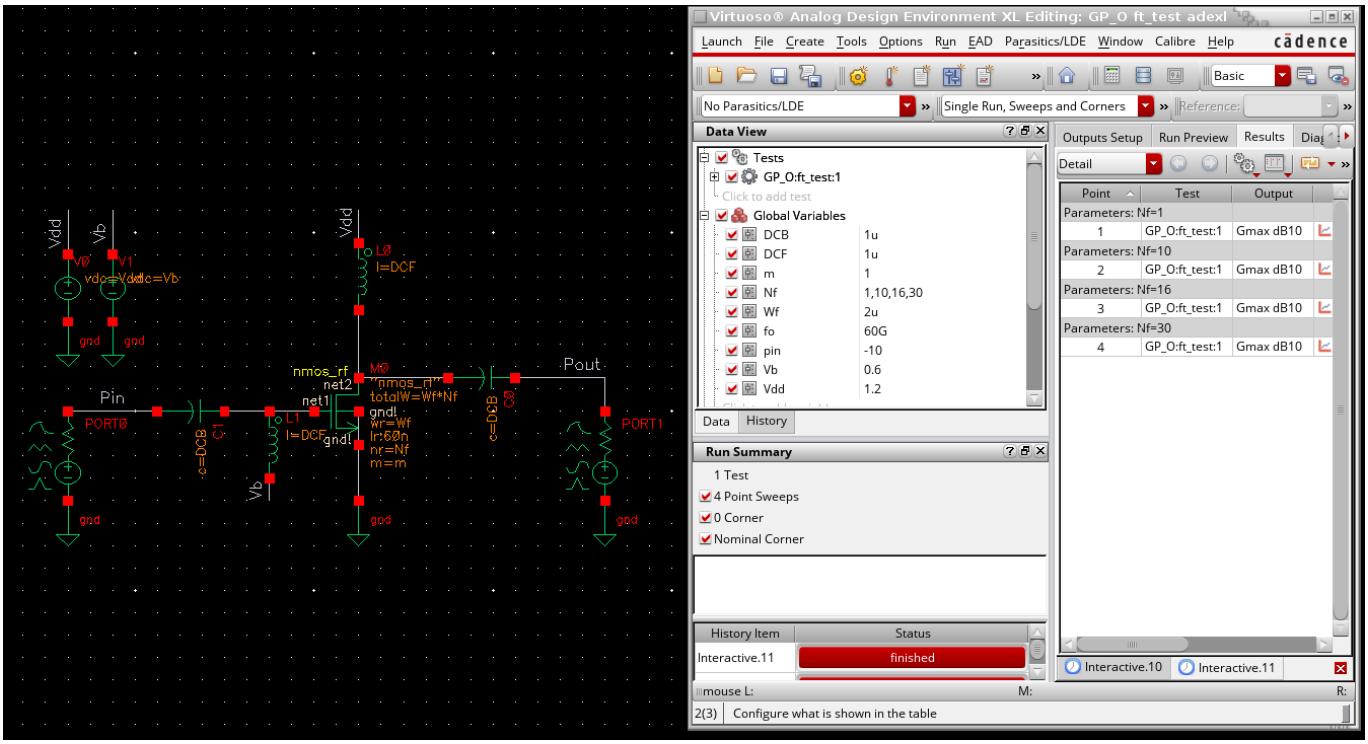


Figure 4.1: f_{max} Testbench

For this device, the length and width were chosen as the default ($L = 60nm$, $W = 2\mu m$). These values were also reached upon after some testing to optimize performance. A multiplier of 1 is used for this test. The device can be later multiplied as needed with no impact on the performance.

Notably, this device has a maximum recommended voltage of around $1.5V$ and an absolute maximum breakdown voltage of around $2.0V$.

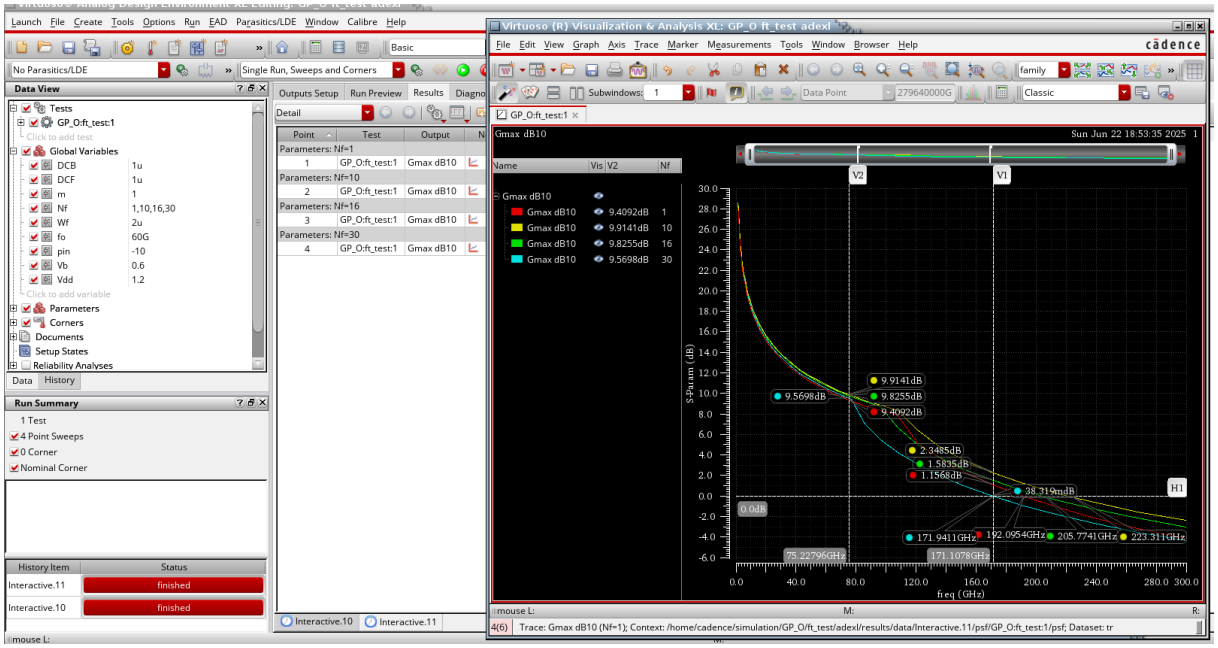


Figure 4.2: f_{max} results for different finger values.

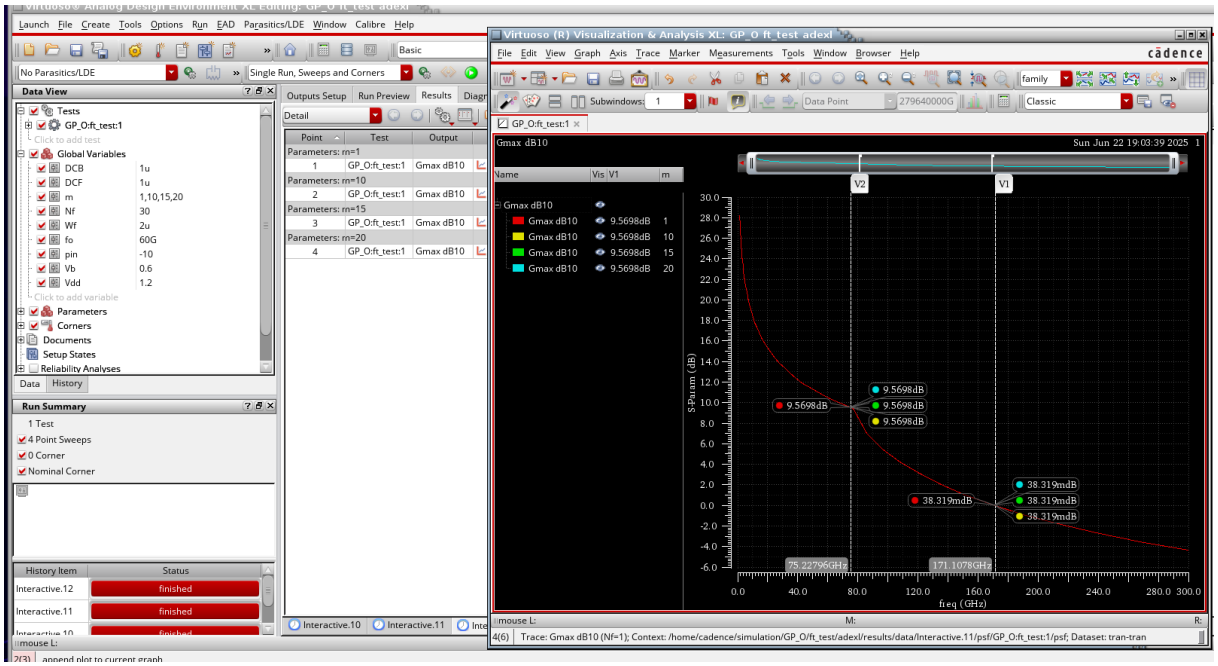


Figure 4.3: f_{max} results for different multiplier values.

The device with $N_f = 30$, while shows the least f_{max}, has very similar gain to the others at up to $\approx 75GHz$, but would source much more power. It was used as the main device for the two designs at Wi-Fi bands. A multiplier of 15 was later chosen; A value that was achieved after some tuning and characterization.

A more detailed methodology for the choice of the transistor size, as a result of chapter 2.5 discussion for class A amplifier equations, is to be shown in the form of a python code in the appendix. The methodology is detailed in chapter 6 of Sorin's Book [3].

4.2 Device DCIV Characterization

The device was then taken to ADS (through a dynamic-link) in order to design the amplifier. Before that, however, characterization is needed to get an idea of the bias points and the currents of the device.

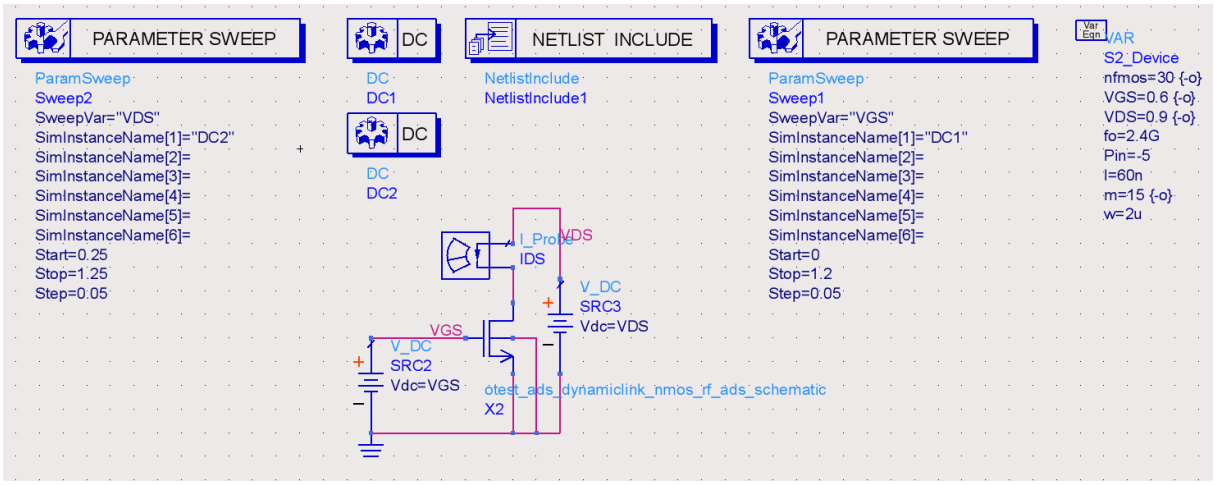


Figure 4.4: DCIV Characterization Testbench. Both V_{GS} & V_{DS} are swept.

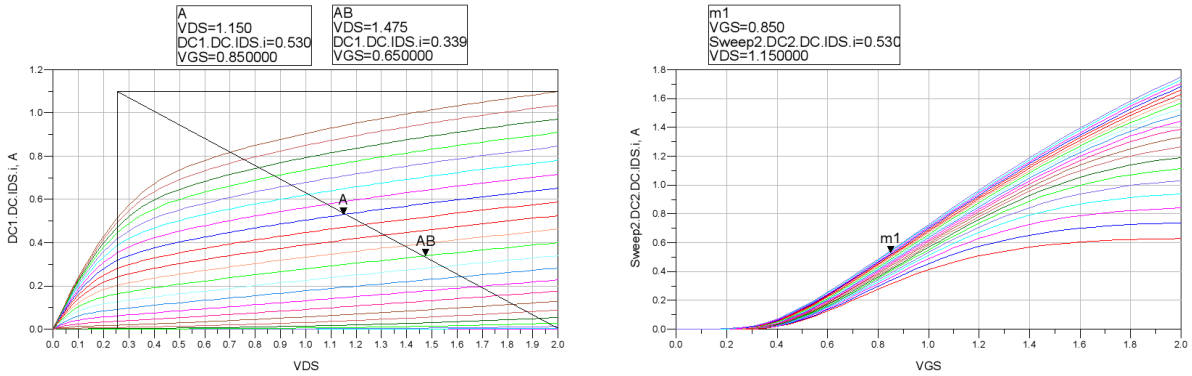


Figure 4.5: DCIV Characterization of the device. Note the SCE & CLM.

The SCE (short-channel effects) & CLM (channel-length modulation) of the device make pinpointing the loadline and bias points more difficult than when use a device with a long channel and much less CLM. These values give an initial idea of the device. **It should be noted that these values will later be tuned down, as the currents are too high.**

4.3 The RF CS Stage

The CS stage, with some modification, would lay the foundation of this work. The CS stage has one transistor configured in a common-source configuration. There are two DC Blocking caps that pass the RF signal of interest while blocking DC components at the gate and drain node. Similarly, there are two DC Feed inductors that help set the DC bias voltage of the MOSFET and passing the supply voltage

(the RF choke) at the gate and drain nodes. The values of these capacitors and inductors are, ideally, very large in order to act as AC & DC switches. Practically, these values have to be sensible and of reasonable quality factors to be integrated on chip. More importantly, a value that is too high - with a limited Q - will have too much losses after factoring in the parasitic resistance of the element (inductors in particular have high losses). For 2.45 GHz and 5.5GHz, the highest inductors with a Q factor of ≥ 20 where found to be in the range of $1nH$ and $0.8nH$ respectively. The selection of these values are touched upon in Appendix B.

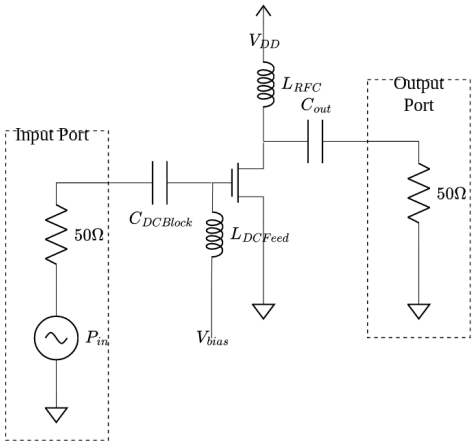


Figure 4.6: RF CS Stage.

Note: the frequency choice of the $2.4GHz$ & $5GHz$ bands were $2.45GHz$ and $5.5GHz$ - roughly targeting the middle of the band. This allows the peak of the gain to be centered over the frequency band of interest. This is also to deal with any small deviations in matching that might occurs when using an accurate model of the PDK devices and after routing which will (at least) slightly affect the frequency at which the S-parameters are optimal.

A degeneration inductor was later added to improve stability. It's value was kept as low as possible in order to preserve the gain and its Q factor was selected to very high (≥ 30 if possible).

During the first phases of the design and during debugging, ideal DC Feeds and blocks were used instead of the finite inductors and caps (with the inductors having losses added on later). However, the DC blocking caps are shunted in this design; they were made to be integrated as a part of the matching network whenever possible. Otherwise, when this proved difficult, the maximum mimcap value of $20pF \implies -j0.145\omega$ was used. It's area was comparatively much smaller than the inductor and thus was used.

The power stage was designed in the following manner for the 2.4 & 5GHz Stages. The pre-driver was much the same, but biased at a lower voltage in order to boost the efficiency while targeting gain and better S-parameters at the cost of a slightly lower $P1dB$.

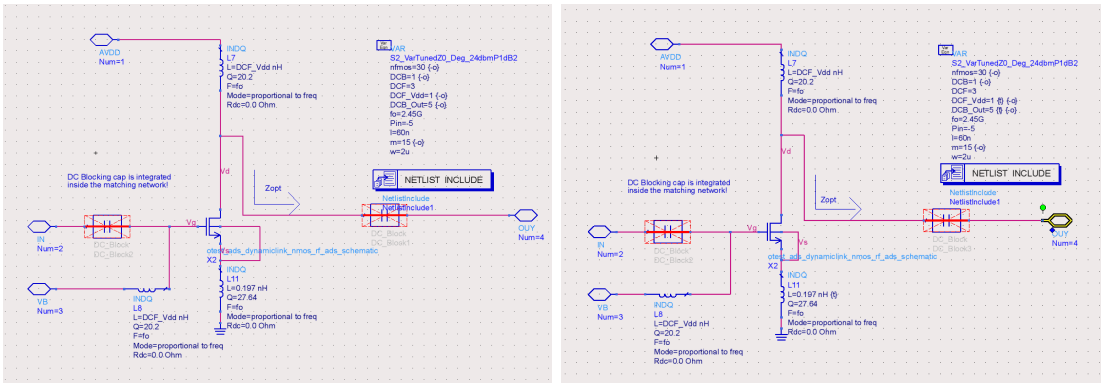


Figure 4.7: The two stages of the 2.4 GHz PA.

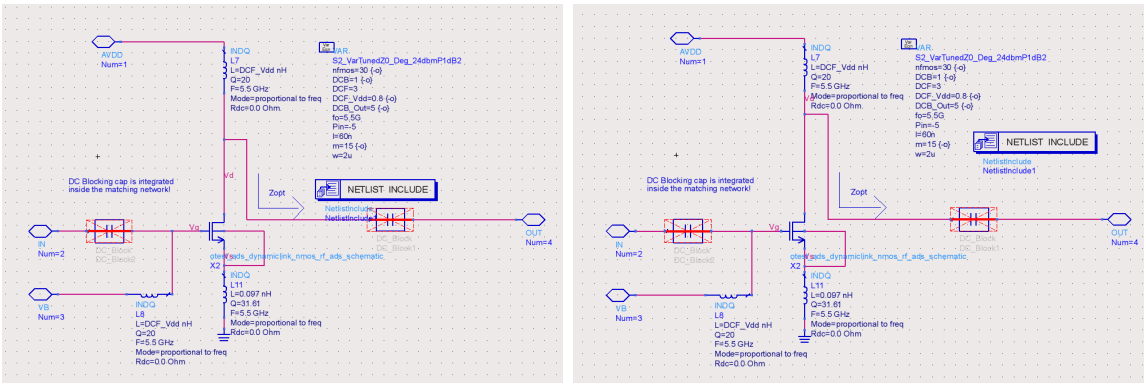


Figure 4.8: The two stages of the 5 GHz PA.

4.4 Impedance Matching

Matching is crucial to get the best performance of any block that works at high enough frequencies, as wire are effectively transmission lines and thus microwave theory is to be used. The following testbench was used to test the PA at different phases of the design.

For power amplifiers, the matching is best done from the output of the last stage and backwards. Note that the output port was first made variable to help in the tuning process until an optimum Z_{opt} value was found. And then the output port was fixed to 50Ω. Note that all matching networks and stages are deactivated in this particular instance.

Note that a more rigouros and systematic approach is outlined later in the appendix that involves loadpull and sourcepull tests.

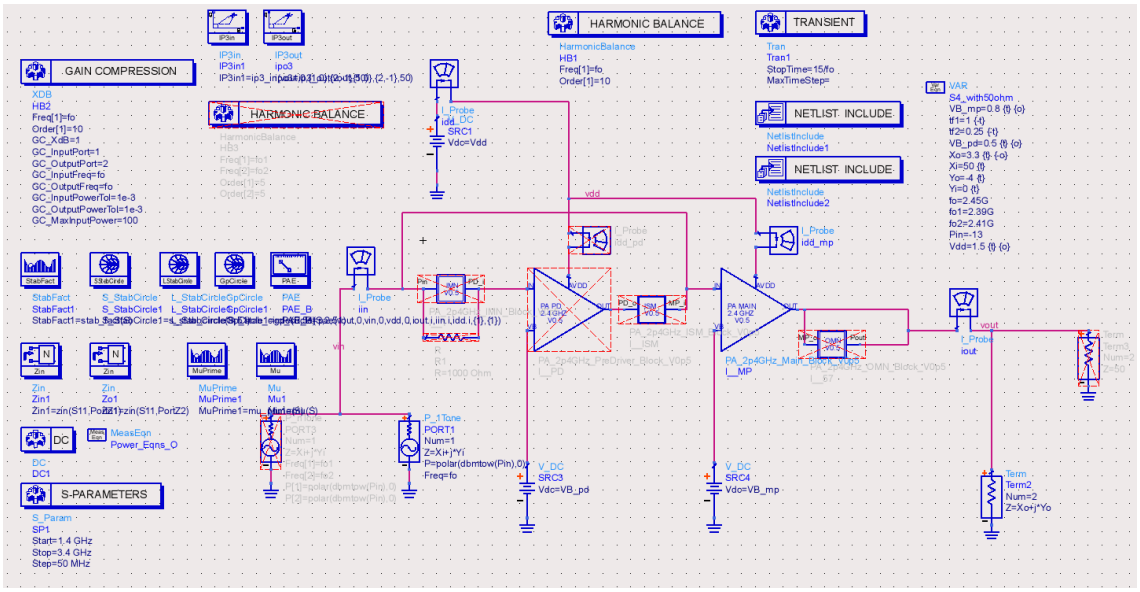


Figure 4.9: Testbench for the Power Stage. Other stages for different designs were tested similarly.

In order to maximize power delivery, R_{opt} was presented to the output of each transistor instead of providing a conjugate match. This directly follows from the above discussion: to get the most power, we want to maximize the swing. The DCIV characterization done above helps us settle upon an initial estimate of the optimal load-line. The above gives us roughly $\frac{1.15}{0.53} \approx 2.2\Omega$. Obviously, this is merely a starting point. The process is shown for the first power amplifier designed: the $2.4GHz$ PA. A very similar procedure was done for the $5GHz$ PA. Ideally, Loadpull, which will be touched upon later, should be used.

Also note that during the first iterations of testing, the blocking caps were turned on until they were eventually replaced with those of the network. Note the weak gain in this stage because there's no input matching network in effect now.

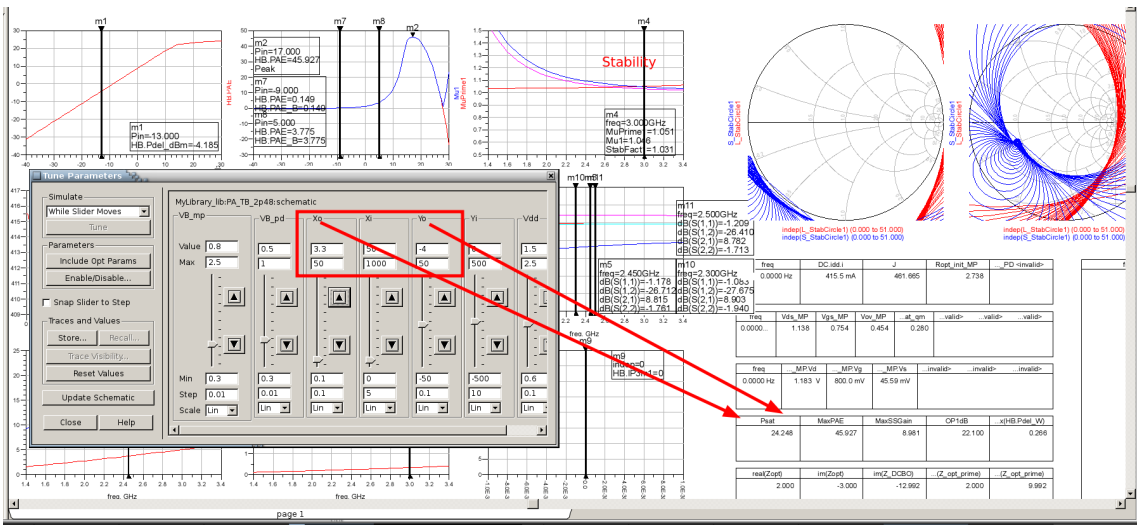


Figure 4.10: Tuning $Z_{opt} = X_o + jY_o$ of the 2.4GHz Power Stage.

The smith chart was then used to "transform" the 50Ω output to the Z_{opt} value.

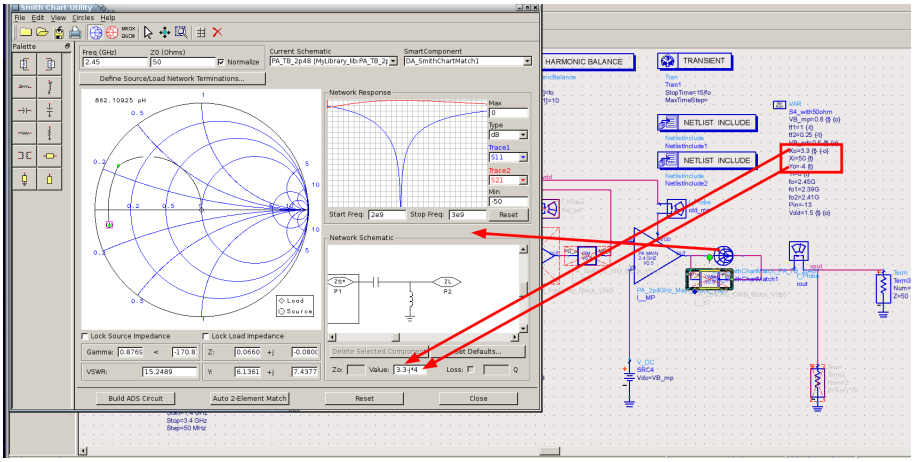


Figure 4.11: Output matching example of the $2.4GHz$ PA last stage.

For the Input matching, a conjugate match is used for the amplifier. Small-signal parameters can be a viable option here because we want to maximize the S_{21} & S_{11} performance of the PA from the input side to ensure maximum gain and the least reflections. Additionally, since the input power is very low (comparatively) and the they are far from the compressive regions of the PA, the small-signal approximation could be used.

The Z_{in} value was found from the testbench for both the PA power stage and input stages. Ideally, this step should be the last. Here, an input impedance of $Z_{in} = 4.7 + j17.6$ is matched to the 50Ω input port (the 50Ω port is used as the source this time).

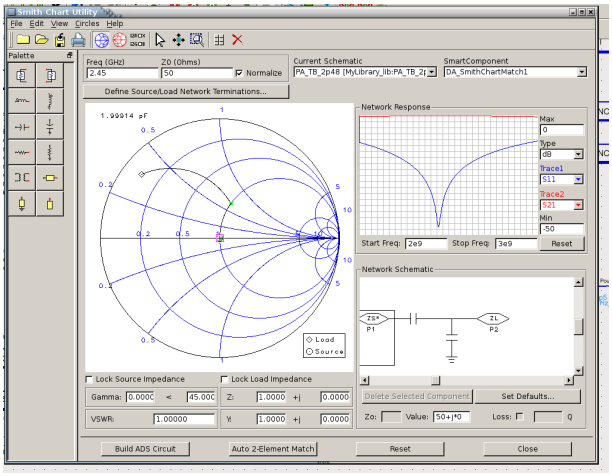


Figure 4.12: Input matching example of the $2.4GHz$ PA input stage.

Similarly, an interstage matching network was used to match the output of one stage to the input of the other. The goal here is to maximize power transfer and limit reflections. Thus, a conjugate match was used. An alternative is to match both the input and output of every stage to 50Ω. This approach, however, would introduce more elements, thus consuming more area as well as introduce more losses associated with the matching networks.

Here, $Z_{opt} = 5.5 - j4$ (taken as source) of the predriver was matched to the input of the power stage $Z_{in} = 3.8 + j15.8$ (taken as load).

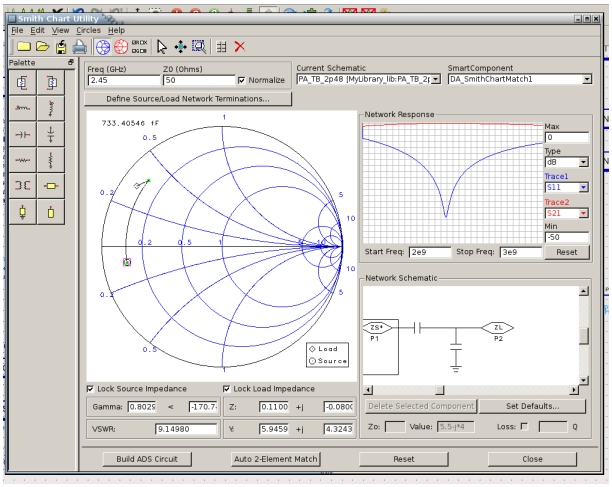


Figure 4.13: Interstage matching example of the $2.4GHz$ two stages PA.

There are two main points that are worth mentioning: 1. DC blocking capacitors were integrated as part of the matching network whenever needed. 2. A fully capacitive network was used whenever possible for any matching stage. All matching was normalized to 50Ω.

4.5 Final Design, Testbench and Preliminary Results

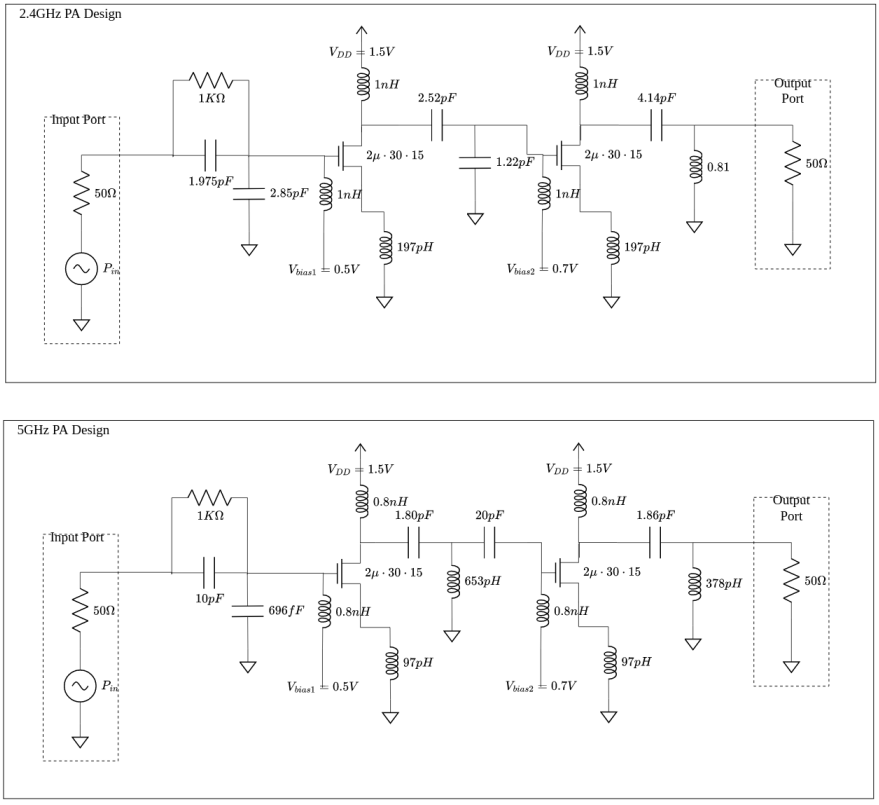


Figure 4.14: The $2.4GHz$ & $5GHz$ PA schematics.

The final designs for each of the stages, as well as the input, output, and interstage matching networks are as shown for the $2.4GHz$ & $5GHz$ designs. Note that the inductors are now modeled with accurate Q-factors for the PDK.

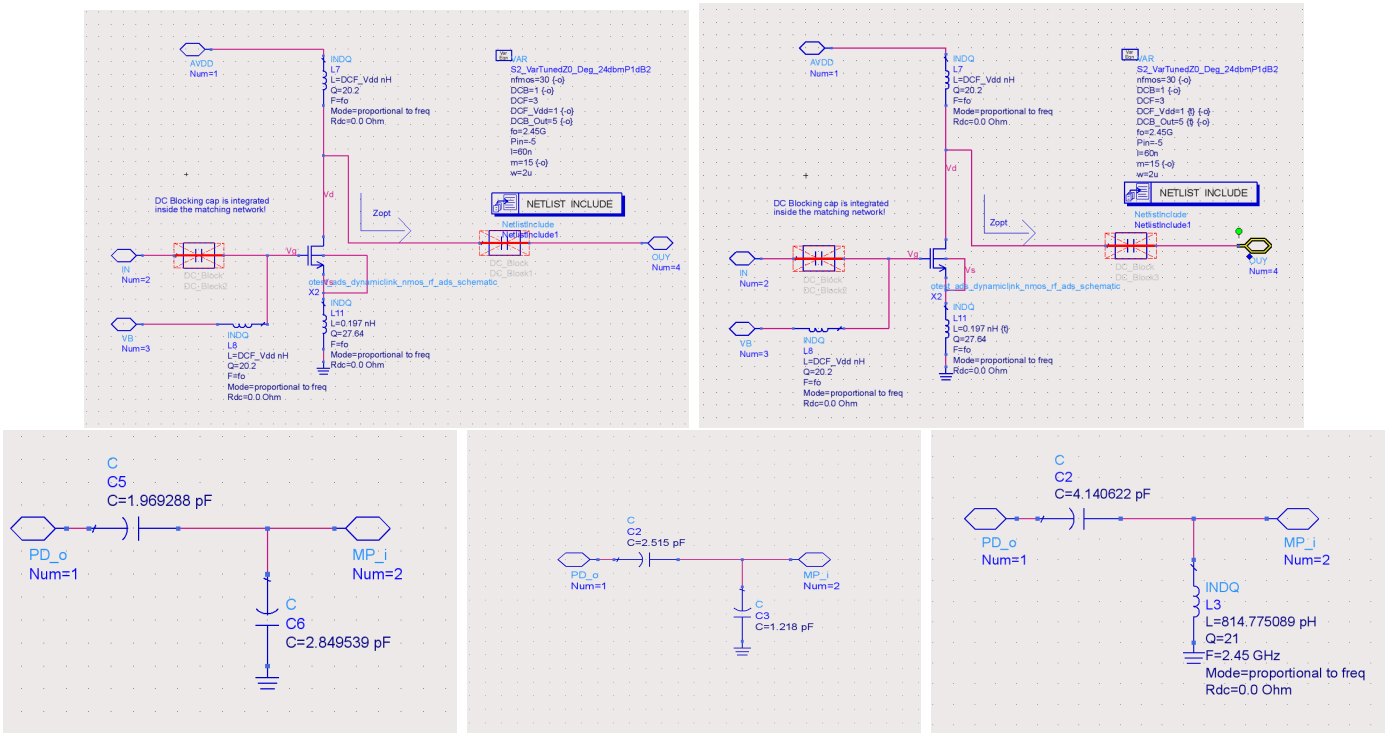


Figure 4.15: The two stages, input, interstage, & output networks of the $2.4GHz$ PA respectively.

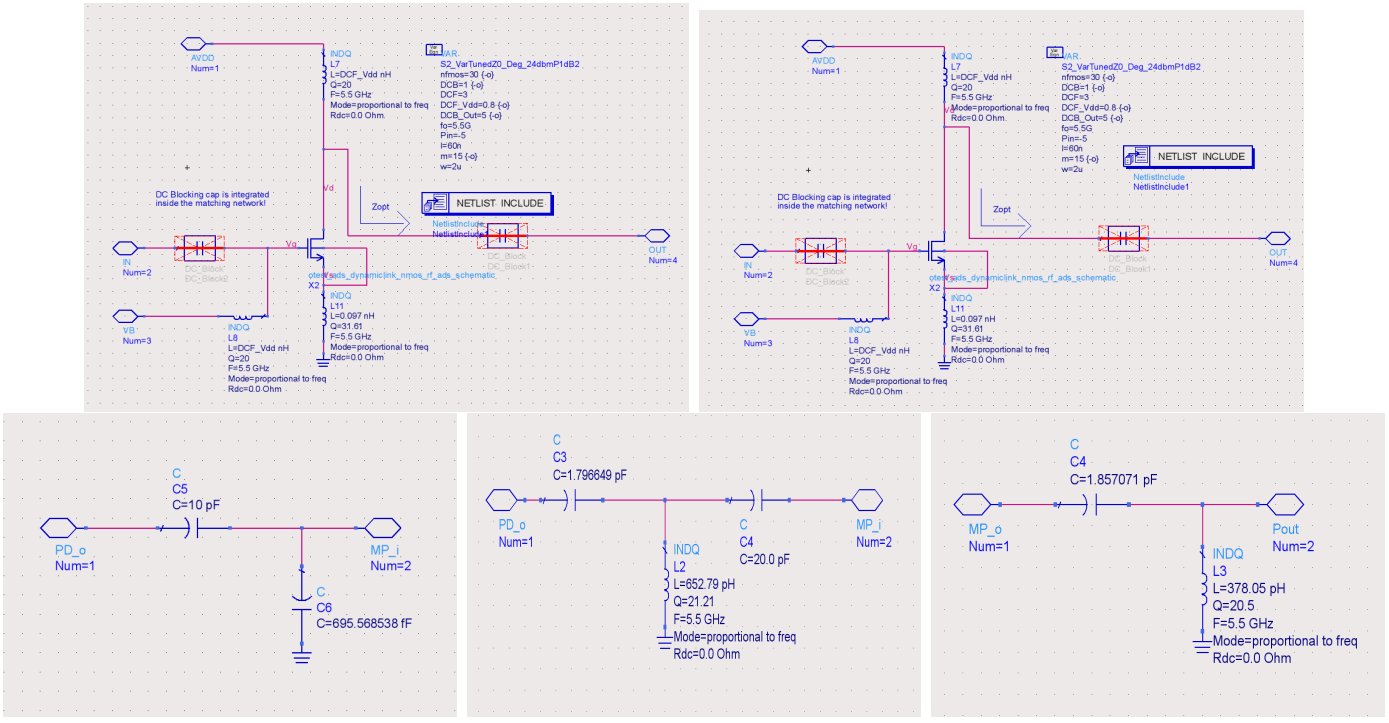


Figure 4.16: The two stages, input, interstage, & output networks of the $5GHz$ PA respectively.

After some simulations for the whole PA from $1Hz$ to $100GHz$, the PA was found to be only

conditionally stable at a low frequency $\leq 10KHz$ for the $5GHz$ design. This was remedified by adding a small $1k$ resistance in parallel to the input of the first matching network. This was also added to the $2.4GHz$ design.

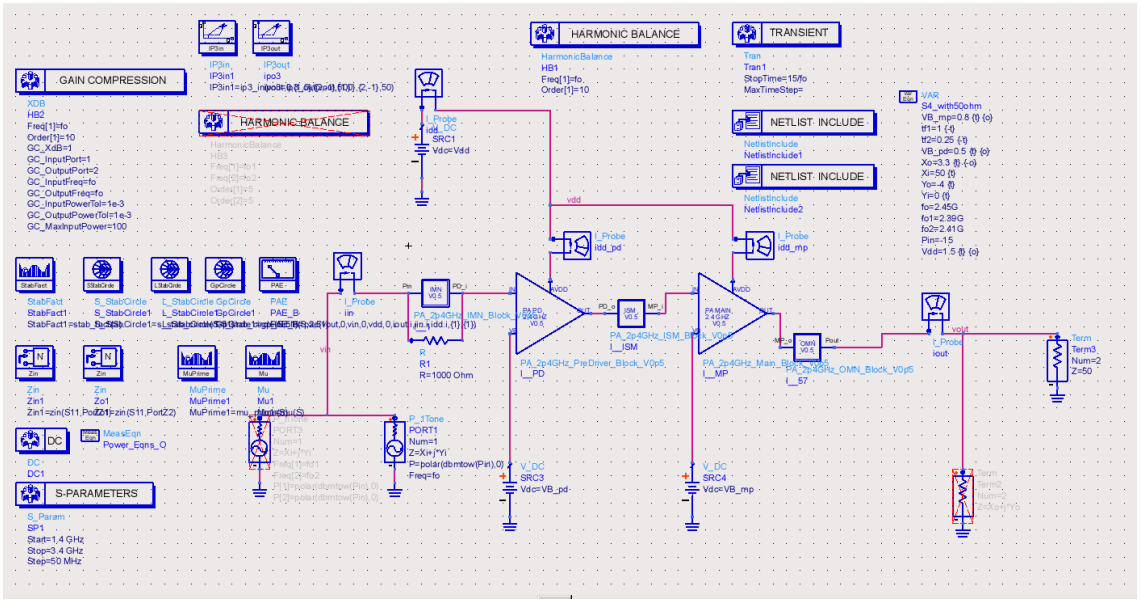


Figure 4.17: The testbench of the $2.4GHz$ PA.

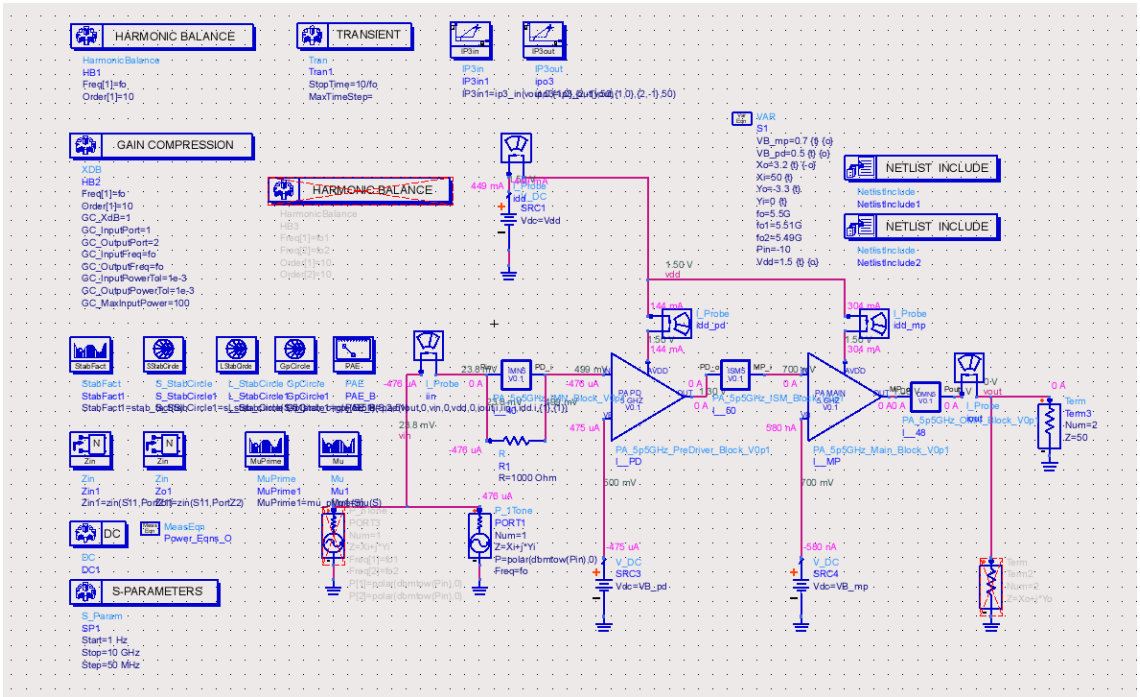


Figure 4.18: The testbench of the $5GHz$ PA.

Note that this PA integrates several tests: DC simulations, S-parameters, Harmonic Balance and XdB analyses for the compression point, and transient analysis to ensure that the drain-source voltages of the transistors are within limits.

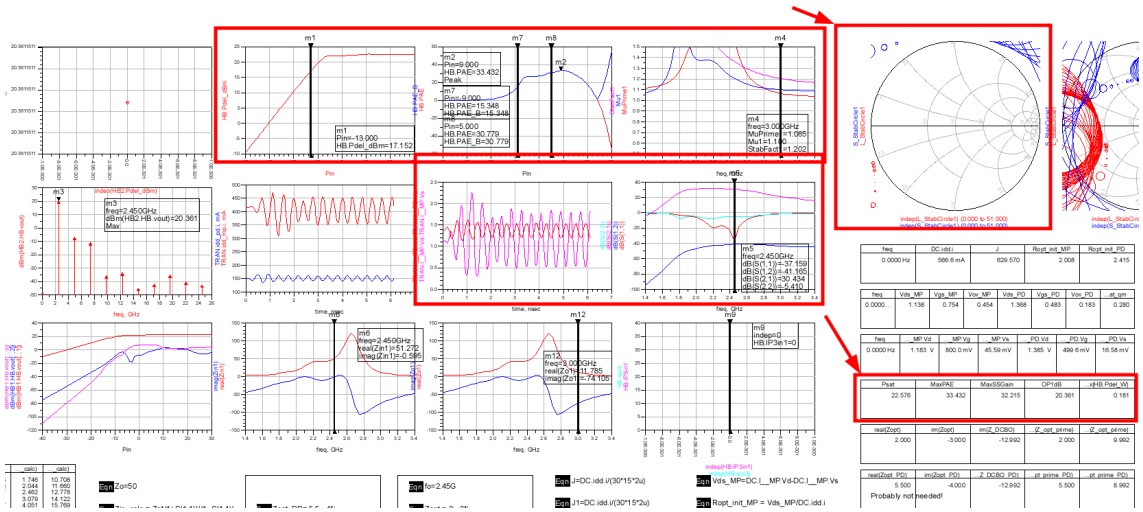


Figure 4.19: ADS Results of the $2.4GHz$ PA.

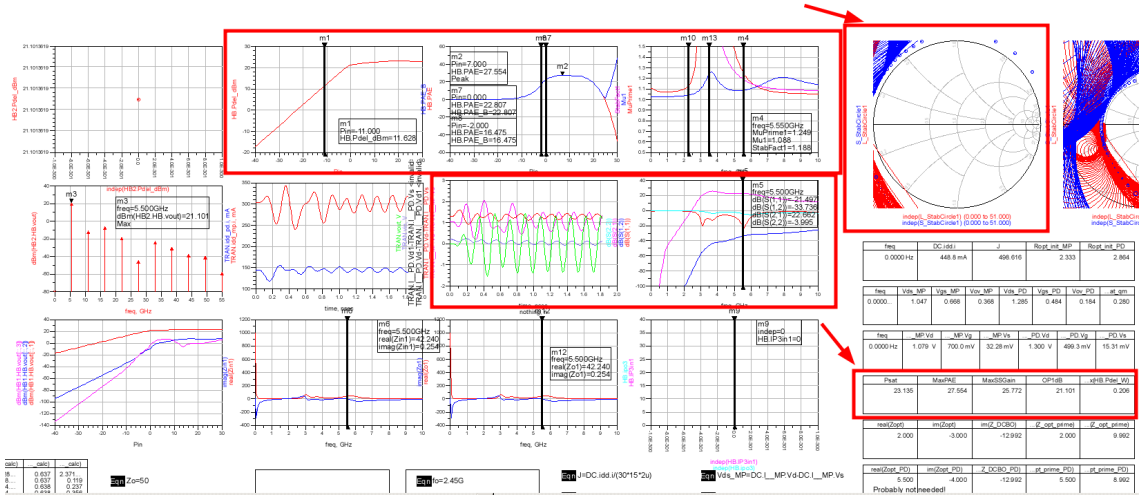


Figure 4.20: ADS Results of the $5GHz$ PA.

The PA is stable (and will soon shown be over the entire range in Cadence Virtuoso as well). The transient responses shows that it only briefly reaches $2V$ in a very brief period of less than $0.1ns$ for the 5GHz design and a relative time frame for the 2.4GHz design.

The small signal gains and overall S-parameters are very satisfactory except for S_{22}, which is expected. Furthermore, the PAE and $P1dB$ are both satisfactory for now.

Also note the K, μ, and μ' factors and stability circles, all of which show stability over this range. (The other circle is an "extended" smith chart, which also shows negatives impedances. The standard smith chart is what concerns us.)

4.6 Final Simulations & Results

Finally, the design was ported to Cadence Virtuoso. A device was also made for the NMOS since this device was to be used for the two designs. These are the final designs:-

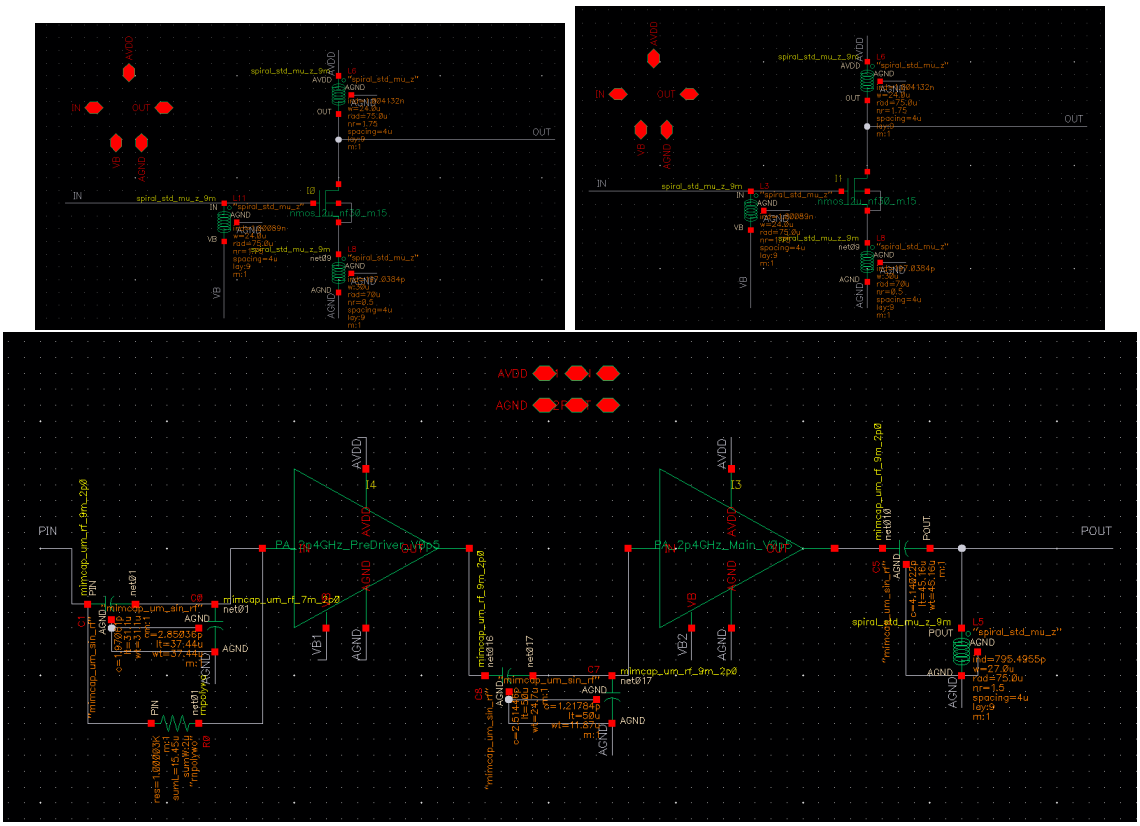


Figure 4.21: The 2.4GHz PA with PDK elements.

The testbench is as follows. The input port was activated for all analyses except for envelope analysis, in which the `wlan` source took over along with the wireless probe.

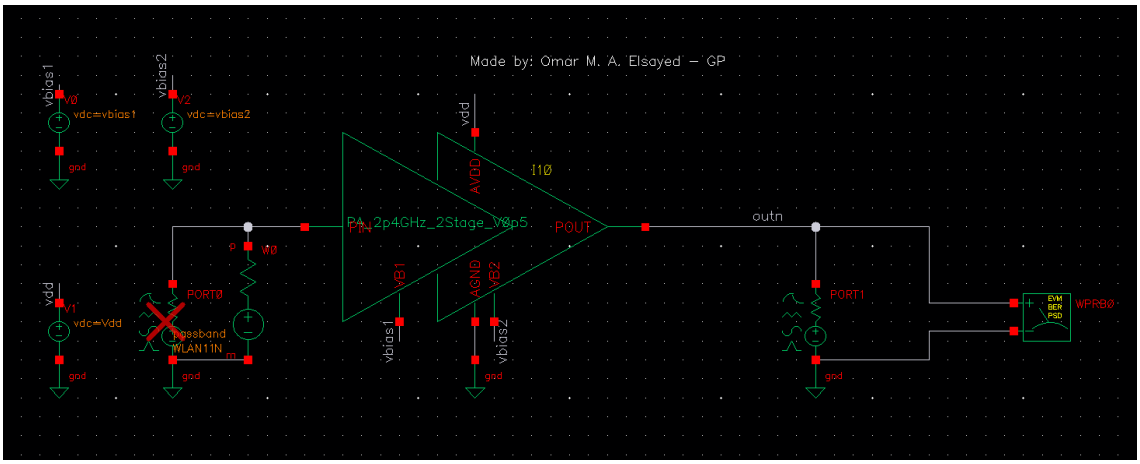


Figure 4.22: Cadence testbench of the 2.4GHz PA.

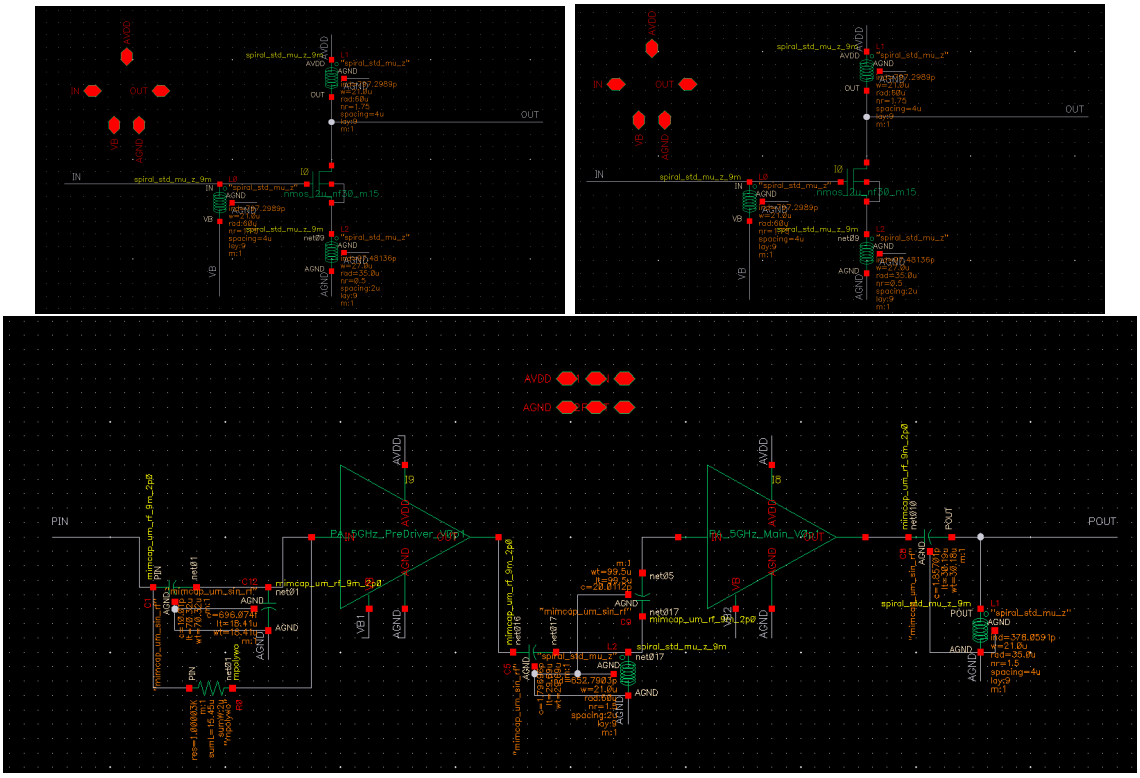


Figure 4.23: The $5GHz$ PA with PDK elements.

The exact concept as the previous testbench. However, since the $5GHz$ band supports 802.11ac, another `wlan` source for it was used instead of only one for 802.11n.

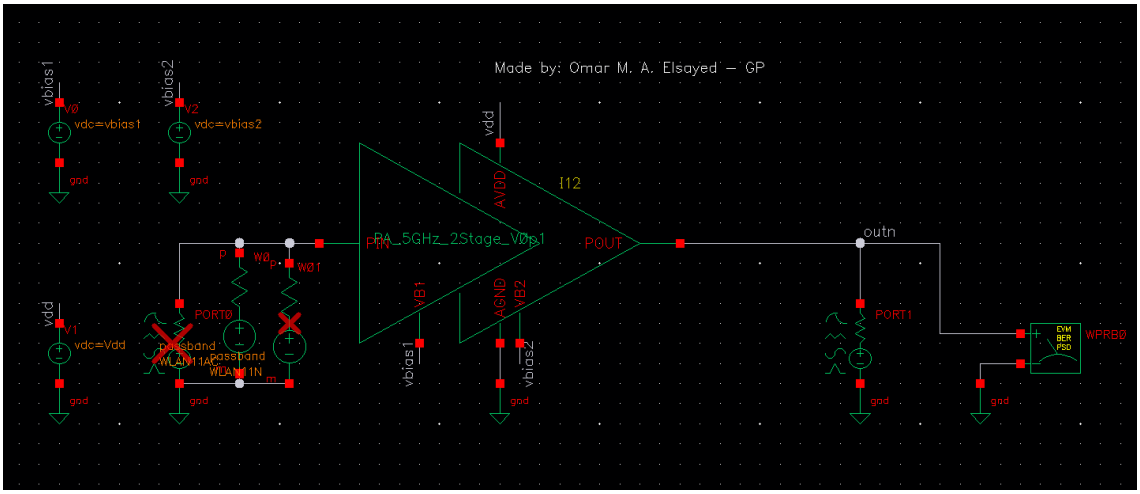


Figure 4.24: Cadence testbench of the $5GHz$ PA.

The following tests are the focus: DC operating point, S-Parameters test (stability will also be checked from $1Hz$ to $100GHz$), harmonic balance analyses: PSS & PAC (PSS to check for compression points, PAE, and conduct a two tone test using PAC), and finally, an ENVLP analysis to check the EVM & ACPR. A corner at 125° was also added.

4.6.1 2.4GHz PA Simulation & Results

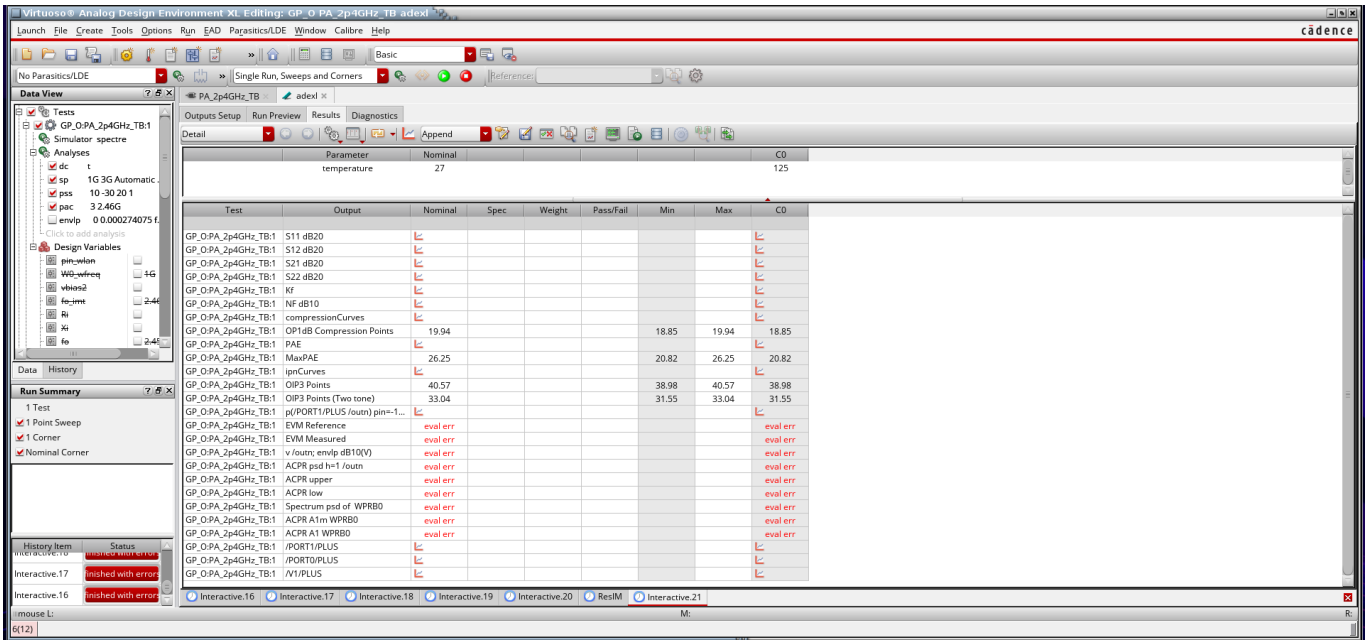


Figure 4.25: $2.4GHz$ PA Summary

Note the max PAE, the Output P1dB, as well as Output IP3 points across the corners.

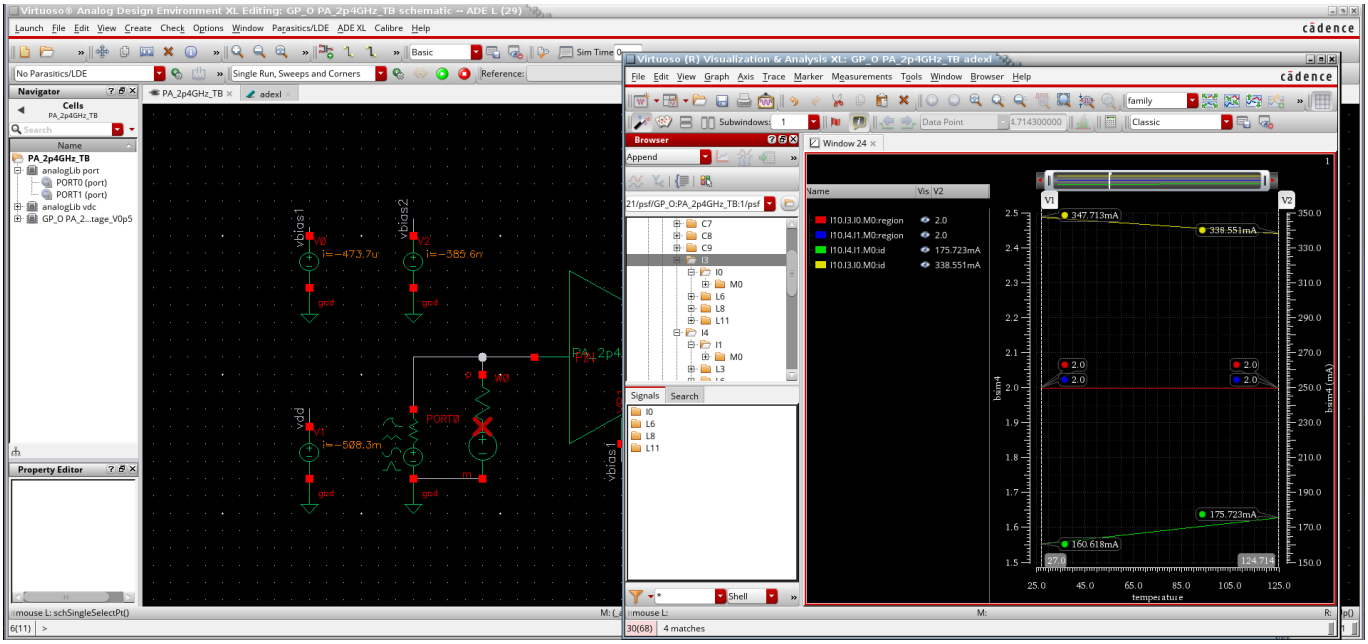


Figure 4.26: $2.4GHz$ PA DC Results

The total DC consumption is $P_{dc} \approx 0.5 * 1.5 = 0.75W$, Note that the DC Bias points were tuned alongside the overall performance to decrease current consumption compared to before.

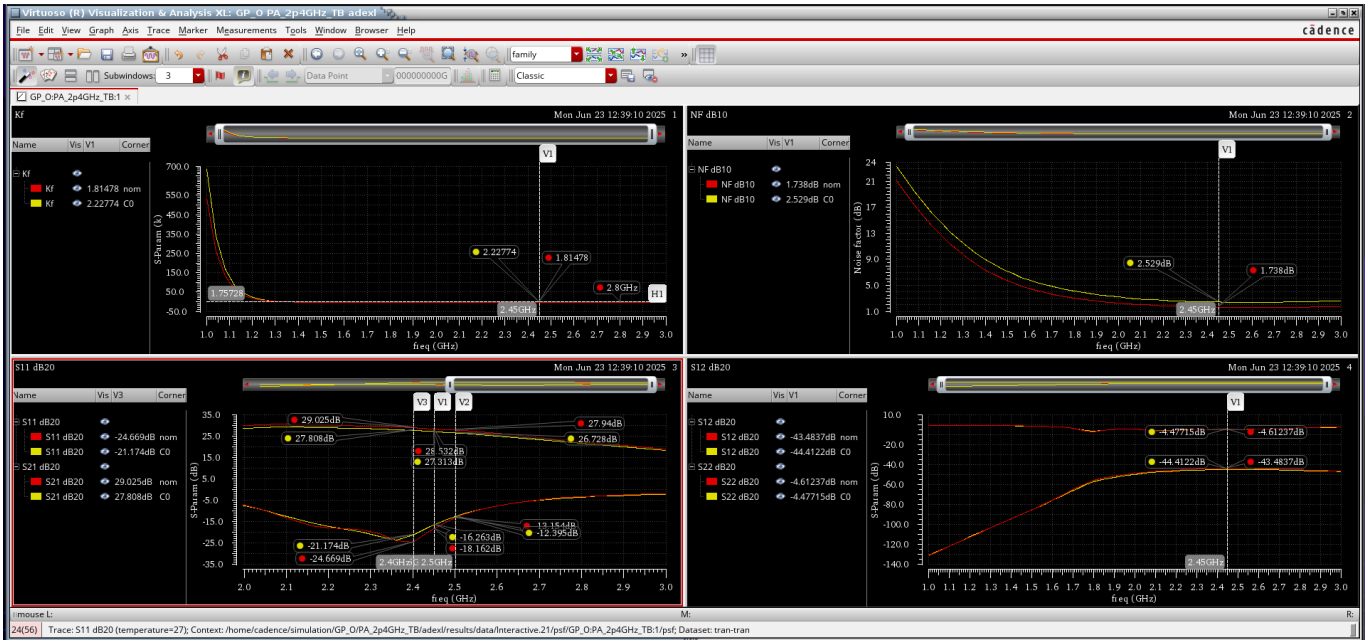


Figure 4.27: $2.4GHz$ PA S-Parameters

The S-parameters are satisfactory for the outlined ranges with little variation.

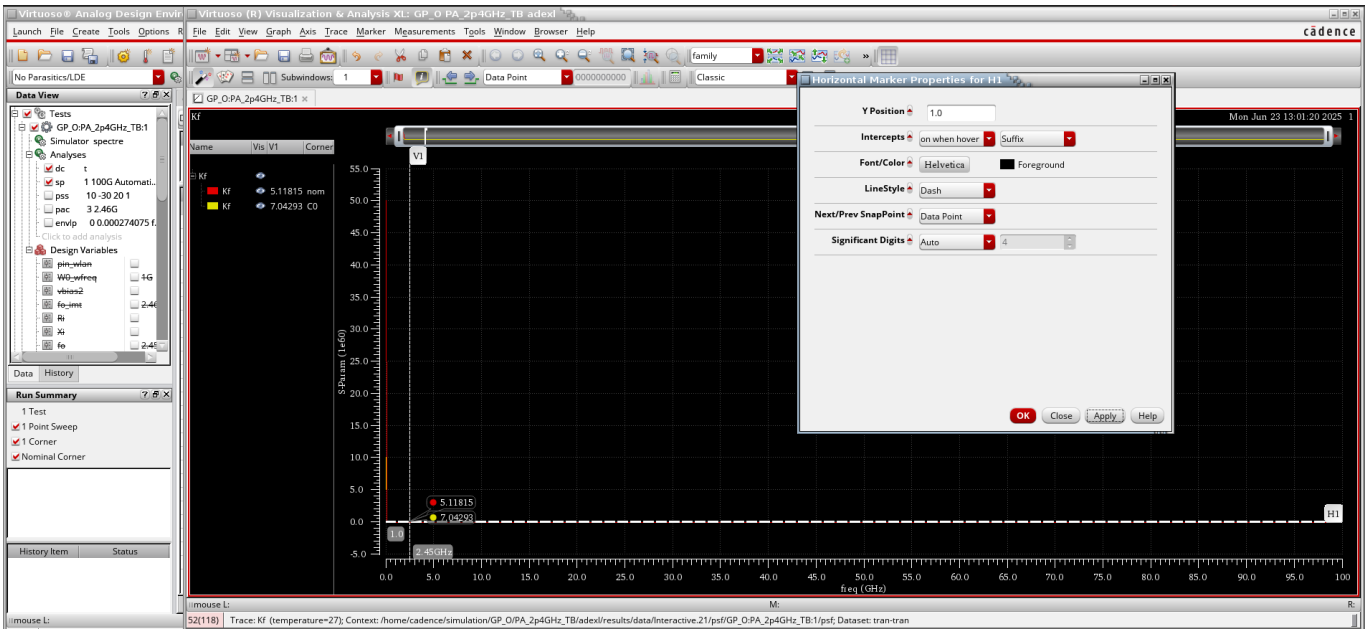


Figure 4.28: $2.4GHz$ PA Stability

The PA is unconditionally stable from $1Hz$ to $100GHz$. Note that the horizontal marker doesn't intersect the curve at $y = 1$.

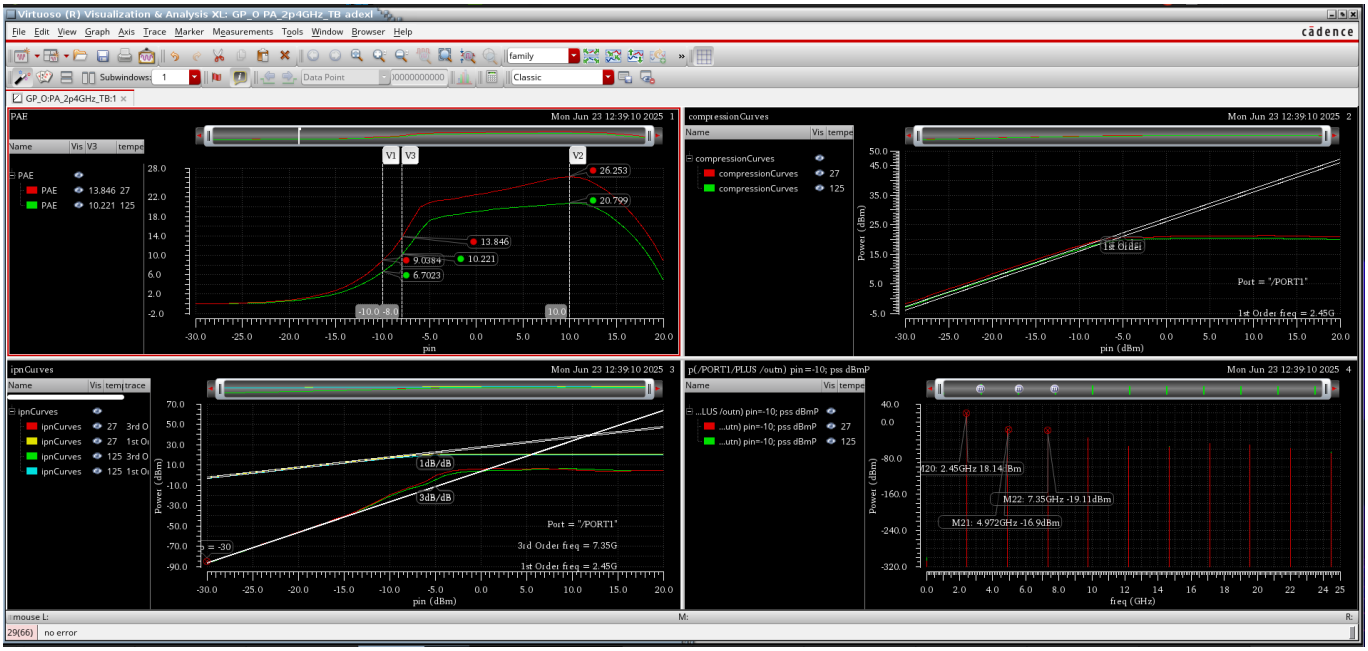


Figure 4.29: $2.4GHz$ PA PAE, P1dB & IP3 Curves

The PA exhibits the shown values above. Also note that the higher order harmonics have very low levels.

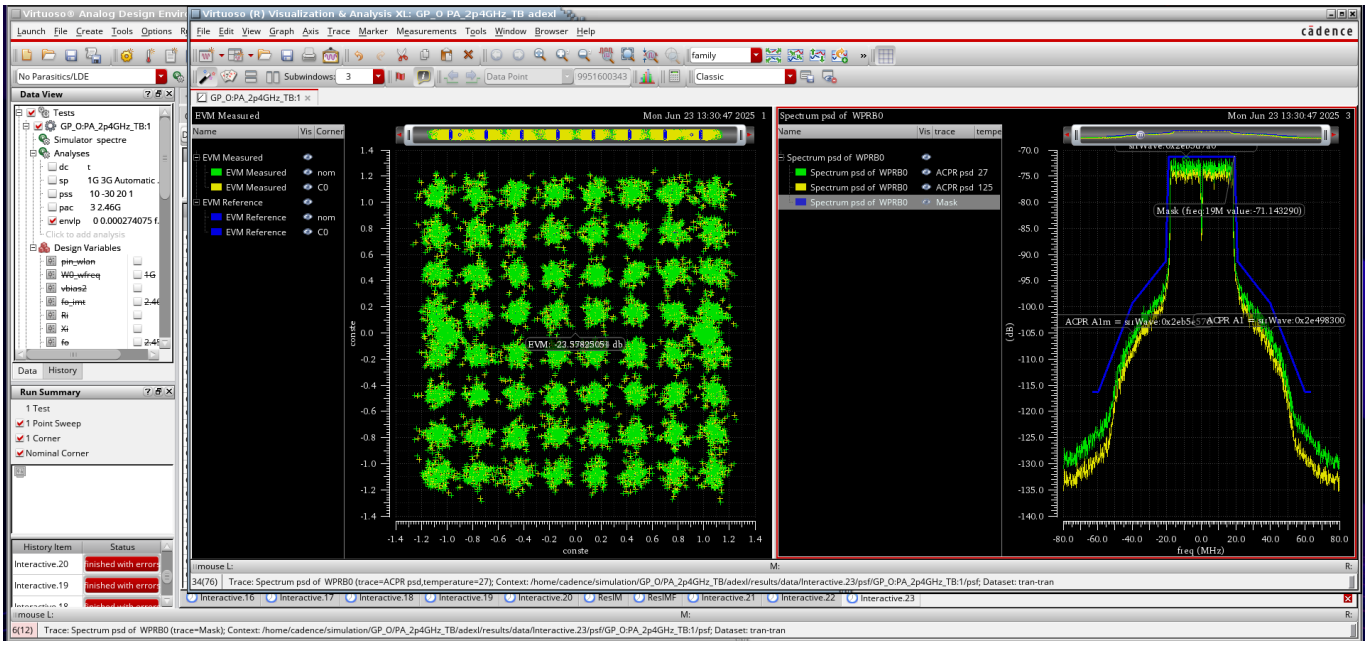


Figure 4.30: $2.4GHz$ PA EVM & ACPR (802.11n)

The PA has achieved $EVM \leq -22dB$ for a $P_{in} = -13dBm \implies P_{out} = 15.4dBm$ while complying with the spectral mask results for an 802.11n signal with $40MHz$ channel width and a short guard band.

4.6.2 5GHz PA Simulation & Results

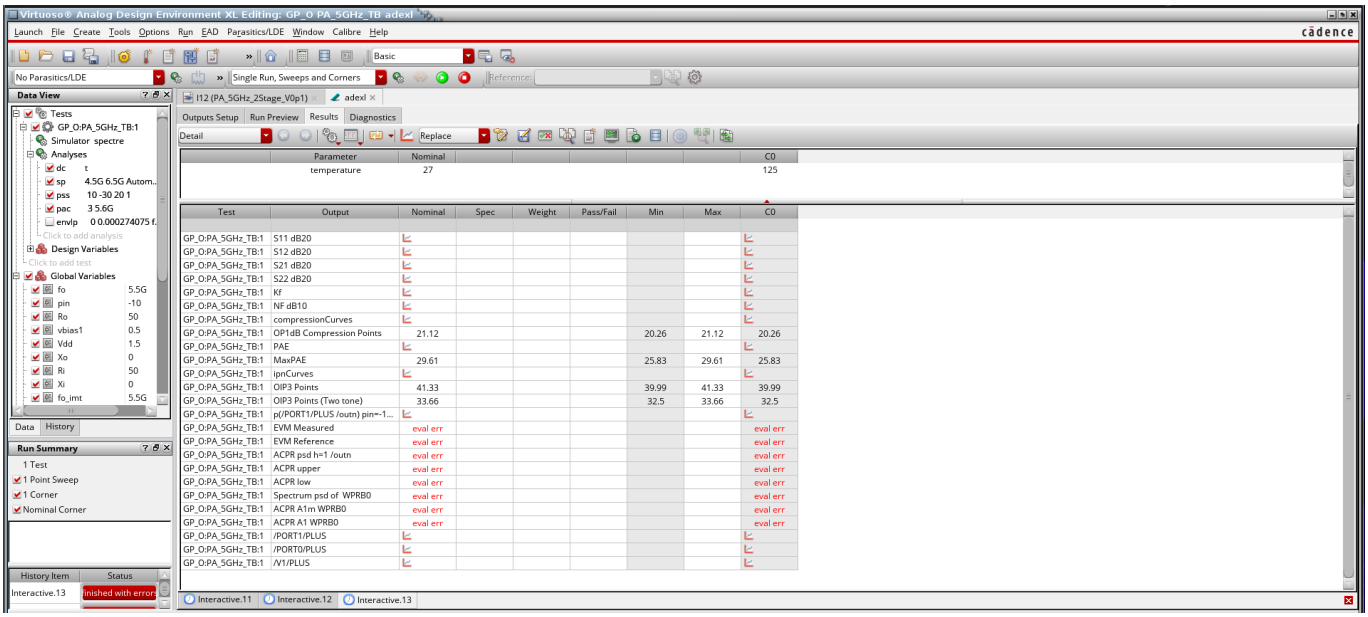


Figure 4.31: $5GHz$ PA Summary

Similarly, note the max PAE, the Output P1dB, as well as Output IP3 points across the corners.

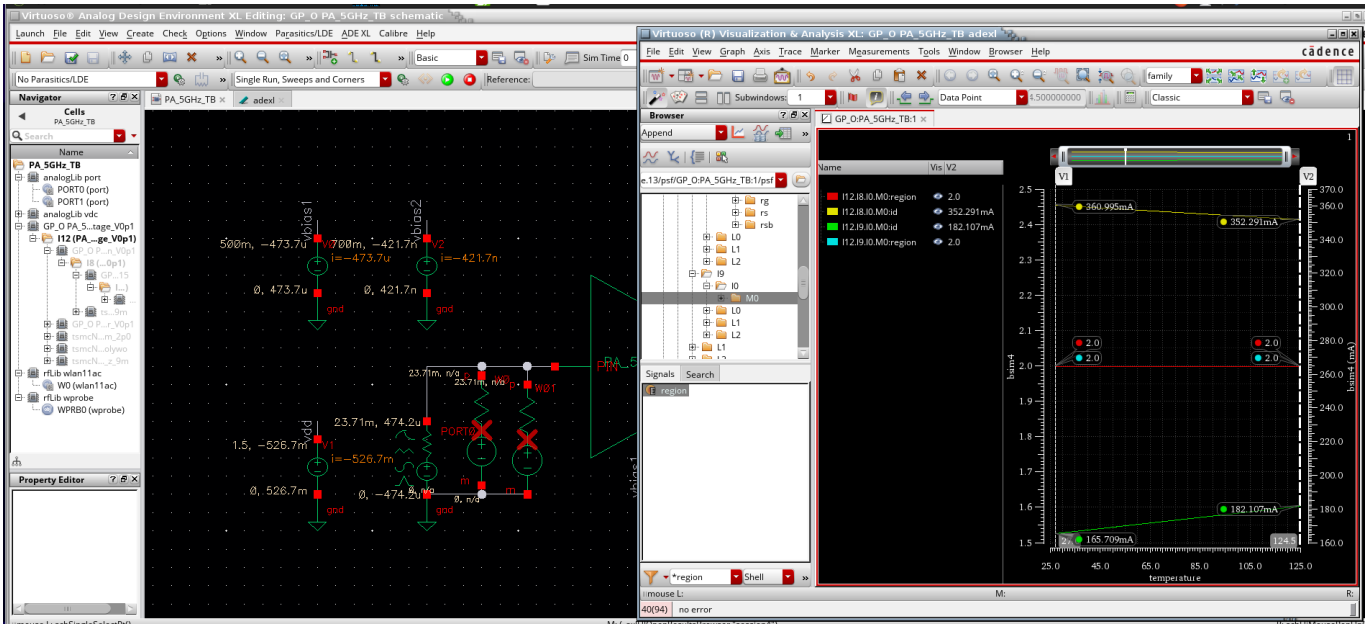


Figure 4.32: $5GHz$ PA DC Results

The total DC consumption is $P_{dc} \approx 0.526 * 1.5 = 0.789W$, Note that the DC Bias points were tuned alongside the overall performance to decrease current consumption compared to before.

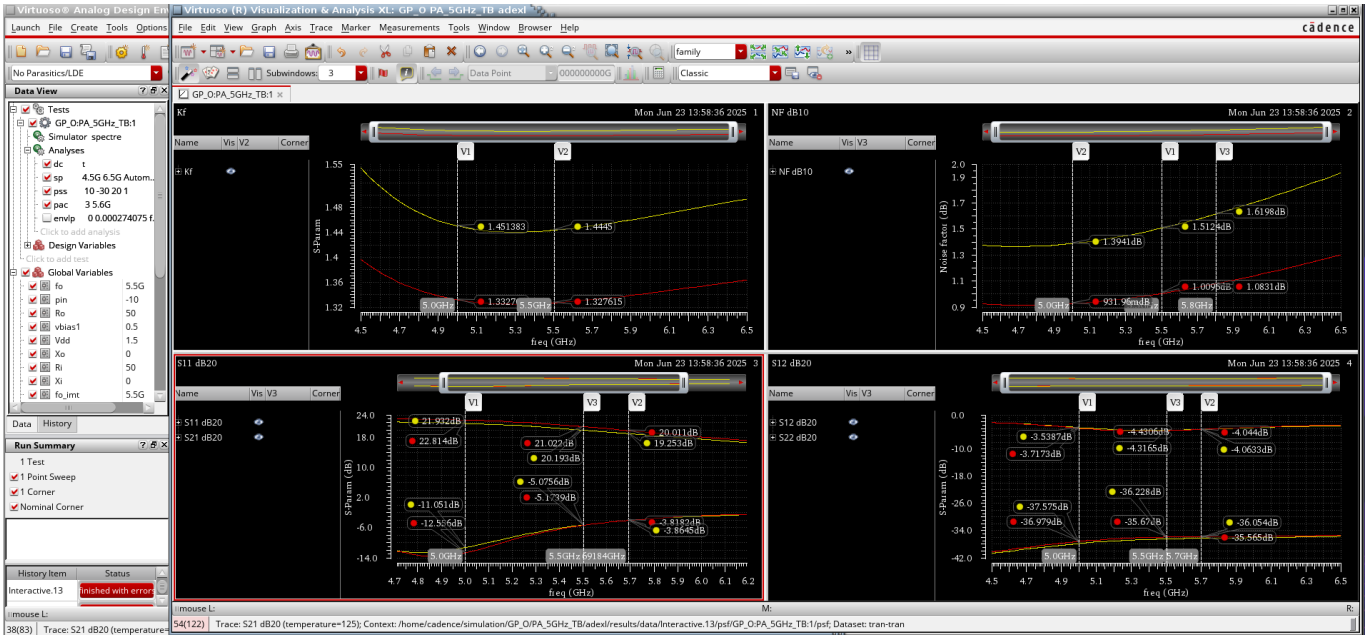


Figure 4.33: $5GHz$ PA S-Parameters

The S-parameters seem to have taken a hit after porting the desgin cadence. This stems from the difference between the capacitor and inductor models in both, particularly after connecting the a bulk-terminal (which was not presence in ADS models) to one of the rails (the ground in the above source). This can be tuned later.

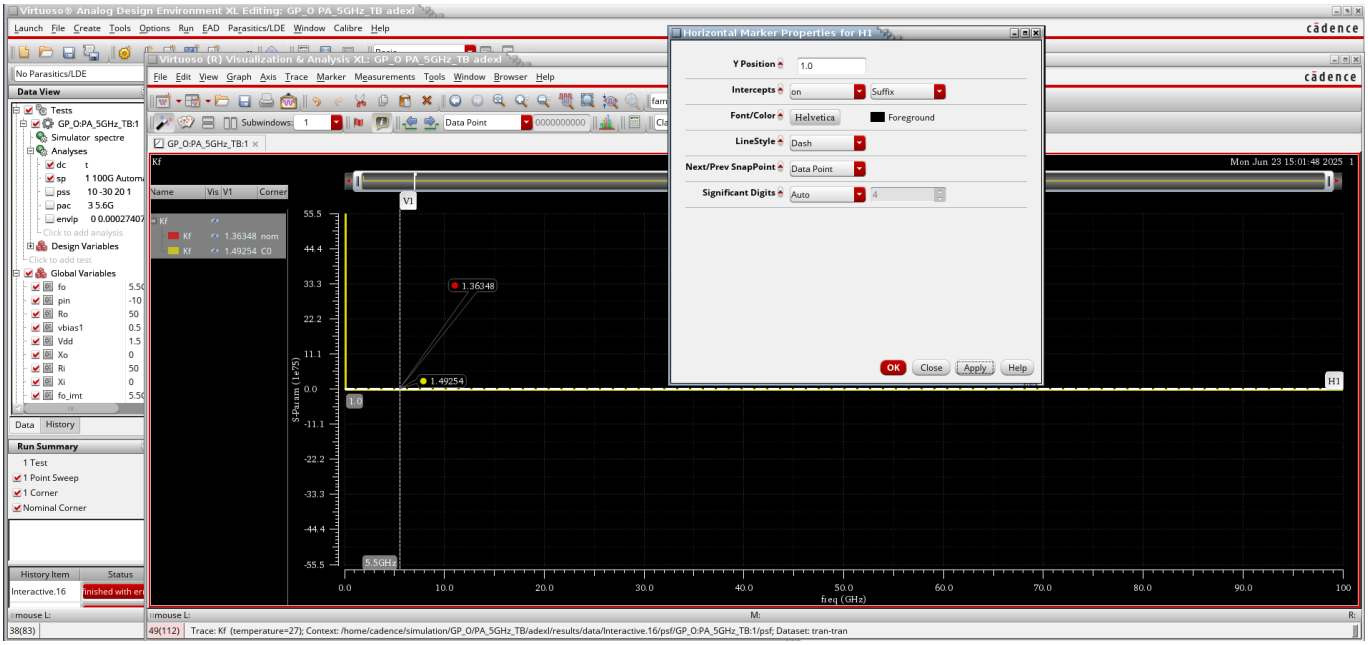


Figure 4.34: $5GHz$ PA Stability

The PA is unconditionally stable from $1Hz$ to $100GHz$. Note that the horizontal marker doesn't intersect the curve at $y = 1$.



Figure 4.35: $5GHz$ PA PAE, P1dB & IP3 Curves

The PA exhibits the shown values above. Also note that the higher order harmonics have very low levels.

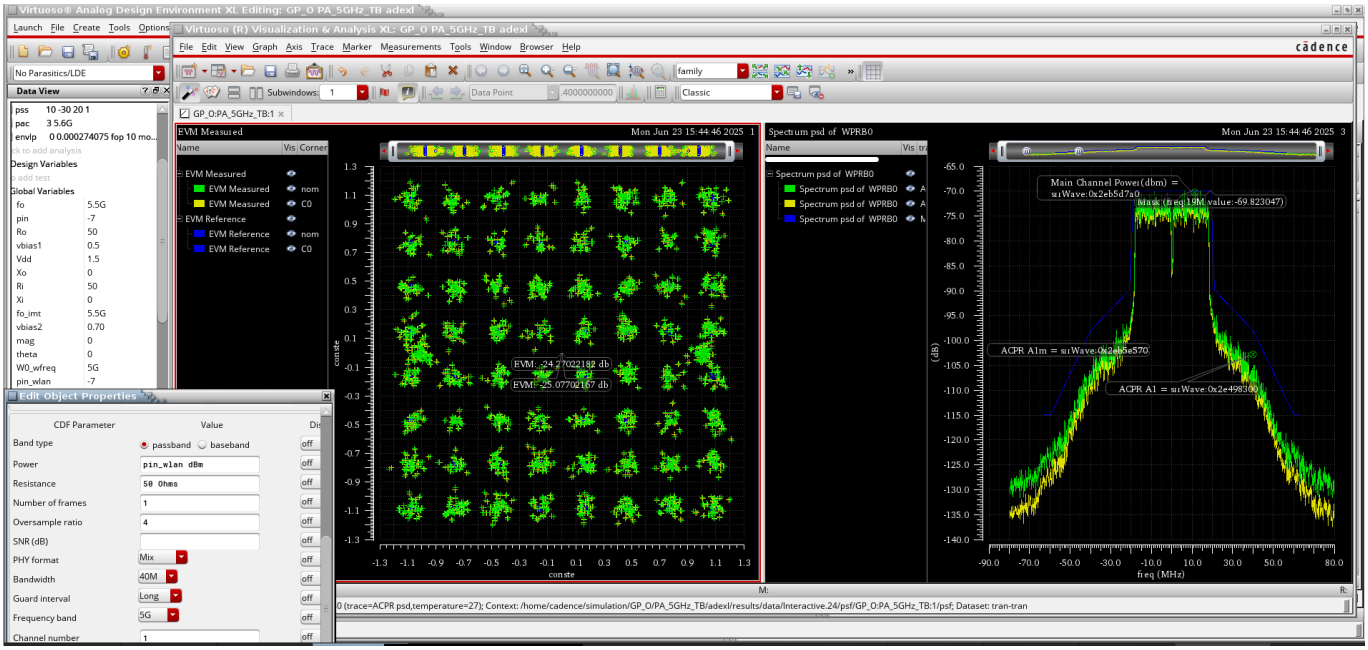


Figure 4.36: $5GHz$ PA EVM & ACPR (802.11n)

The PA has achieved $EVM \leq -22dB$ for a $P_{in} = -7dBm \implies P_{out} \approx 13.9dBm$ (for the nominal corner) while complying with the spectral mask results for an 802.11n signal with $40MHz$ channel width and a short guard band.

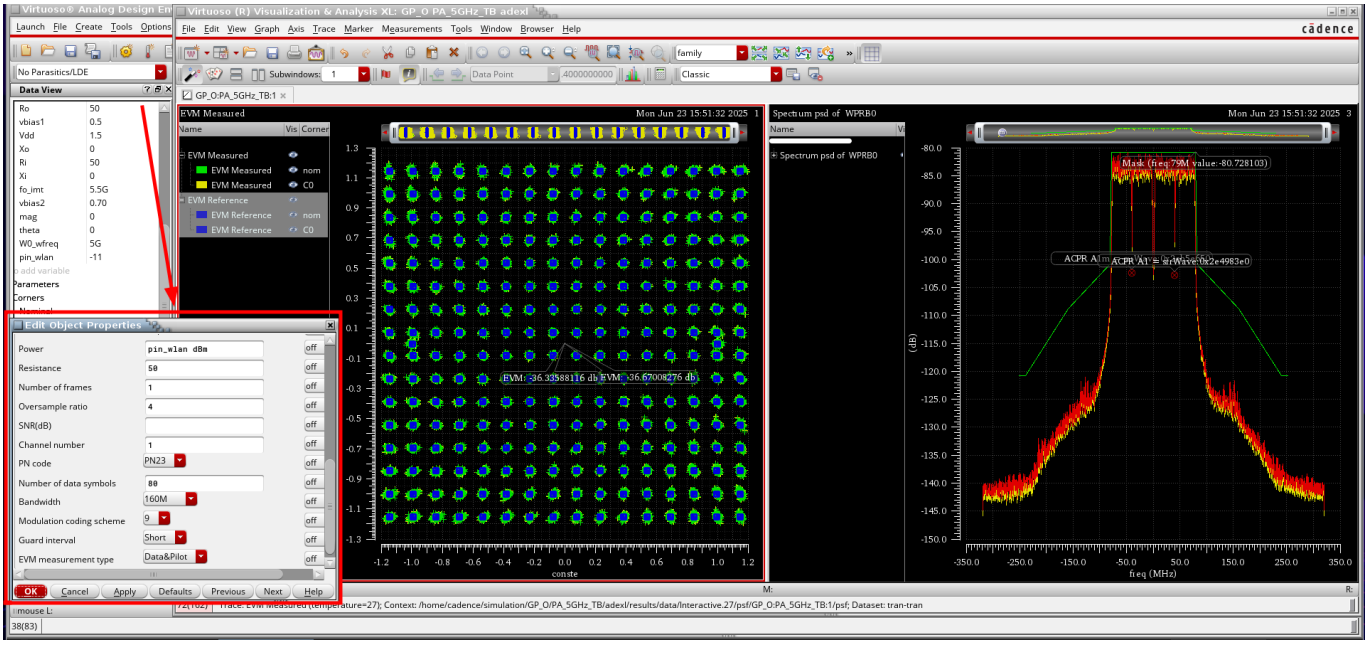


Figure 4.37: $5GHz$ PA EVM & ACPR (802.11ac)

The PA has achieved $EVM \leq -35dB$ for a $P_{in} = -11dBm \implies P_{out} \approx 10.0dBm$ (for the nominal corner) while complying with the spectral mask results for an 802.11ac signal with $160MHz$ channel width and a short guard band (MCS9, 256-QAM).

The PA was further modified to account for the impact of these bulk connections. A summary of the performance and the S parameters are as shown in the next figures with much better performance.

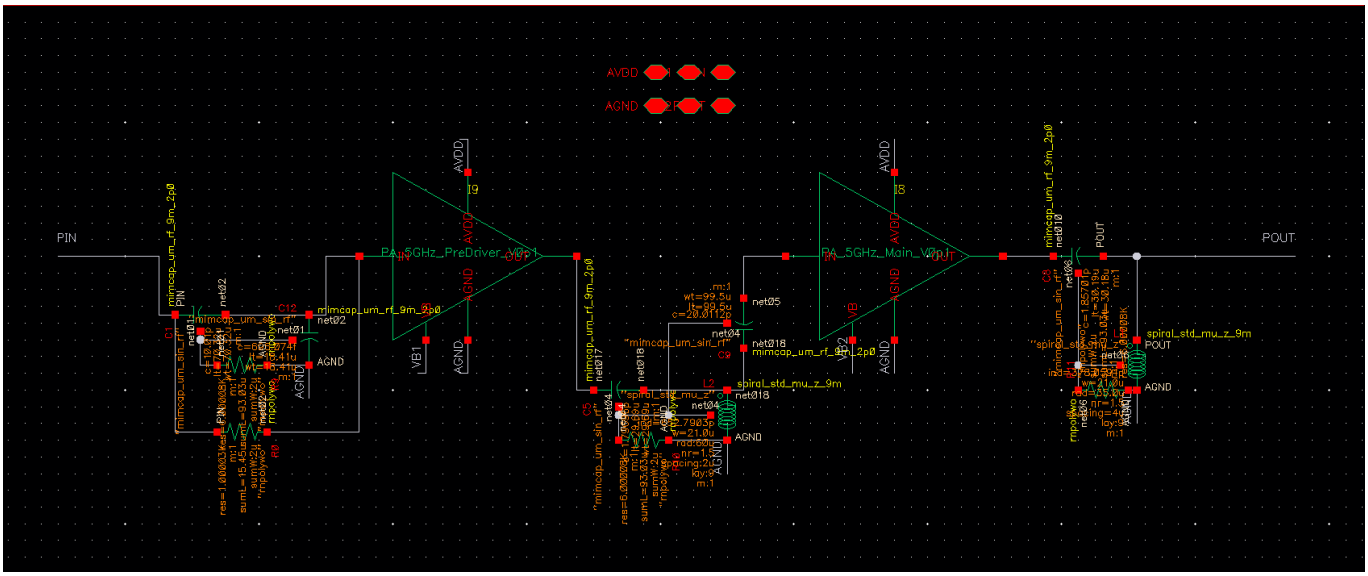


Figure 4.38: The $5GHz$ PA (modified).

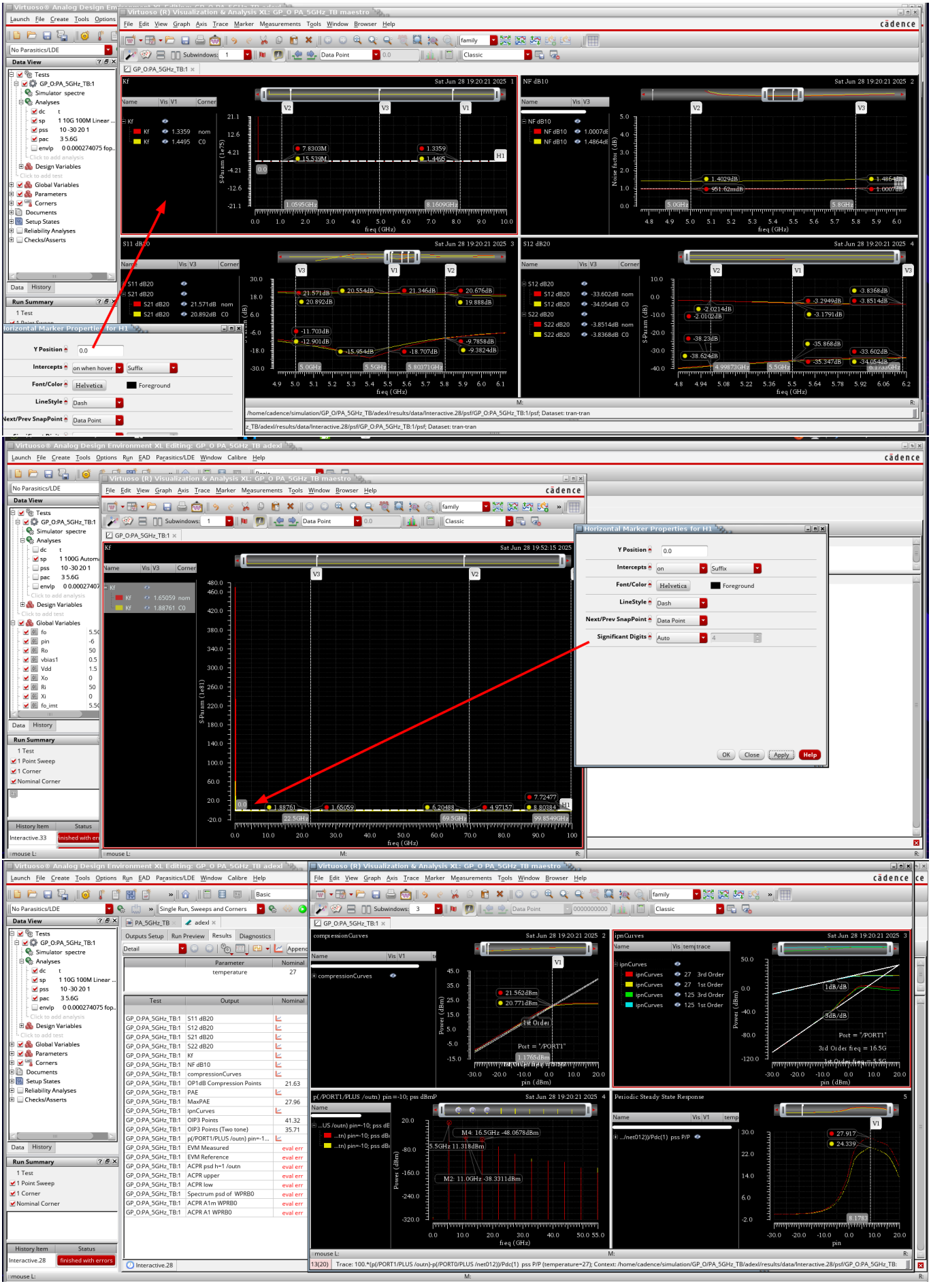


Figure 4.39: $5GHz$ PA SP & HB (modified).

The s-parameters are improved substantially over last time. The pas was also, once again, unconditionally stable from $1Hz \rightarrow 100GHz$ (albeit only shown above is up to $10GHz$. The HB analyses also seem satisfactory $OP1dB \geq 20dBm$ & $PAE \geq 24\%$ for both corners.

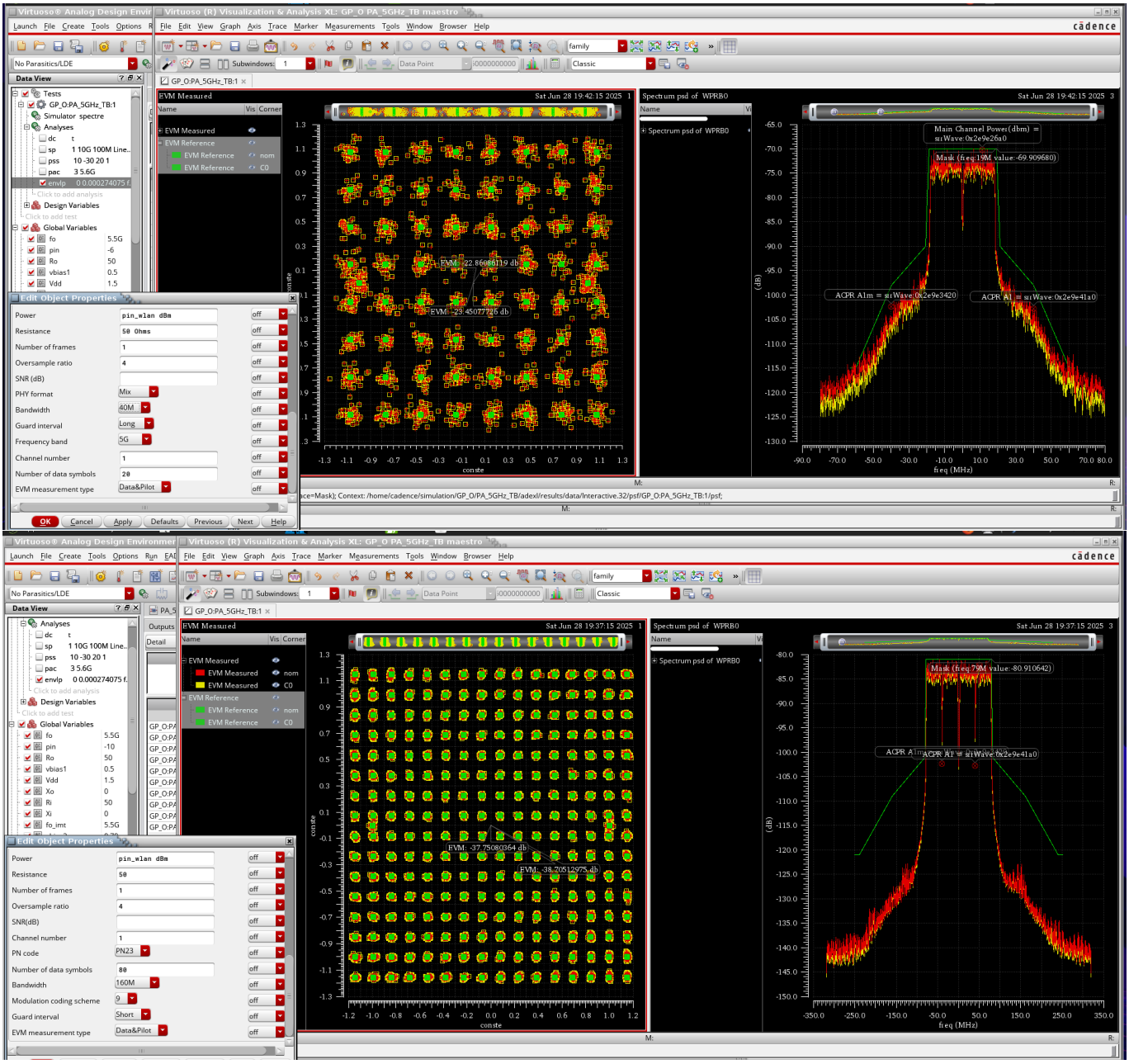


Figure 4.40: $5GHz$ PA envelope analyses (802.11n & 802.11ac) (modified).

This time, the PA has achieved $EVM \leq -23dB, -37dB$ for a $P_{in} = -6dBm, -10dBm \implies P_{out} \approx 15.3dBm, 11.3dBm$ for 802.11n & 802.11ac respectively (for the nominal corner) while complying with the spectral mask results.

4.6.3 Results & Comparisons

A summary of the results is now shown in the following table. It should be mentioned that up to this point, only current simulations are compared. Obviously, these results would worsen after layout, but this should provide a good base for tuning. The two PAs comply with the spectral masks of the 802.11n and 802.11ac (for the $5GHz$ PA) standards. DC Power is around $0.75W$ & $0.79W$ with unconditional stability from $1Hz$ to $100GHz$. The rest of the results are satisfactory as well and shown above.

Metric	**Design Target**	$2.4GHz$ **PA**	$5GHz$ **PA**
Frequency band	$2.4GHz$ *or* $5GHz$	$2.4GHz \rightarrow 2.5GHz$	$5GHz \rightarrow 5.8GHz$
Linear P_{OUT}	10 to 15dBm	$15.4dBm$ (802.11n)	$15.3dBm$ (802.11n) & $11.3dBm$ (802.11ac, MCS9)
P_{1dB}	Typically $\approx 5dB \rightarrow 7dB$ higher than the Linear P_{Out}	$\approx 18.9 \rightarrow 19.9dBm$	$\approx 20.8 \rightarrow 21.6dBm$
EVM	$\leq -22dB$ for 64-QAM, $\leq -35dB$ for 256-QAM, $\leq -43dB$ for 1024-QAM.	$\leq -22dB$ (802.11n)	$\leq -22dB. - 37dB$ (802.11n, 802.11ac MCS9)
PAE_{MAX}	**Maximize** ($\geq 20\%$)	$\geq 20\%$	$\geq 24\%$
Small-signal gain	$\geq 20dB$	$\geq\approx 27.3dB$ @ $2.4GHz$	$\geq\approx 20.9dB$ @ $5GHz$

Table 4.1: Achieved Specs (Schematic with PDK elements.)

A comparison with other similar works is now warranted. Note that the nominal corner is used.

Metric	**Tech**	**Freq (GHz)**	**Pout (dBm)**	**Max PAE**	**Psat (dBm)**	**P1dB (dBm)**	**EVM (dB)**	**Gain (dB)**	**Area (mm^2)**	**Vdd (V)**
[26]	65nm	2.45	9.4 & 17.4	-	-	17	-28.4 @ 9.4dBm	-	1×1	3.3
[27]	65nm	2.4	-	25%	31.5	-	-25 @ 25.5dBm	12	1.3×0.9	3.3
[31]	65nm	2.24	-	26.8%	26.1	25.4	-	21.8	1	2.2
This Work	65nm	2.4 - 2.5	15.4	$\geq 26\%$	21.5	19.9	-22 @ $15.4dBm$	≥ 27.9	-	1.5

Table 4.2: Comparison with other CMOS PAs (2.4 GHz)

Metric	**Tech**	**Freq (GHz)**	**Pout (dBm)**	**Max PAE**	**Psat (dBm)**	**P1dB (dBm)**	**EVM (dB)**	**Gain (dB)**	**Area (mm^2)**	**Vdd (V)**
[32]	55nm	5	17.6	-	-	-	-35	20.6	0.76×1.26	5
[33]	65nm	5	21.0	51%	-	-	-	26	-	-
[35]	180nm	5 - 5.8	22.6	19.8%	23.1	18.6	-	-	-	-
This Work	65nm	5 - 5.8	11.3 & 15.3	$\geq 28\%$	23.3	21.6	-37.8 @ 11.3dBm	≥ 20.7	-	1.5

Table 4.3: Comparison with other CMOS PAs (5 GHz)

5 Power Amplifier Layout

5.1 Device Layout, Verification & Parasitic Extraction

The device is shown, with its respective layout. These first iterations could be tuned for future work.

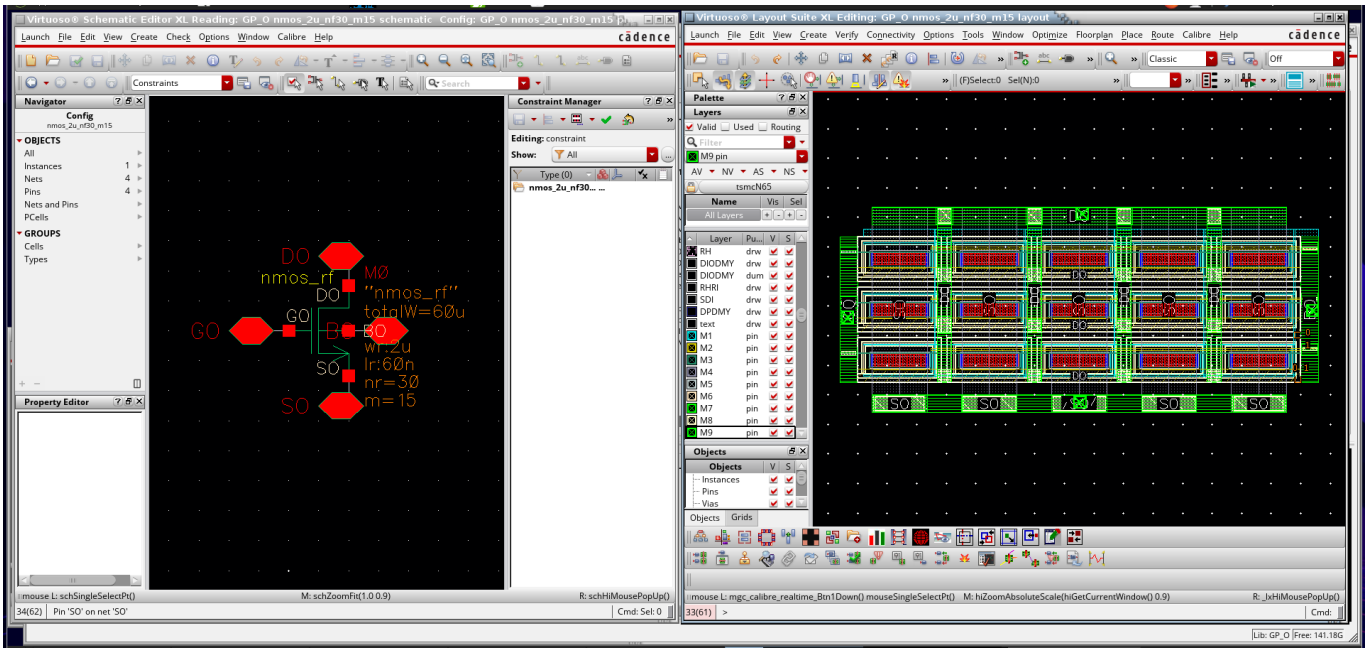


Figure 5.1: Transistor schematic & layout

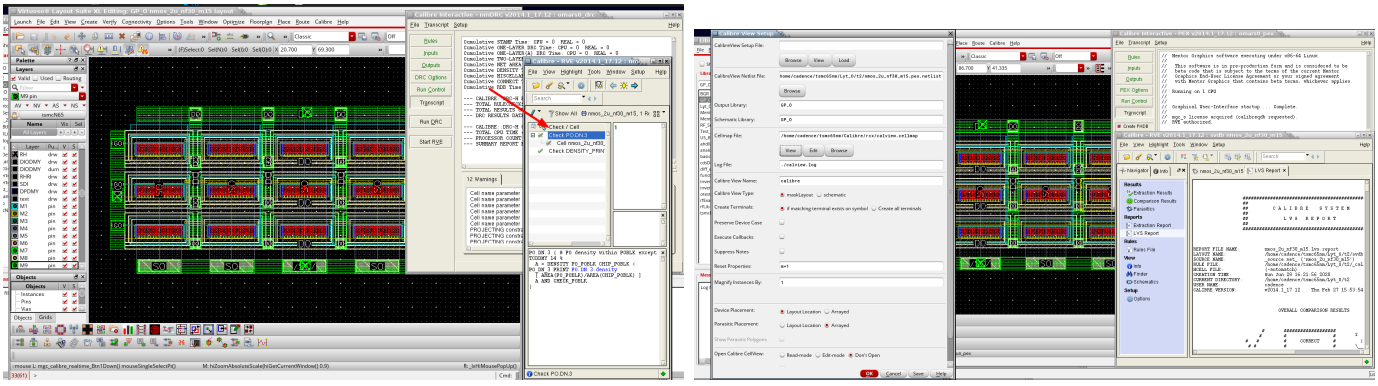


Figure 5.2: Transistor DRC, LVS, & PEX

Only density DRC errors are present. The comparison for LVS is correct and so is the parasitic extraction (the PEX analysis above is dependent on a correct LVS).

5.2 PA Layout, Verification & Parasitic Extraction

Then, the Power Amplifier layout was done for the following two PAs.

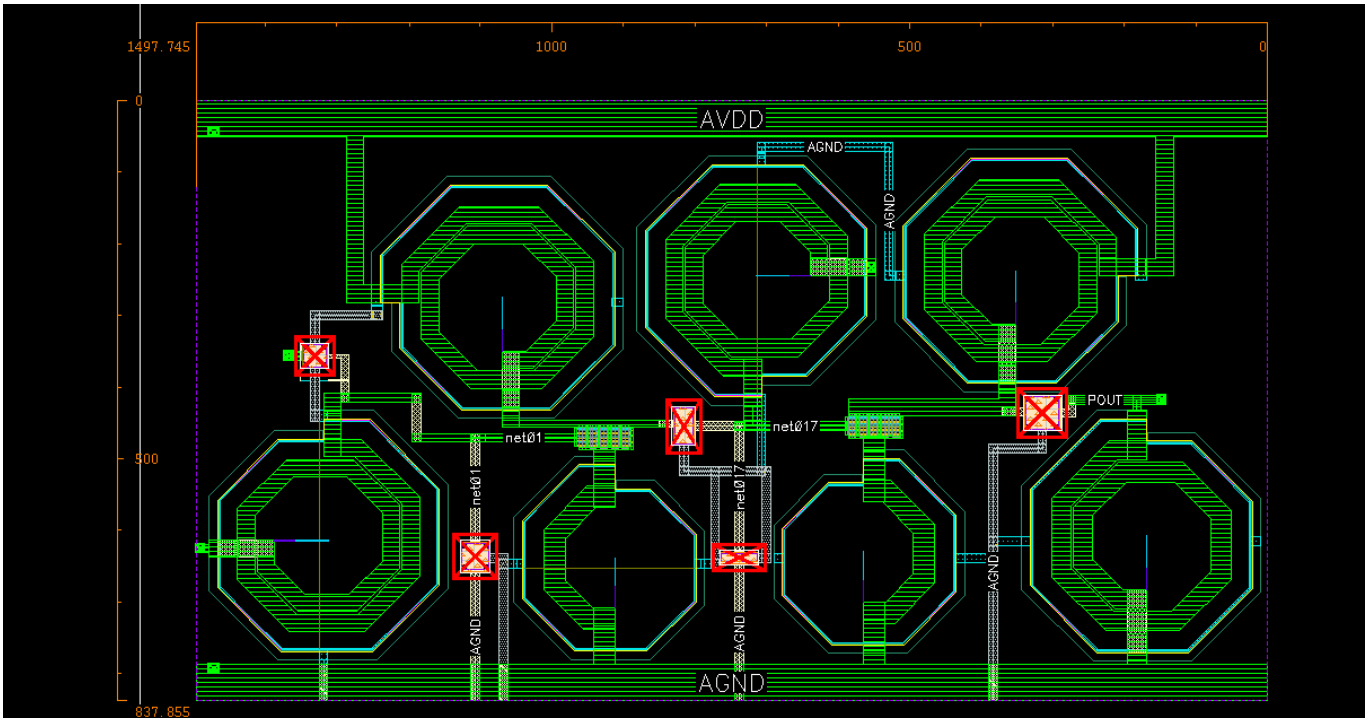


Figure 5.3: $2.4GHz$ PA Layout

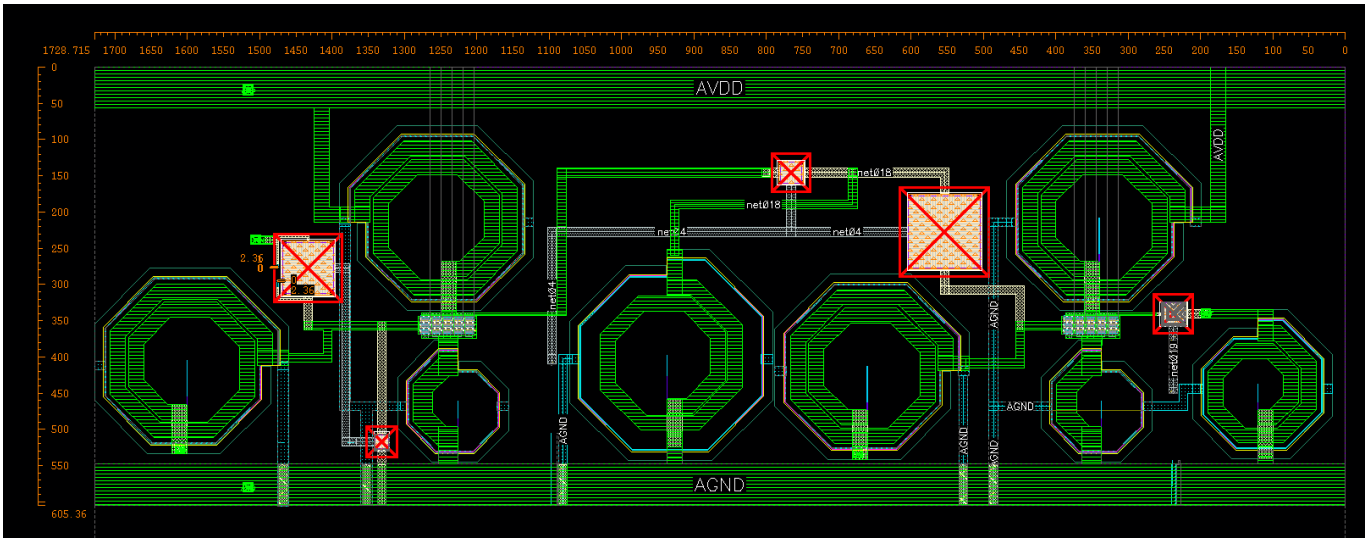


Figure 5.4: $5GHz$ PA Layout

There were grid and density errors that were waived/fixed. The grid errors come from the PDK elements or other software-related technicalities. However, since this step is not crucial and fabrication is out of scope, the focus was resolving minium width, spacing, enclosure, & extension errors. The LVS comparison and the PEX was successful for both PAs. The two PAs occupy areas around $1.498mm \cdot 0.838mm \approx 1.255mm^2$ & $1.729mm \cdot 0.605mm \approx 1.046mm^2$ respectively.

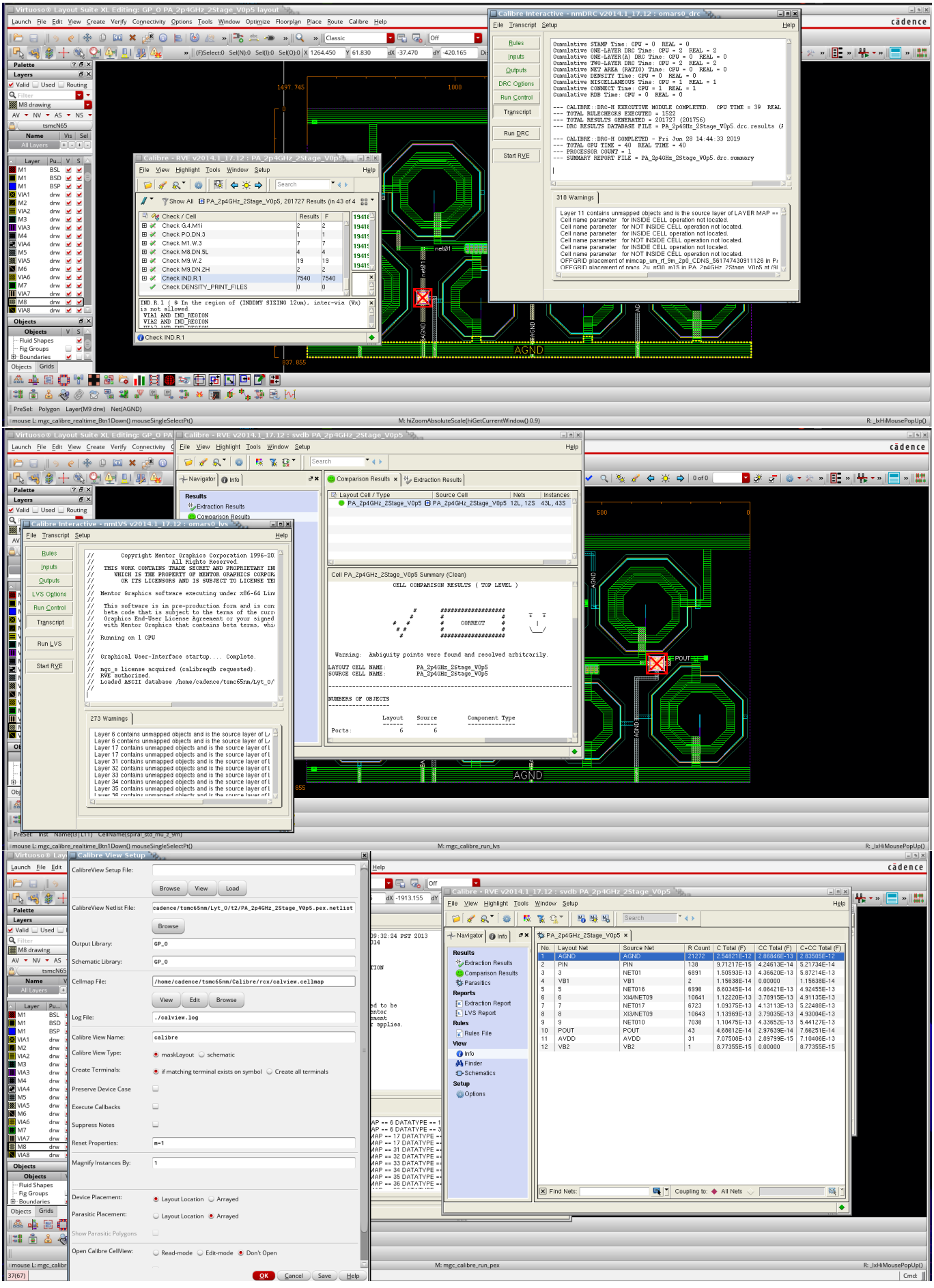


Figure 5.5: $2.4GHz$ PA Layout DRC, LVS, & PEX

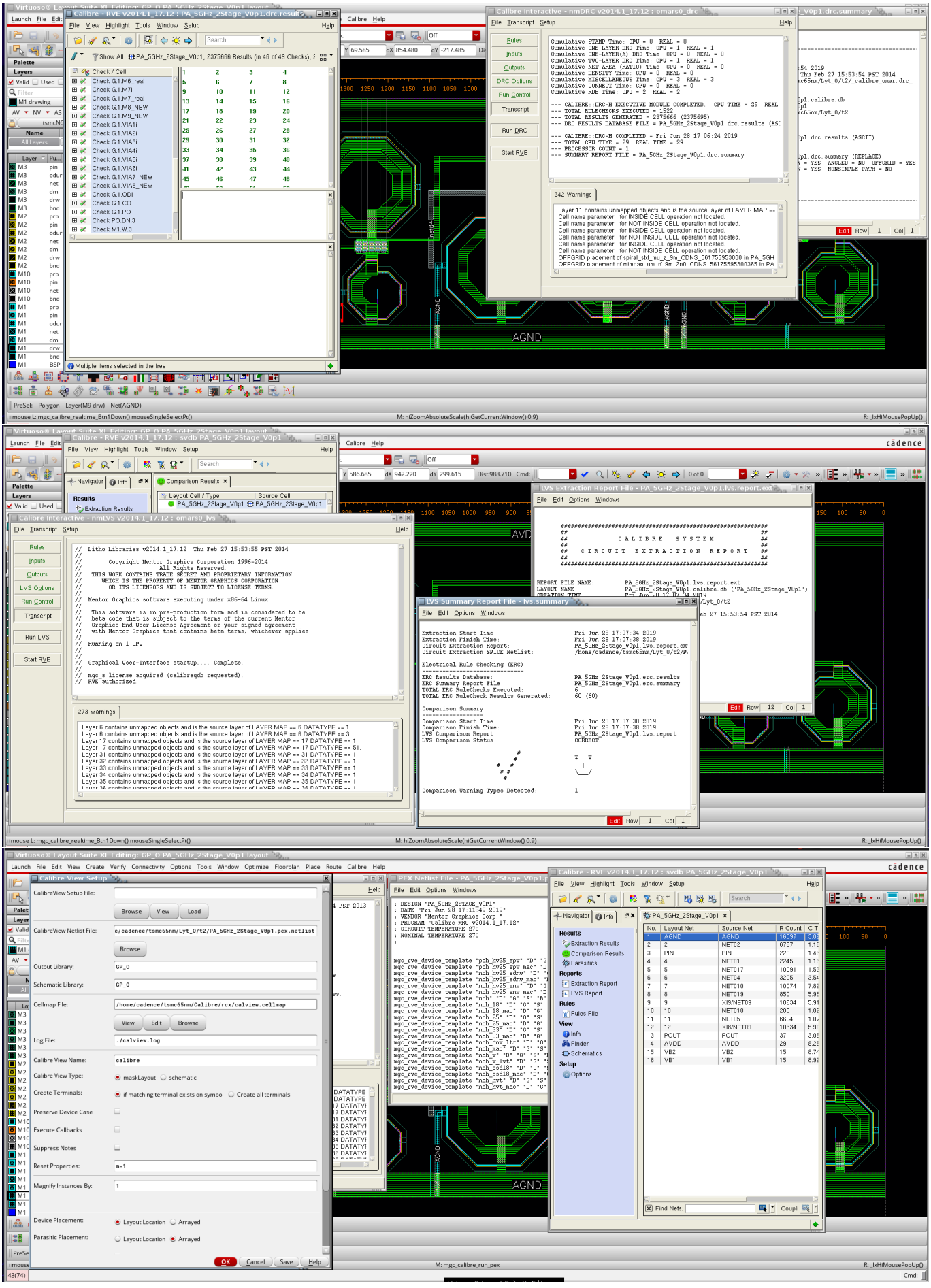


Figure 5.6: $5GHz$ PA Layout DRC, LVS, & PEX

5.3 Inductor EM Co-Simulation

All of the inductors used were highly characterized by the PDK (as they were taken from the finder). Nonetheless, the DC feed inductor to the gate was EM-simulated with the guard rings removed, which would help conserve space later on if added. (This step is done for demonstration purposes. Technical difficulties because of the software as well as other challenges made simulating everything difficult.)

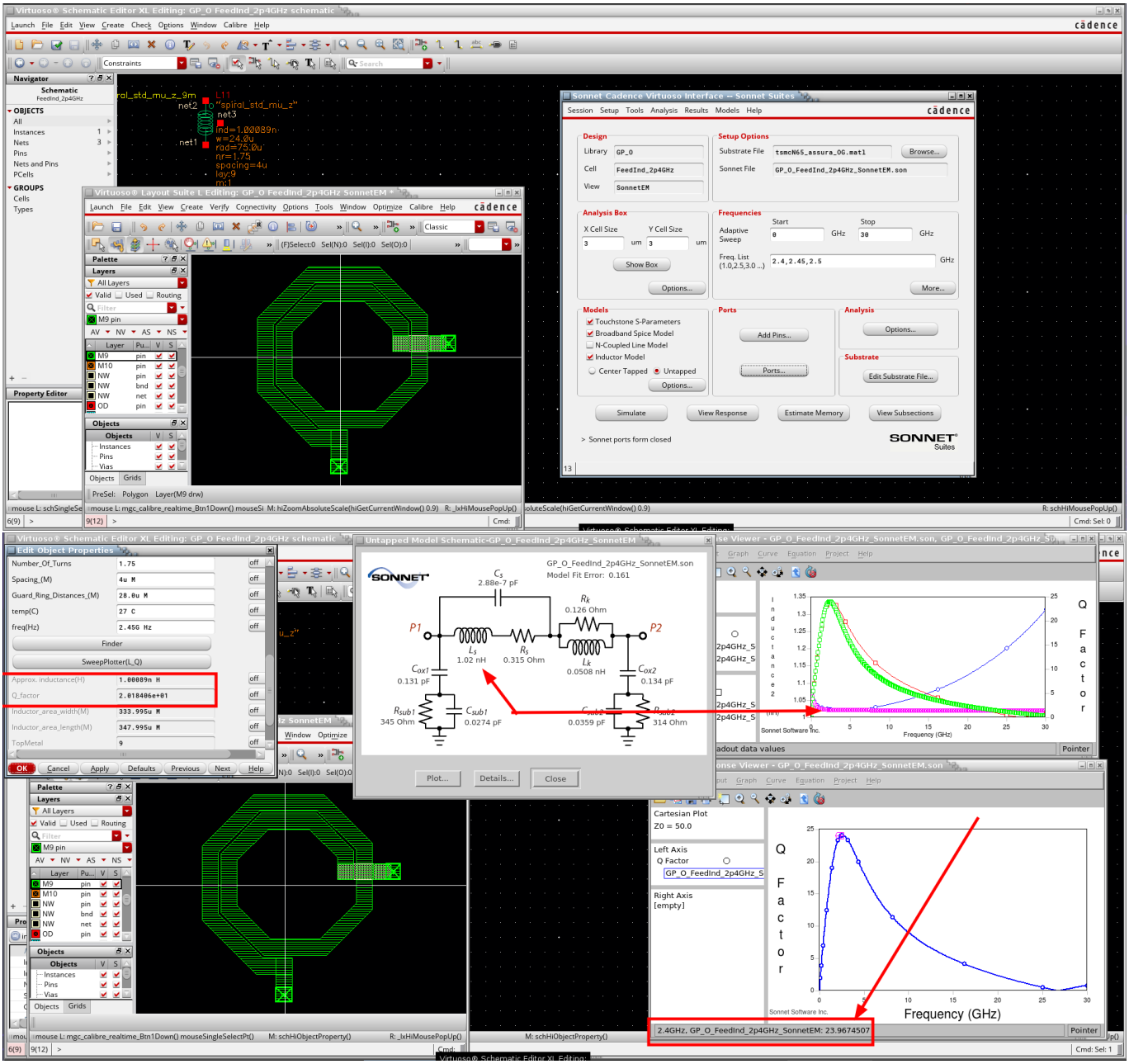


Figure 5.7: $1nH$ Feed Inductor EM Simulation & Results (For the $2.4GHz$ PA)

The $2.4GHz$ inductor results show that the Q factor of the inductor and the inductance are close or nearly identical to the those of the PDK. The $5GHz$ inductor is less accurate, but the deviation is not excessive. More accuracy is to be expected if a smaller cell size and finer sweeps were to be made.

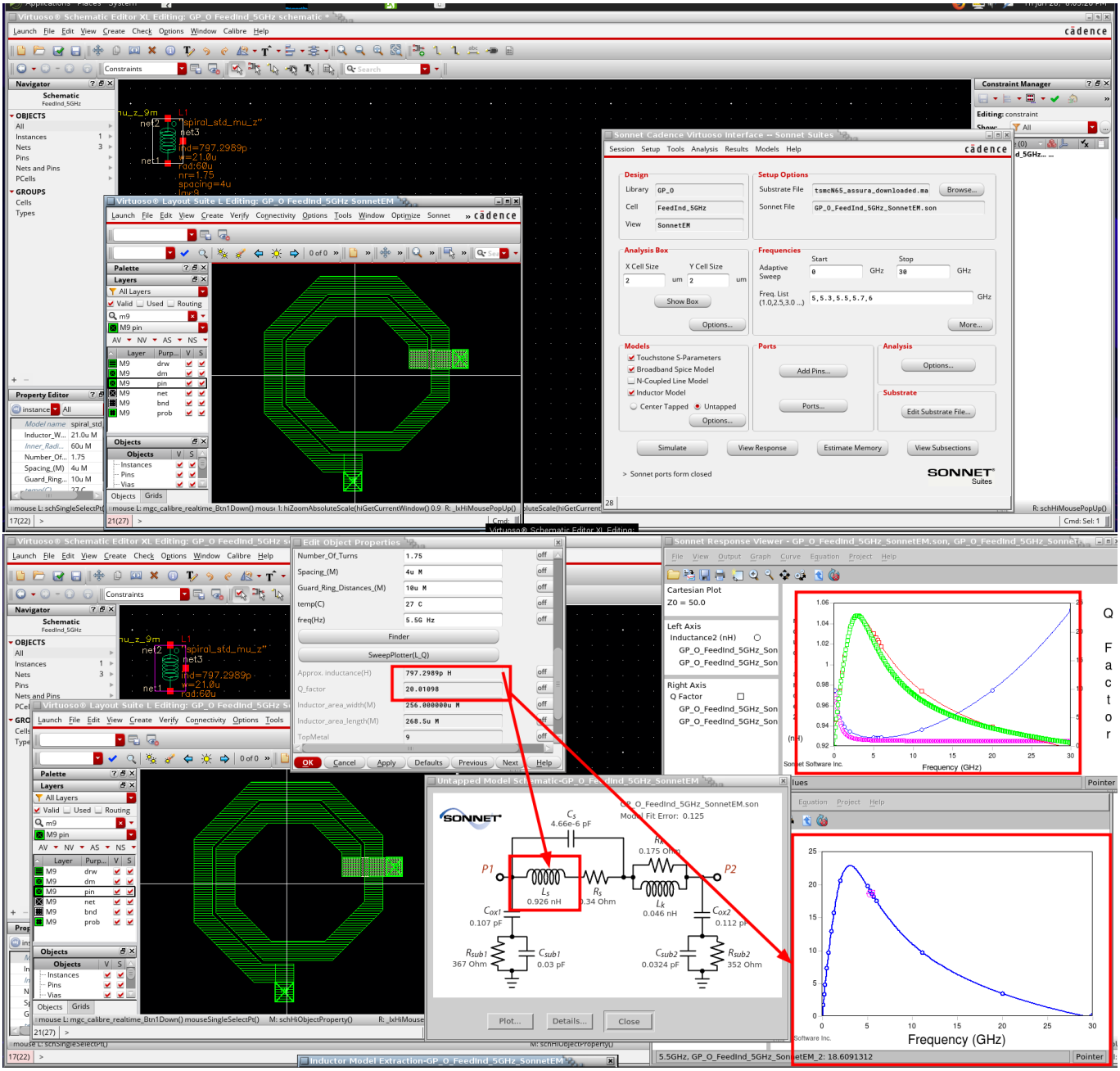


Figure 5.8: $0.8nH$ Feed Inductor EM Simulation & Results (For the $5GHz$ PA)

It should be mentioned, however, that both the inductor, the routing tracks, and the entire board should be EM simulated as whole. This step was shown in order to be consistent with the flow and to showcase what to do with elements that are not well characterized by the PDK (e.g. inductors operating at $f \geq 20GHz$, custom transmission lines, etc).

5.4 Post-Layout Simulation

Finally, the post-layout results were in for the first iterations of the layout for both PAs. The $2.4GHz$ showcases somewhat satisfactory results even after taking a major hit to the gain. Nonetheless, it's still above $20dB$.

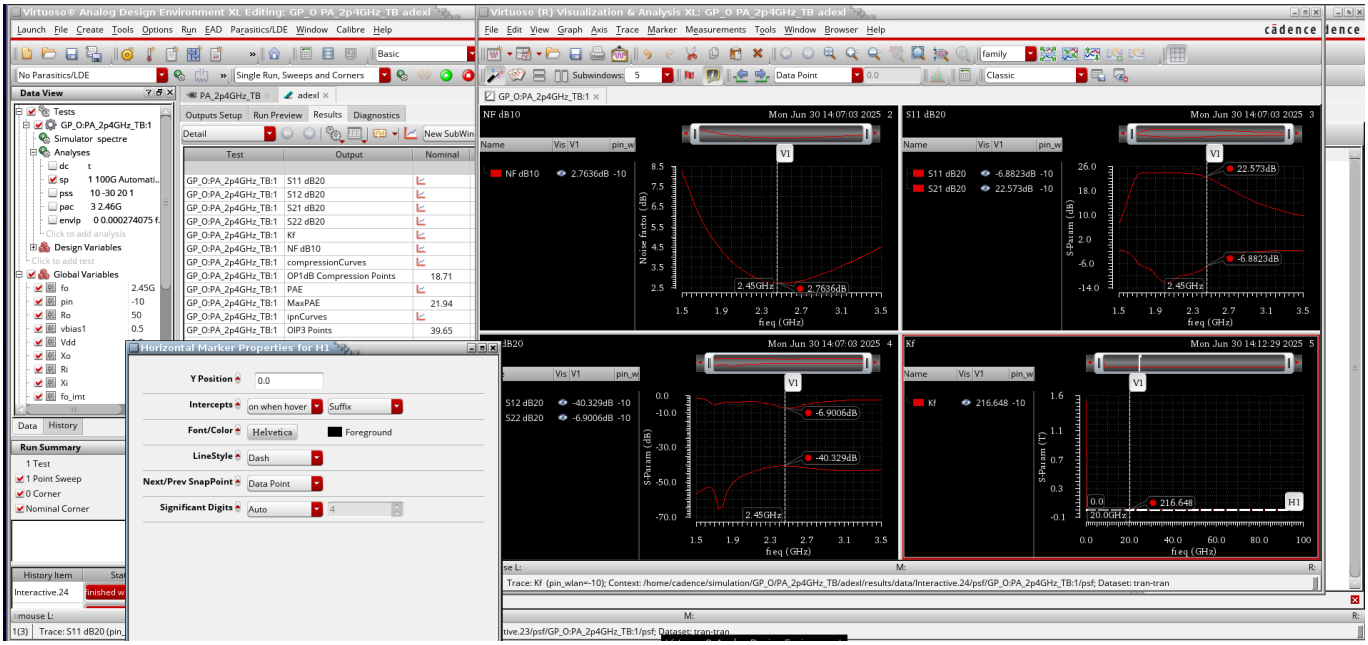


Figure 5.9: $2.4GHz$ PA Post-Layout Summary

For the $5GHz$ PA, it's obvious the tuning and optimizations are imperative to optimize this major hit the S-parameters of this PA took.

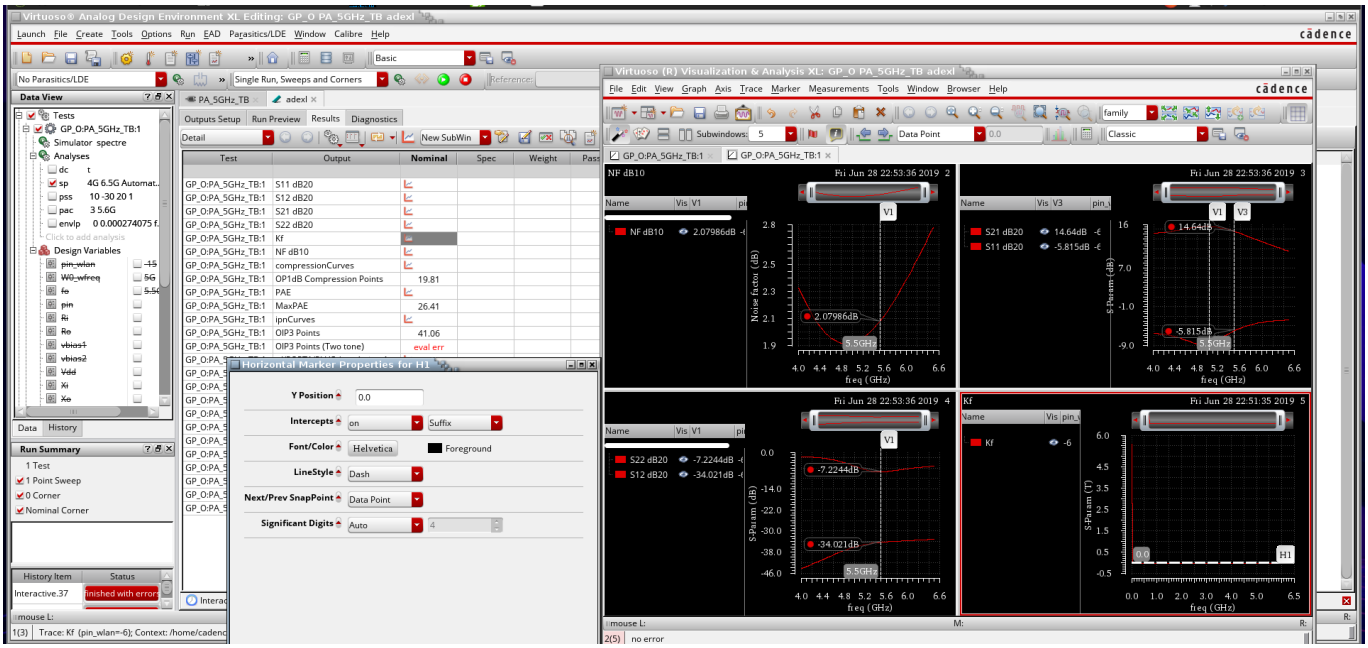


Figure 5.10: $5GHz$ PA Post-Layout Summary

Another post-layout simulation run has shown that the main culprit behind this hit is the device itself, where a simulation was done using only the extracted device without extracting the entire PA with its passives. The device by itself has caused the most performance degradation post-layout. The transistor, due to the very high number of multipliers present, likely introduced many new parasitics. Retuning the matching network will likely offset some of this loss. Another approach is to create a new

device with much less multipliers, such that the device post-layout does not contribute the same amount of parasitics.

The following table summarize post-layout results. Note that in the above figure, only the nominal corner was simulated because of technical issues. The envelope analysis also had convergence problems, likely due to the very large size of the extracted circuit. Nonetheless, an idea could be formed of the possible EVM at what P_{out} by inspecting the compression points.

Metric	Tech	Freq (GHz)	Pout (dBm)	Max PAE	Psat (dBm)	P1dB (dBm)	EVM (dB)	Gain (dB)	Area (mm^2)	Vdd (V)
[26]	65nm	2.45	9.4 & 17.4	-	-	17	-28.4 @ 9.4dBm	-	1 × 1	3.3
[27]	65nm	2.4	-	25%	31.5	-	-25 @ 25.5dBm	12	1.3 × 0.9	3.3
[31]	65nm	2.24	-	26.8%	26.1	25.4	-	21.8	1	2.2
This Work	65nm	2.4 - 2.5	-	21.9%	20.1	18.7	-	$\geq$ 22	1.26	1.5

Table 5.1: Post-Layout Comparison with other CMOS PAs (2.4 GHz)

Metric	Tech	Freq (GHz)	Pout (dBm)	Max PAE	Psat (dBm)	P1dB (dBm)	EVM (dB)	Gain (dB)	Area (mm^2)	Vdd (V)
[32]	55nm	5	17.6	-	-	-	-35	20.6	0.76 × 1.26	5
[33]	65nm	5	21.0	51%	-	-	-	26	-	-
[35]	180nm	5 - 5.8	22.6	19.8%	23.1	18.6	-	-	-	-
This Work	65nm	5 - 5.8	-	26.4%	21.8	19.8	-	$\geq$ 13.4	1.05	1.5

Table 5.2: Post-Layout Comparison with other CMOS PAs (5 GHz)

6 Conclusion & Future Work

We can conclude from the above discussions that a power amplifier for both the $2.4GHz$ & $5GHz$ bands was designed and successfully simulated, creating a blueprint for a future PA for use in 802.11n & 802.11ac applications with room for improvement and optimization. The $2.4GHz$ & $5GHz$ PAs both show output $P1dB \geq 18.9dBm, 20.8dBm$, max PAE of $\geq 20\%, 24\%$, a gain of $26.7dB, 19.9dB$, unconditional stability from $1Hz$ to $100GHz$, while complying with the spectral mask and achieving low enough EVMs of $\leq -22dB$ for 802.11n, and $\leq -37dB$ for 802.11ac for the 5GHz PA. All at the worst-case corner. The post-layout simulation shows a minimal degradation in the $P1dB$ & PAE, but a substantial gain degradation over the band by around $8dB$. Nonetheless, due to the performance outlined above, the proposed two power amplifiers could be viable options for operation at the mentioned frequencies for WiFi applications with IoT & mobile devices after tuning.

Future work & suggestions

There are several recommendations for future work and suggestions:

- Use power-boosting topologies (e.g., stacked FETs) to decrease current while preserving output power or even increasing it. This also has several positive implications on the layout (lower track widths needed, lower voltage headroom wasted on parasitics, etc).
- Optimize the layout and use techniques such as common centroid to get the best performance and make up for fabrication imperfections. The layout could be extensively tuned from this point.
- Simulate more corners while taking more parameters into consideration (Voltage & Process variation, etc). Monte Carlo analyses are also crucial if such a chip is to be fabricated.
- Integrate with other blocks to see the overall performance of the transmitter & transceiver.
- Limit the maximum V_{ds} on each transistor to around $1.5V$ to improve reliability and extend the lifespan of the device.
- Conduct EM Co-Simulation for the entire PA.
- Use MATLAB and other tools to create blocks such as digital predistorters after characterizing the PA.
- Investigate higher frequency designs and utilize microstrip lines as well as other wideband matching techniques.

7. Appendices

Appendix A: Python Code (Sorin Class A Methodology)

This code is best used as a Jupyter notebook. It's a skeleton that allows for many modification. It also includes simple functions to calculate reactances at certain frequencies (which could be used to get a quick estimate of the parasitic resistance according to the Q "on the go").

```
# Created by Omar. M. A. Elsayed
# These values are meant to give an initial point for the sizing of an NMOS
# for a Class A amplifiers. This methodology is in line with Sorin's HF mmWave
# Class PA methodology. The values can be edited, however, and the insights
# taken from this initial point could be used to also design class AB amps.

# Best used as a Jupyter Notebook!

# ===== # initialization # ===== #

from math import *
# import sympy as sy, numpy as np, pandas as pd # Commented out. Not needed for now.

# ===== # suffix # ===== #
m=1e-3
u=1e-6
n=1e-9
p=1e-12
f=1e-15

K=1e3
M=1e6
G=1e9

# ===== # Parameters # ===== #
# Useful for calculating reactances if needed
fo=60*G
wo=60*pi*fo

# Functions
def dbm_to_linear_mW(dbm_value): #dBm to mW
    res=10**(dbm_value/10)
    return res

def linear_si_to_dbm(lin_value): #W to dBm
    res=10*log10(lin_value/m)
    return res

def cap_reactance(capv, fo):
    wo=2*pi*fo
    res=1/(wo*capv)
    print(res)
    return res

def ind_reactance(indv, fo):
```

```
    wo=2*pi*fo
    res=(wo*indv)
    print(res)
    return res

# ===== # Requirements # ===== #
# ================================> Set these!
# You can change the default values below as needed

req_p1db_dbm = 21 # In dBm
req_p1db = dbm_to_linear_mW(req_p1db_dbm)
req_p1db_si = req_p1db*m

print("Required P1dB: ", req_p1db_dbm, "dBm")
print("Required P1dB: ", req_p1db, "mW")
print("Required P1dB: ", req_p1db_si, "W")

print("\n")

# ===== # Transistor Sizing & Biasing # ===== #

Vdd=1.2 # Not used for now
Vk=0.25
Vmax=1.5

Vswing = Vmax-Vk # Voltage swing in V

Vd_bias = (Vmax+Vk)/2

print("Vswing: " ,  Vswing, "V")
print("Vd_bias: ", Vd_bias, "V")

print("\n")

# ===== # Width calculations from Current Density # ===== #

# SI: No suffix
# There's a distinction between the swing and the DC current. One can set the Idc to be simply equal to 0.5/Iswing

Jopt_App = 0.40*m/u # App/m Set as needed.
# Default value of 0.40mA/um is used in line with
# the optimum current density value of P1dB-efficiency tradeoff

Jopt_mApp_um = 0.40 #(mApp/um)
print("Jswing:", Jopt_App, "A/m")
print("Jswing:", Jopt_mApp_um, "mA/um")

Jopt_idc = 0.30*m/u # Set as needed
Jopt_idc_mA_um = 0.30 # Set as needed

print("Jdc: " ,Jopt_idc, "A/m")
print("Jdc: " ,Jopt_idc_mA_um,"mA/um")


Iswing= 4*(req_p1db_si)/(Vd_bias-Vk)
Iswing_m = Iswing/m

print("Iswing:" ,Iswing,"A")
print("Iswing:" ,Iswing_m,"mA/um")
```

```
print("\n")

# ===== # Sizing # ===== #

TW_u = Iswing_m/Jopt_mApp_um #Iswing = Jopt_mApp_um * W_u

Id_bias_m=TW_u*Jopt_idc_mA_um
Id_bias = Id_bias_m*m

Id_bias_sens_m= Iswing_m/2

print("Total Width:", TW_u, "um")
print("")
print("Idc:", Id_bias_m, "mA")
print("Idc (if taken as half Iswing):", Id_bias_sens_m, "mA")

print("\n")

Wnf_u = 2 # Set finger width as desired
Wnf = Wnf_u*u

nf = 30 # Set number of fingers as desired

print("Finger Width:", Wnf_u, "um")
print("Number of Fingers:", nf)

mul_calc=TW_u/(Wnf_u*nf)
mul_calc_corrected=ceil(TW_u/(Wnf_u*nf))

print("Multiplier:", mul_calc)
print("Multiplier (ceiled):", mul_calc_corrected)

Corrected_TW_u = mul_calc_corrected*nf*Wnf_u
Arbirary_TW_u = 15*30*2 #Up to the user to try different things

print("Total Width (corrected):", Corrected_TW_u, "um")
print("Other Arbitrary Width:", Arbirary_TW_u, "um (ignore if not set")

chosen_TW_u = Corrected_TW_u
Jswing = Jopt_mApp_um # Set the same as above. User is free to try different values

print("\n")


# ===== # Final Power # ===== #
P1dBr_mW = Jswing*chosen_TW_u*(Vswing)/8
P1dBr_si = P1dBr_mW*m
P1dBr_dbm = linear_si_to_dbm(P1dBr_si)
R_opt = Vswing/Iswing

print("P1dB:", P1dBr_mW, "dBm")
print("P1dB:", P1dBr_si, "W")
print("P1dB:", P1dBr_dbm, "mW")
print("Ropt:", R_opt, "Ohm")
```

Appendix B: Inductor Selection

The choice of the inductor could be facilitated by finding out the typical ranges of the inductors in the PDK and their Qs. The PDK Inductor finder characterizes inductors for frequencies up to $20GHz$. For 2.45 GHz and 5.5GHz, the highest inductors with a Q factor of ≥ 20 where found to be in the range of $1nH$ and $0.8nH$ respectively. These translate to reactance values of $j15.39\Omega$ & $j27.65\Omega$. While these values are low, for the drain inductor, the R_opt is almost an entire order of magnitude lower! At the same time, the parasitic resistance are $R = \frac{\omega L}{Q} = 0.77, 1.38\Omega$ respectively. The inductor

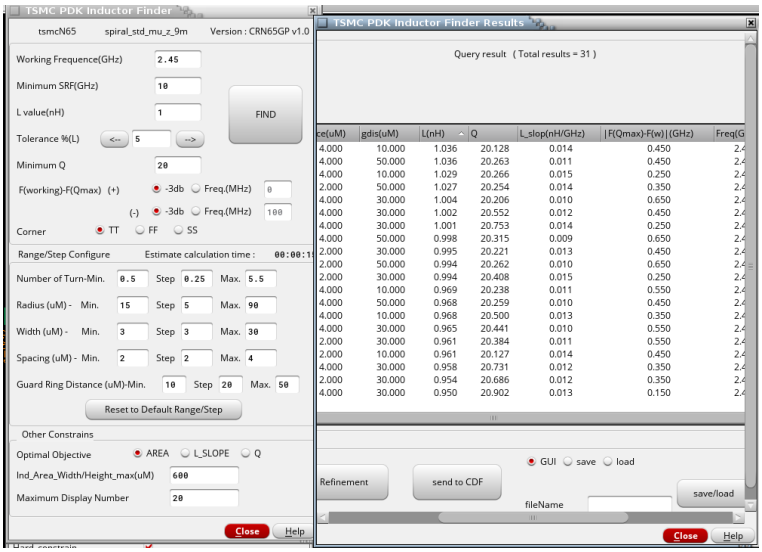


Figure 7.1: PDK Inductor Example

Other factors also play in the inductor selection in this project: its area, and the number of turns. Sometimes, it's desirable to have a number of turns with 0.5 increments to ease routing for the degeneration inductor and DC feed for example, as the rails are vertically aligned with them.

For higher designs—such as in mmWave bands—the inductor would have to be created by the designer and simulated with an EM solver. Alternatively, microstrip lines can be used instead when the frequency is high enough.

$$c = \lambda f \implies \lambda = \frac{c}{f} \tag{7.9}$$

So for 60GHz for example, $\lambda = 5mm$ only! And thus a quarter wave transformer would have a length of $1.25mm$ and small inductance and capacitance values could be realized. E.g. for an electrical length $\beta l = 9°$, a $0.125mm$ microstrip line is needed, a substantial size improvement over the big inductors of the 2.4 & 5GHz designs!

Appendix C: Loadpull & Sourcepull for a 60GHz PA Stage

A more systematic approach to finding the output impedance, however, is to conduct **loadpull analysis**. This approach not only finds the optimal load impedance in a less painful manner, but it also allows the designer to make the efficiency-linearity tradeoffs as desired. This is shown for, example, in the design of this stacked-FET 60GHz PA output stage:-

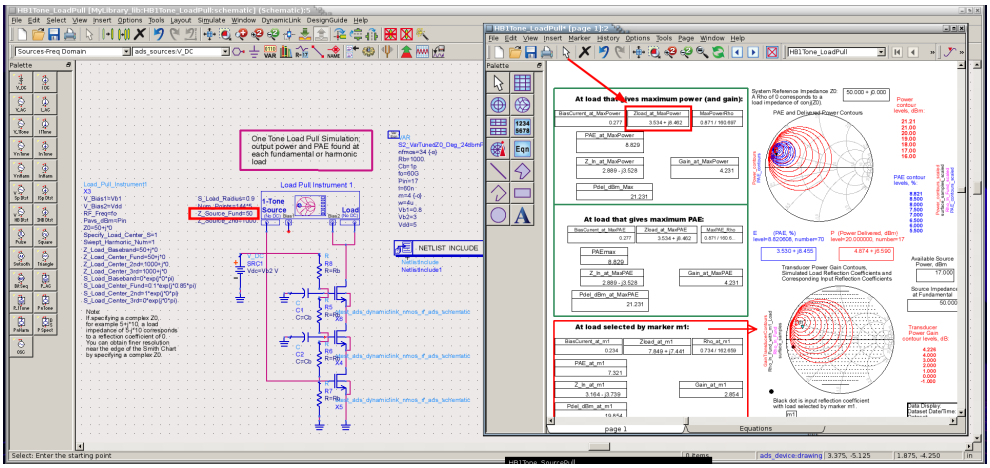


Figure 7.2: Loadpull for a 60 GHz PA output stage.

In the above example note that $Z_{load} \approx 3.5 + j8.5\Omega$ at a $Z_{source} = 50\Omega$. Loadpull simulations are important because they depend on Harmonic Balance Simulators that could capture the non-linear effects of transistors.

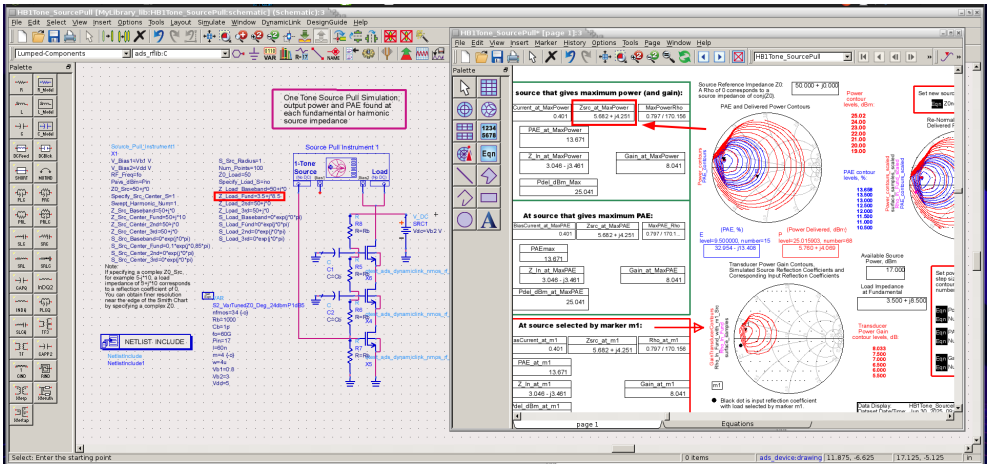


Figure 7.3: Sourcepull for a 60 GHz PA.

Similarly, from the input side, a simple conjugate match at the input side may not be sufficient sometimes, especially if a broadband design is warranted at very high frequencies. And thus **sourcepull** might be a better alternative. Taking the previous Z_{load} value, a sourcepull test will be conducted to find the optimal Z_{source}. Note that $Z_{source} \approx 5.7 + j4.3\Omega$ and the vastly improved $P1dB$ & gain.

This is an iterative process where both loadpull and sourcepull analyses are done until we converge to a single value. So loadpull analysis will be conducted once more.

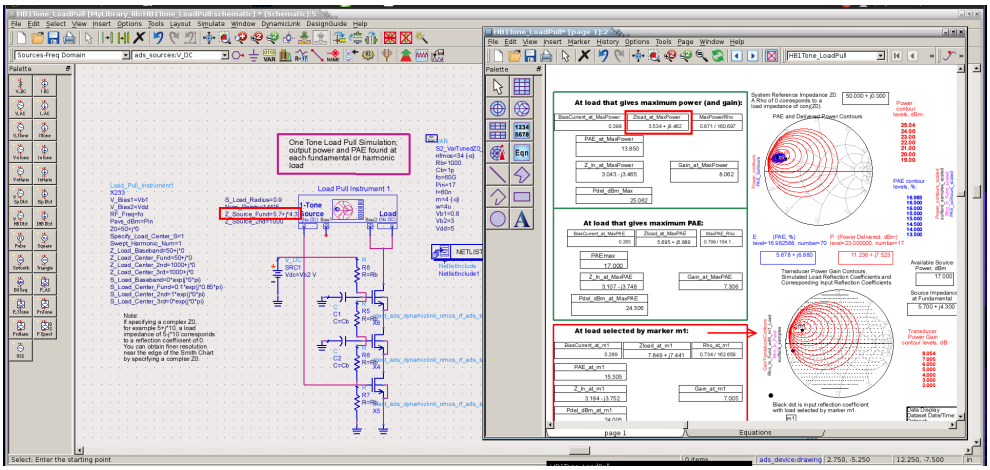


Figure 7.4: Loadpull (second run) for a 60 GHz PA with the new Z_{source}.

$Z_{load} \approx 3.5 + j8.5\Omega$. As discussed above, in the design of such a stage, transmission line matching will likely provide the best results.

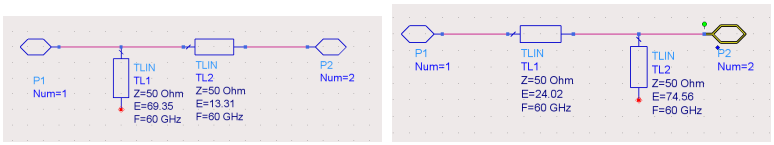


Figure 7.5: Transmission Line matching. Input: Left. Output: Right.

Note the broadband performance, a characteristic of transmission line matching. A similar testbench was used to the previous but with passives. This is merely a demonstration.

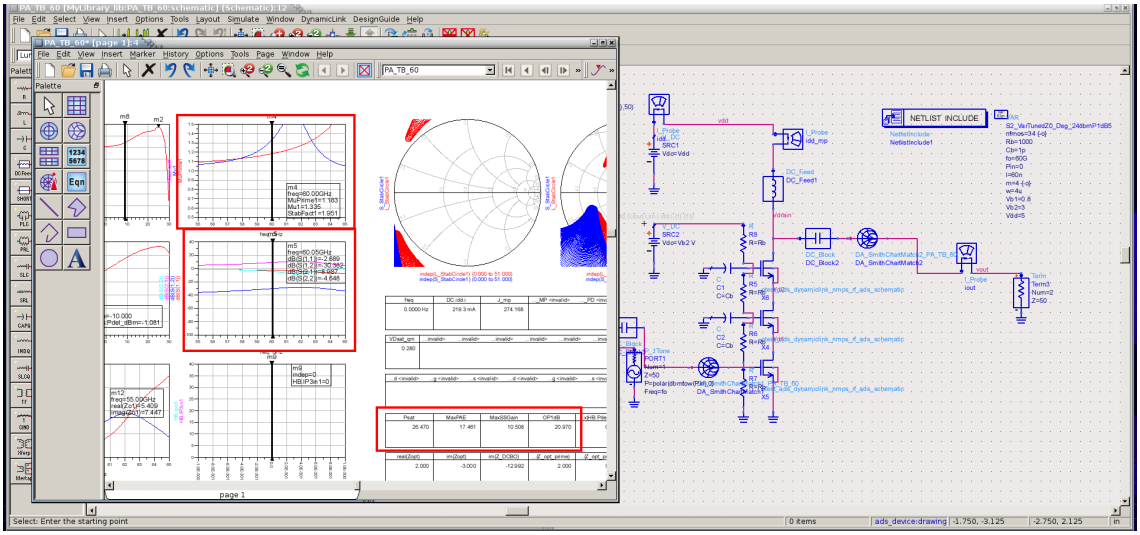


Figure 7.6: $60GHz$ PA output stage testbench & initial results.

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